

84C654349	TITLE SHEET	SECT. A0	TITLE SHEET	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
CONT. ON SH.2 SH. NO.1				

THE WABTEC CORPORATION
2901 EAST LAKE ROAD, ERIE, PA. 16531

DRAWING NO. 84C654349
DRIVE SYSTEM SCHEMATIC
NHL 240T HIGH ALTITUDE HAUL TRUCK
GE 240 AC QUAD CHOPPER
INVERTEX II

MAIN GROUP MODEL: 17KG560D
RETARDER GRID MODEL: 17EM146
AC WHEEL MODEL: 5GEB25A
AC ALTERNATOR MODEL: 5GTA41

0	1	2	3	4	5	6	7	8	9
A	SARIKA A 24APR2023 TC13			TITLE: NHL 240 TAC QC II SCHEMATIC	MADE BY: MATHLV	DATE: 06APR2022	APPROVED BY: R.LUDWIGSON		SECT. A0
0	MATHLV 06APR2022 TC13			FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	ISSUED BY: MATHLV	DATE: 06APR2022	06APR2022		9



ACES

84C654349
CONT. ON SH.2 SH. NO.1

84C654349
CONT. ON SH.3 SH. NO.2

REVISION SHEET

SECT. A1

REVISION SHEET

THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.

REV	DATE	CO/ECR	AUTHOR	SHEETS AFFECTED	APPROVAL	COMMENTS
0	04-06-2022	TEAMCENTER13	MATHI.V	ALL SHEETS	RUSS LUDWIGSON	INITIAL RELEASE
A	24-02-2023	TEAMCENTER13	SARIKA.A	A0,A1,A2,A3,A11,A12,V1,V3,V4	RUSS LUDWIGSON	A0: UPDATED CORRECT AC ALTERNATOR AND MAIN GROUP MODEL PART NUMBER A1: UPDATED REVISION HISTORY TABLE SHEET. A2 & A3: UPDATED REVISION NUMBERS COLUMN IN INDEX SHEET. A11: REMOVED POT REFERENCE. ACCELERATION PEDAL RETARD SPEED CONTROL SIGNALS 17FB173 SLOT 5 & FULL RETARD REQUEST FROM 17FB104 SLOTS, REMOVED DIGITAL INPUTS FROM 17FB104 SLOT 7. A12: ADDED OTHERS IN ENGINE TO DSC DATA IN CAN DATA SHEET. V1: CHANGED THE PART NUMBER FOR CM1,CM2,CM3 & CM4. V3 & V4: CHANGED THE PART NUMBERS FOR PHASE MODULES IN DEVICE TABLE.

0123456789

A SARIKA.A 24APR2023 TC13

0 MATHI.V 06APR2022 TC13

REVISIONS

TITLE: NHL 240 TAC QC II SCHEMATIC
FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE

MADE BY: MATHI.V
ISSUED BY: MATHI.V

DATE: 06APR2022
DATE: 06APR2022

APPROVED BY:
R.LUDWIGSON
06APR2022


ACES

SECT. A1
9
84C654349
CONT. ON SH.3 SH. NO.2

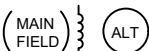
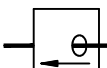
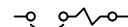
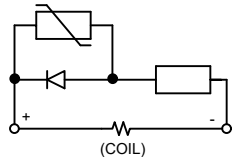
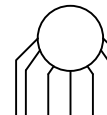
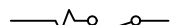
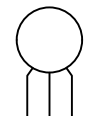
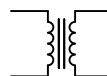
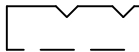
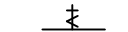
84C654349		INDEX SHEET	SECT. A2	INDEX SHEET	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
CONT. ON SH.4	SH. NO.3				

SEC	SECTION_LEFT	SECTION_RIGHT	SHT	REV
A0	TITLE SHEET	TITLE SHEET	1	A
A1	REVISION SHEET	REVISION SHEET	2	A
A2	INDEX SHEET	INDEX SHEET	3	A
A3	INDEX SHEET	INDEX SHEET	4	A
A4	SYMBOLS	SYMBOLS	5	0
A5	SYMBOLS	SYMBOLS	6	0
A6	BLOCK DIAGRAM: STATIC EXCITER, ALTERNATOR & RECTIFIER	BLOCK DIAGRAM: STATIC EXCITER, ALTERNATOR & RECTIFIER	7	0
A7	BLOCK DIAGRAM: FDR,GRR,VAM3, AND CAPACITOR CHARGE LIGHTS #1 & #2	BLOCK DIAGRAM: FDR,GRR,VAM3, AND CAPACITOR CHARGE LIGHTS #1 & #2	8	0
A8	BLOCK DIAGRAM: GRID BLOWER MOTOR, RETARD GRIDS, AND CHOPPERS	BLOCK DIAGRAM: GRID BLOWER MOTOR, RETARD GRIDS, AND CHOPPERS	9	0
A9	BLOCK DIAGRAM: INVERTERS #11 AND #12 AND WHEEL MOTOR	BLOCK DIAGRAM: INVERTERS #11 AND #12 AND WHEEL MOTOR	10	0
A10	BLOCK DIAGRAM: INVERTERS #21 AND #22 AND WHEEL MOTOR	BLOCK DIAGRAM: INVERTERS #21 AND #22 AND WHEEL MOTOR	11	0
A11	BLOCK DIAGRAM: DRIVE SYSTEM CONTROL	BLOCK DIAGRAM DRIVE SYSTEM CONTROL	12	A
A12	CAN DATA	CAN DATA	13	A
B	ALTERNATOR CIRCUITS	ALTERNATOR CIRCUITS	14	0
B1	RECTIFIER	DC LINK AND GROUND CIRCUITS	15	0
B2	RETARD CHOPPER MODULE #1	RETARD CHOPPER MODULE #2	16	0
B3	RETARD CHOPPER MODULE #3	RETARD CHOPPER MODULE #4	17	0
B4	GRID MOTOR PROTECTION	GRID BLOWER MOTORS & RETARD GRIDS	18	0
B5	INVERTER #11: LEFT WHEEL	INVERTER #11: LEFT WHEEL	19	0
B6	INVERTER #12: LEFT WHEEL	INVERTER #12: LEFT WHEEL	20	0
B7	INVERTER #21: RIGHT WHEEL	INVERTER #21: RIGHT WHEEL	21	0
B8	INVERTER #22: RIGHT WHEEL	INVERTER #22: RIGHT WHEEL	22	0
B9	WHEEL MOTORS	WHEEL MOTORS	23	0
B10	WHEEL MOTORS	WHEEL MOTORS	24	0
D	INVERTER #11 TMC: CPU AND FIBER OPTIC I/O	INVERTER #11 TMC: CPU AND FIBER OPTIC I/O	25	0
D1	INVERTER #12 TMC: CPU AND FIBER OPTIC I/O	INVERTER #12 TMC: CPU AND FIBER OPTIC I/O	26	0
D2	INVERTER #11 & #12 PHASE A, B, C CURRENT SENSOR MODULES	INVERTER #11 & #12 PHASE A, B, C CURRENT SENSOR MODULES	27	0
D3	INVERTER #11 & #12 TMC: PHASE CURRENT I/O	INVERTER #11 & #12 TMC: PHASE CURRENT I/O	28	0
D4	INVERTER #11 LINK AND INVERTER #11 & #12 PHASE VOLTAGE I/O	BLANK	29	0
D5	INVERTER #11 & #12 TMC: SPEED SENSOR I/O	INVERTER #11 & #12 TMC: SPEED SENSOR I/O	30	0
D6	INVERTER #21 TMC: CPU AND FIBER OPTIC I/O	INVERTER #21 TMC: CPU AND FIBER OPTIC I/O	31	0
D7	INVERTER #22 TMC: CPU AND FIBER OPTIC I/O	INVERTER #22 TMC: CPU AND FIBER OPTIC I/O	32	0
D8	INVERTER #21 & #22 PHASE A, B, C CURRENT SENSOR MODULES	INVERTER #21 & #22 PHASE A, B, C CURRENT SENSOR MODULES	33	0
D9	INVERTER #21 & #22 TMC: PHASE CURRENT I/O	INVERTER #21 & #22 TMC: PHASE CURRENT I/O	34	0
D10	INVERTER #21 LINK AND INVERTER #21 & #22 PHASE VOLTAGE I/O	BLANK	35	0
D11	INVERTER #21 & #22 TMC: SPEED SENSOR I/O	INVERTER #21 & #22 TMC: SPEED SENSOR I/O	36	0
D12	INVERTER #11 & #12 GATE DRIVERS & POWER CONVERTER	INVERTER #11 & #12 GATE DRIVERS & POWER CONVERTER	37	0

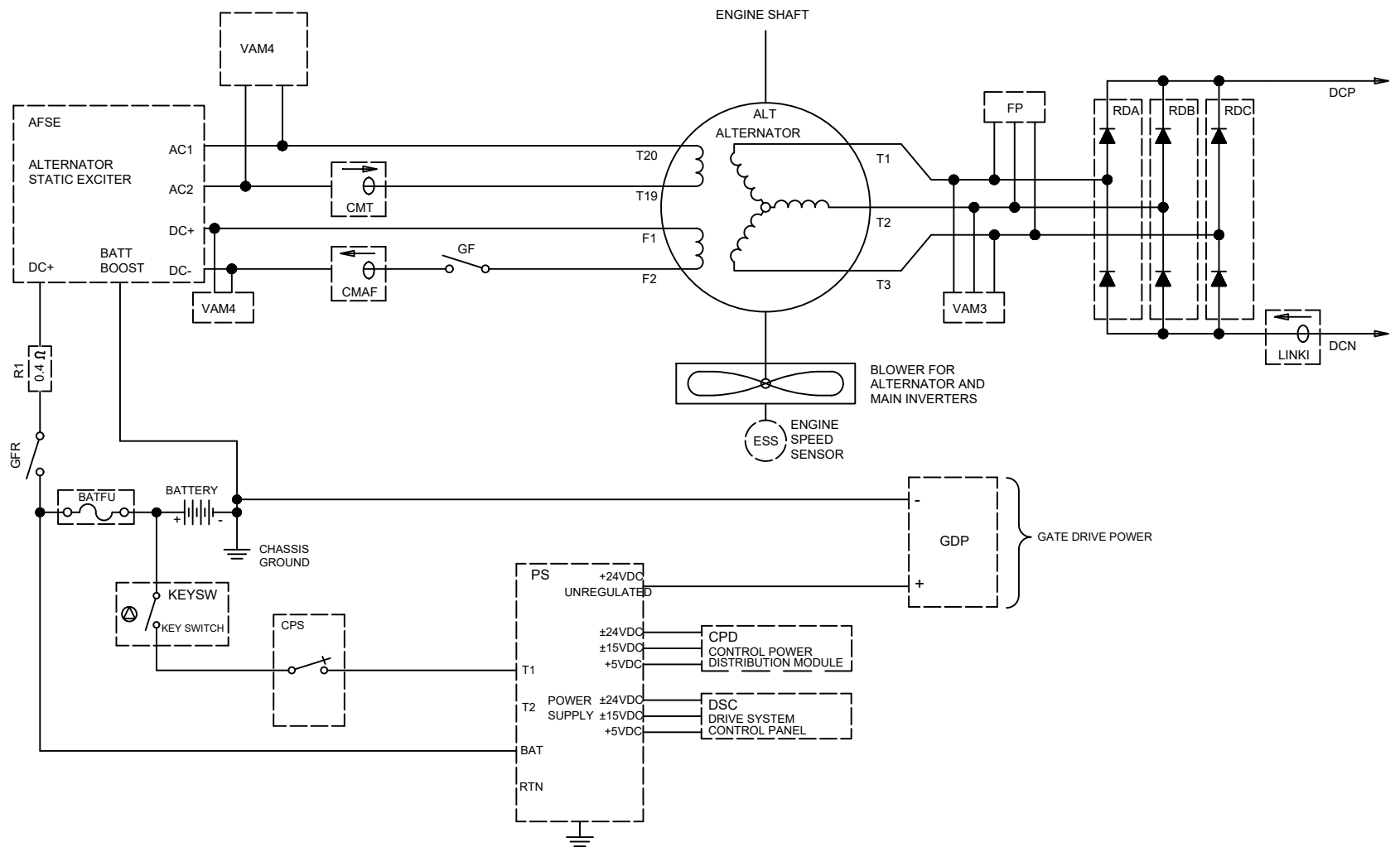
SEC	SECTION_LEFT	SECTION_RIGHT	SHT	REV
D13	INVERTER #21 & #22 GATE DRIVERS & POWER CONVERTER	INVERTER #21 & #22 GATE DRIVERS & POWER CONVERTER	38	0
D14	CAPACITOR CHARGED RESISTOR PANELS & LIGHTS	CAPACITOR CHARGED RESISTOR PANELS & LIGHTS	39	0
D15	CONTROL POWER DISTRIBUTION	CONTROL POWER DISTRIBUTION	40	0
D16	DSC ANALOG CARD: CURRENT SENSOR	DSC ANALOG CARD: CURRENT SENSOR	41	0
D17	DSC ANALOG CARD: CURRENT SENSOR	DSC ANALOG CARD: CURRENT SENSOR	42	0
D18	DSC ANALOG CARD: CURRENT SENSOR	DSC ANALOG CARD: CURRENT SENSOR	43	0
D19	DSC ANALOG CARD: ALT. PHASE, LINK & ALT. FIELD VOLTAGE MODULES	DSC ANALOG CARD: ALT. PHASE, LINK & ALT. FIELD VOLTAGE MODULES	44	0
D20	CONTROL POWER DISTRIBUTION	DSC ANALOG CARD	45	0
D21	DSC ANALOG CARD	DSC ANALOG CARD	46	0
D22	DSC ANALOG CARD	DSC ANALOG CARD	47	0
D23	DSC ANALOG CARD: NON DIRECTIONAL FRONT SPEED SENSORS	DSC ANALOG CARD: SPEED SENSOR INPUTS	48	0
D24	DSC ANALOG CARD	DSC ANALOG CARD	49	0
D25	DSC CPU: SERIAL I/O	DSC CPU: SERIAL I/O	50	0
D26	DSC CPU: CAN	DSC CPU: CAN	51	0
D27	ETHERNET CARD AND CONNECTIONS	ETHERNET CARD AND CONNECTIONS	52	0
D28	ETHERNET CARD AND CONNECTIONS	ETHERNET CARD AND CONNECTIONS	53	0
D29	DSC DIGITAL I/O CARD: #1 OUTPUTS	DSC DIGITAL I/O CARD: #1 OUTPUTS	54	0
D30	DSC DIGITAL I/O CARD: #1 OUTPUTS	DSC DIGITAL I/O CARD: #1 OUTPUTS	55	0
D31	DSC DIGITAL I/O CARD: #2 OUTPUTS	DSC DIGITAL I/O CARD: #2 OUTPUTS	56	0
D32	DSC DIGITAL I/O CARD: #2 OUTPUTS	DSC DIGITAL I/O CARD: #2 OUTPUTS	57	0
D33	DSC DIGITAL I/O CARD: #1 INPUTS	DSC DIGITAL I/O CARD: #1 INPUTS	58	0
D34	DSC DIGITAL I/O CARD: #1 INPUTS	DSC DIGITAL I/O CARD: #1 INPUTS	59	0
D35	DSC DIGITAL I/O CARD: #2 INPUTS	DSC DIGITAL I/O CARD: #2 INPUTS	60	0
D36	DSC DIGITAL I/O CARD: #2 INPUTS	BLANK	61	0
E	LOW VOLTAGE POWER CIRCUIT	LOW VOLTAGE POWER CIRCUIT	62	0
E1	POWER SUPPLY	POWER SUPPLY	63	0
E2	POWER SUPPLY	POWER SUPPLY	64	0
E3	CONTROL POWER SUPPLY CONNECTOR (PS2)	BLANK	65	0
E4	CONTROL POWER DISTRIBUTION	CONTROL POWER DISTRIBUTION	66	0
E5	CONTROL POWER DISTRIBUTION	CONTROL POWER DISTRIBUTION	67	0
E6	CONTROL POWER DISTRIBUTION	DSC ANALOG CARD / DID, PEDAL, RETARD LEVER, & RSC CIRCUITS	68	0
K	CONTROL AREA LIGHTING	CONTROL AREA LIGHTING	69	0
P	CONTROL GROUP AREA LOCATIONS: FRONT VIEW	CONTROL GROUP AREA LOCATIONS: FRONT VIEW	70	0

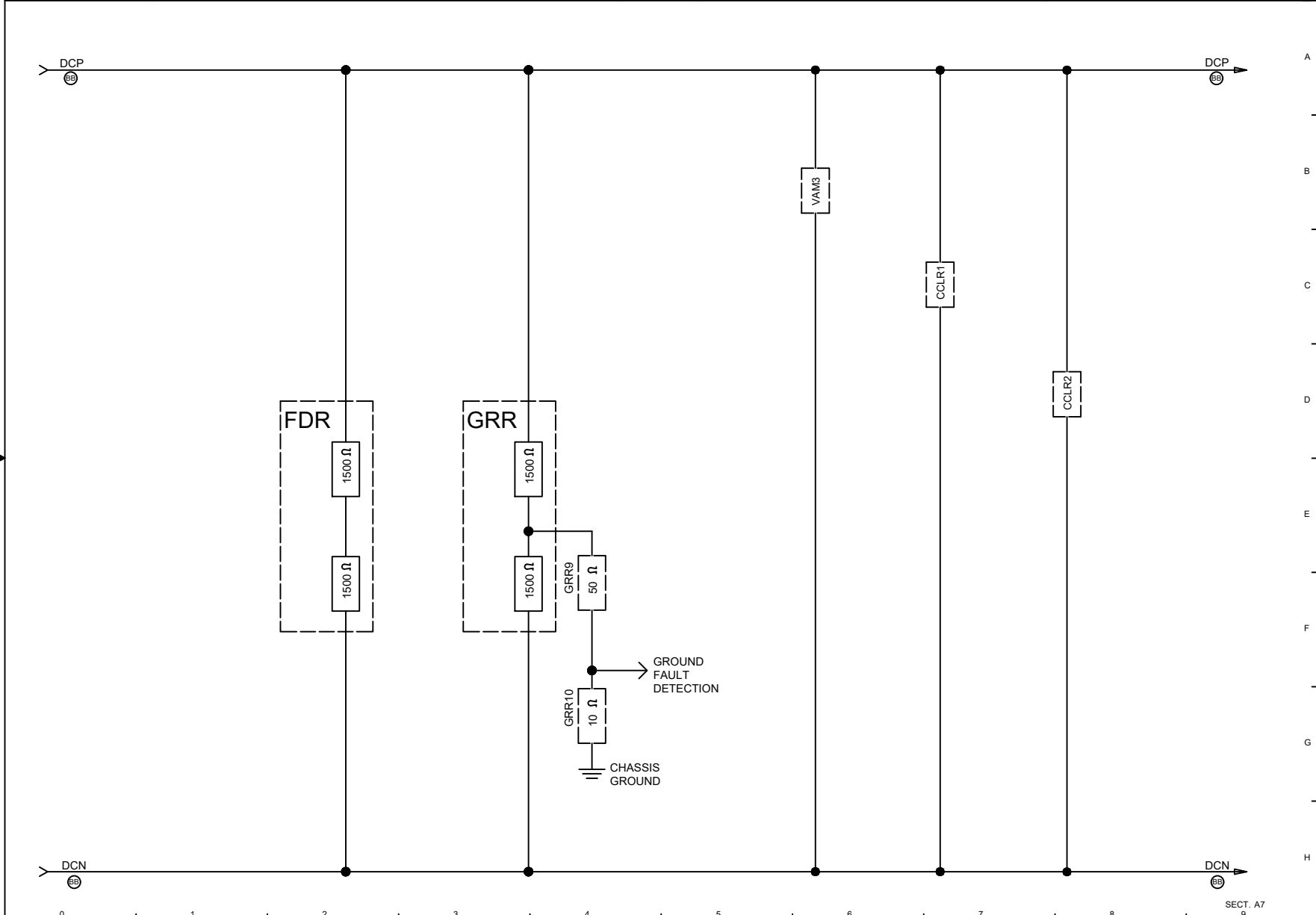
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A	SARIKA.A 24APR2023 TC13					REVISIONS	TITLE: NHL 240 TAC QC II SCHEMATIC			MADE BY: MATHI.V		DATE: 06APR2022		APPROVED BY: R.LUDWIGSON				84C654349			
0	MATHI.V 06APR2022 TC13						FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE			ISSUED BY: MATHI.V		DATE: 06APR2022		06APR2022				CONT. ON SH.4		SH. NO.3	
																		ACES			

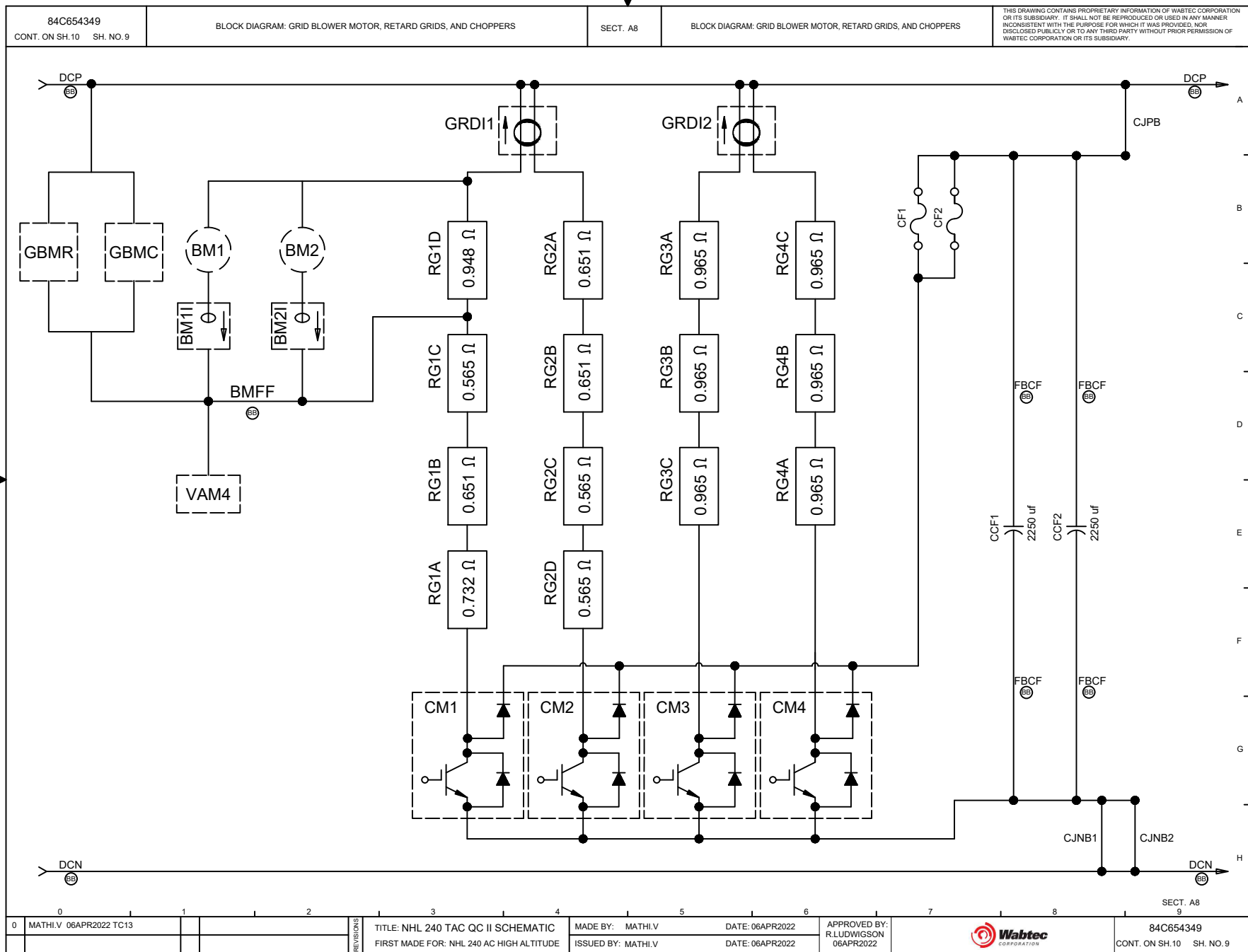
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CONT. ON SH.5 SH. NO.4									
SEC	SECTION_LEFT	SECTION_RIGHT	SHT	REV					
P1	CONTROL GROUP AREA LOCATIONS: LEFT VIEW	CONTROL GROUP AREA LOCATIONS: LEFT VIEW	71	0					
P2	CONTROL GROUP AREA LOCATIONS: BACK VIEW	CONTROL GROUP AREA LOCATIONS: BACK VIEW	72	0					
P3	CONTROL GROUP AREA LOCATIONS: CA1	CONTROL GROUP AREA LOCATIONS: CA1	73	0					
P4	CONTROL GROUP AREA LOCATIONS: CA2	CONTROL GROUP AREA LOCATIONS: CA2	74	0					
P5	CONTROL GROUP AREA LOCATIONS: CA3	CONTROL GROUP AREA LOCATIONS: CA3	75	0					
P6	CONTROL GROUP AREA LOCATIONS: CA4	CONTROL GROUP AREA LOCATIONS: CA4	76	0					
P7	CONTROL GROUP AREA LOCATIONS: CA5	CONTROL GROUP AREA LOCATIONS: CA5	77	0					
P8	CONTROL GROUP AREA LOCATIONS: CA6	CONTROL GROUP AREA LOCATIONS: CA6	78	0					
P9	CONTROL GROUP AREA LOCATIONS: CA7	CONTROL GROUP AREA LOCATIONS: CA7	79	0					
P10	CONTROL GROUP AREA LOCATIONS: CA8	CONTROL GROUP AREA LOCATIONS: CA8	80	0					
P11	DRIVE SYSTEM CONTROL PANEL	DRIVE SYSTEM CONTROL PANEL	81	0					
R	RECEPTACLE AND PLUG DIAGRAMS	RECEPTACLE AND PLUG DIAGRAMS	82	0					
R1	RECEPTACLE AND PLUG DIAGRAMS	RECEPTACLE AND PLUG DIAGRAMS	83	0					
R2	RECEPTACLE AND PLUG DIAGRAMS	RECEPTACLE AND PLUG DIAGRAMS	84	0					
R3	RECEPTACLE AND PLUG DIAGRAMS	RECEPTACLE AND PLUG DIAGRAMS	85	0					
R4	RETARD GRID RESISTORS (17EM146)	RETARD GRID RESISTORS (17EM146)	86	0					
R5	LAMINATED BUS BARS	BLANK	87	0					
R6	INTERFACE CABLE LIST & WIRING GUIDELINES	INTERFACE CABLE LIST & WIRING GUIDELINES	88	0					
R7	FIBER OPTIC TABLE	FIBER OPTIC TABLE	89	0					
S	TERMINAL BOARD INDEX	TERMINAL BOARD INDEX	90	0					
S1	DEVICE TABLE	DEVICE TABLE	91	0					
S2	PANEL-PLUG PIN LOCATION	PANEL-PLUG PIN LOCATION	92	0					
S3	PANEL-PLUG PIN LOCATION	PANEL-PLUG PIN LOCATION	93	0					
S4	PANEL-PLUG PIN LOCATION	PANEL-PLUG PIN LOCATION	94	0					
T	RELAY AND CONTACTOR LOCATION	RELAY AND CONTACTOR LOCATION	95	0					
U	BATTERY POSITIVE WIRE ROUTING	BATTERY POSITIVE WIRE ROUTING	96	0					
U1	+24 VDC POWER SUPPLY WIRING	+24 VDC POWER SUPPLY WIRING	97	0					
U2	-24 VDC POWER SUPPLY WIRING	-24 VDC POWER SUPPLY WIRING	98	0					
U3	+5 VDC POWER SUPPLY WIRING	BLANK	99	0					
U4	+15 VDC POWER SUPPLY WIRING	-15 VDC POWER SUPPLY WIRING	100	0					
U5	GROUND WIRE ROUTING	GROUND WIRE ROUTING	101	0					
U6	GROUND WIRE ROUTING	GROUND WIRE ROUTING	102	0					
U7	GROUND WIRE ROUTING	GROUND WIRE ROUTING	103	0					
V	DEVICE TABLE	DEVICE TABLE	104	0					
V1	DEVICE TABLE	DEVICE TABLE	105	A					
V2	DEVICE TABLE	DEVICE TABLE	106	0					
V3	DEVICE TABLE	DEVICE TABLE	107	A					
V4	DEVICE TABLE	DEVICE TABLE	108	A					
V5	DEVICE TABLE	DEVICE TABLE	109	0					

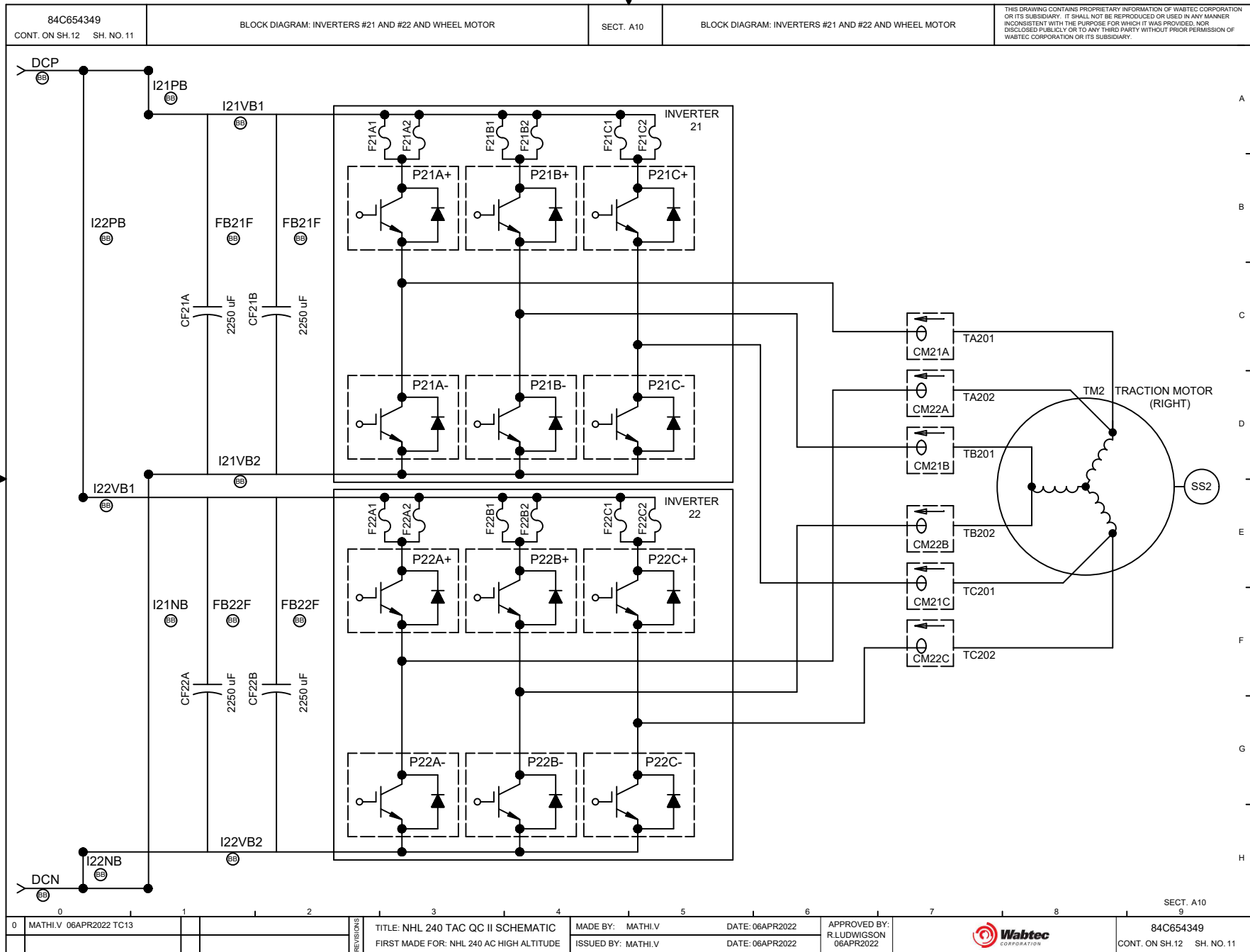
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CONT. ON SH.6 SH. NO. 5									
 FUSE		 RESISTOR		 DIODE		 SCR		A	
 DC MOTOR		 CAPACITOR (NON - POLARIZED)		 CAPACITOR (POLARIZED)					
 AC MOTOR		 CAPACITOR (POLARIZED)		 IGBT (INSULATED GATE BIPOLAR TRANSISTOR)		 METER		B	
 (TERTIARY)		 INDICATING LIGHT OR LED (COLOR - RED)		 TRANSISTOR					
 ALTERNATOR		 VARIABLE RESISTOR		 INDICATOR LIGHT		 DC CURRENT SENSOR		C	
 CIRCUIT BREAKER		 FIXED TAP RESISTOR		 VARISTOR MOV TYPE					
 SHUNT COIL RELAY OR CONTACTOR		 RES/DIODE/ MOV SUPPRESSION MODULE (SHOWN WITH COIL)		 CURRENT TRANSFORMER		 SPEED SENSOR (ACTIVE)		D	
 CONTACTOR (WITH BLOW - OUT COIL)		 TEMPERATURE SENSOR		 SPEED SENSOR (PASSIVE)					
 SINGLE BREAK INTERLOCK OR CONTACT (SHOWN NORMALLY CLOSED)		 INDUCTOR		 TRANSFORMER		 ELECTRONIC CARD		E	
 DOUBLE BREAK INTERLOCK OR CONTACT (SHOWN NORMALLY OPEN)		 HIGH VOLTAGE INDICATOR LIGHT							
 DOUBLE THROW INTERLOCK OR CONTACT								F	
 PRESSURE SWITCH									
 HAND - OPERATED SWITCH								G	
 HAND - OPERATED SWITCH (SPRING RETURN)									
								H	

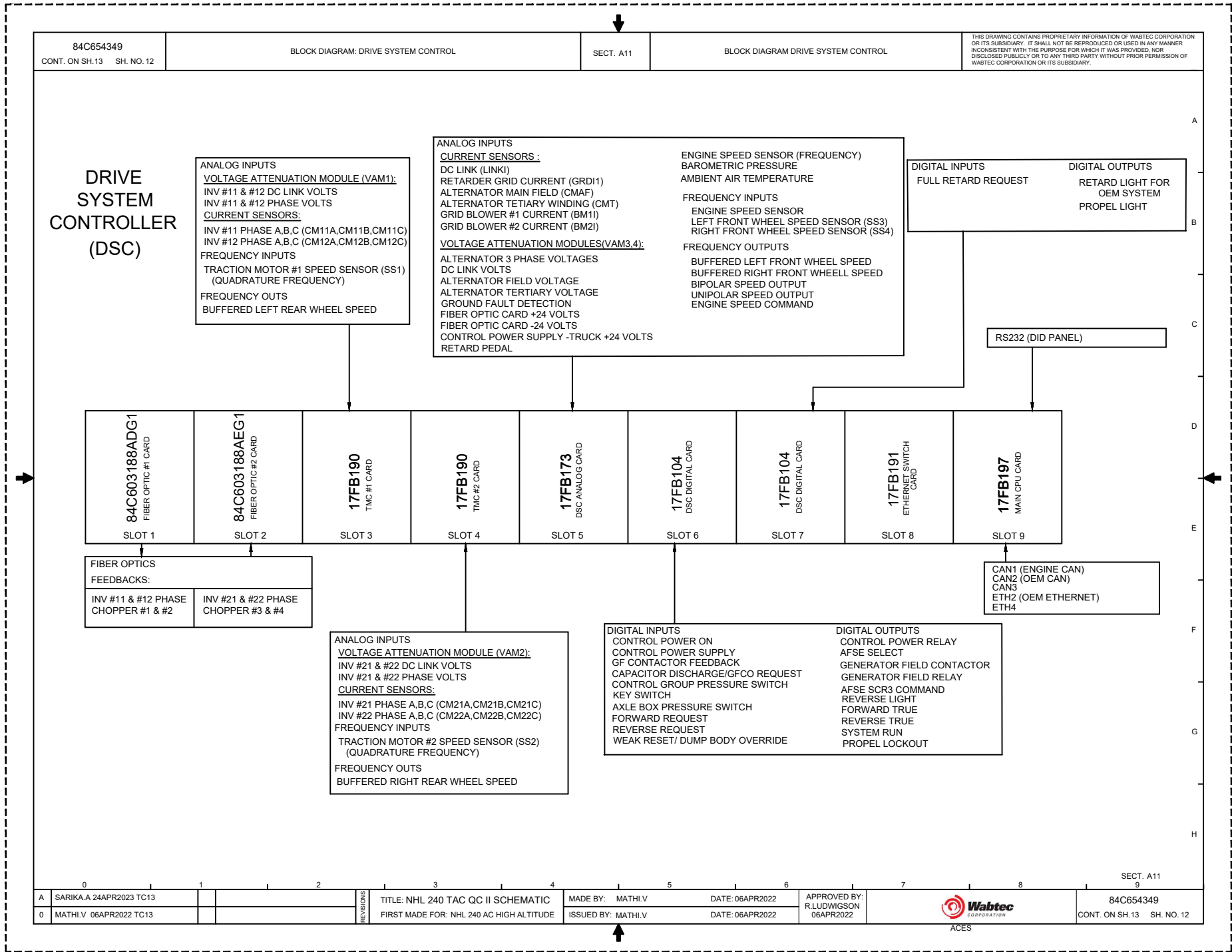
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CONT. ON SH.7		SH. NO. 6							
		BATTERY				BUS BAR CONNECTION		A	
		BOUNDARY OF A REPLACEABLE UNIT WITHIN A CONTROL GROUP SUCH AS ELECTRONIC CARD OR PANEL				ELECTRONIC PANEL CONNECTOR PIN		B	
		BOUNDARY OF A CONTROL GROUP OR MECHANICAL GROUPING				NO CIRCUIT CONNECTION		C	
		TRUCK BODY COMMON				CIRCUIT CONNECTION		D	
		ELECTRONIC PANEL GROUND				INSULATED CONNECTION POINT		E	
		INDICATES DEVICE OR COMPONENT NOT FURNISHED BY G.E. CO.				TERMINAL BOARD CONNECTION POINT		F	
		REMOVABLE BUS BAR				WIRES TWISTED TOGETHER		G	
		BUS BAR				TWISTED SHIELDED WIRE		H	
		(B3) BUS BAR CONTINUES ON OTHER PAGE				WIRE FOLD BACK			
		DEVICE LINK (WIRE IS CONTINUED TO DEVICE DSCBP, CONNECTION POINT 52, AT SECTION D, ROW D, COLUMN 3)				WIRE CUT BACK			
				WIRE HAS NO CONNECTION ON ANOTHER END WIRE CUT BACK OR FOLDED BACK SEE CROSS REFERENCE LINK SHEET					











84C654349	CAN DATA	SECT. A12	CAN DATA	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
CONT. ON SH.14 SH. NO.13				

CAN DATA

NHL TO DSC

SERVICE BRAKE
 PARK BRAKE
 LOAD BRAKE
 RETARD SPEED CONTROL COMMAND
 DIRECTION INPUT- NEUTRAL
 ENGINE WARM-UP
 ENGINE RUN ENABLE(ENGINE KILL)
 ENGINE CRANKING
 GVW(GROSS VEHICLE WEIGHT)
 NO PAY LOAD
 OVERLOAD
 DUMP BODY DOWN
 REST REQUEST
 DATA STORE
 LAMP TEST
 ACCELERATION PEDAL
 RETARD PEDAL
 RSC POTENTIOMETER

ENGINE TO DSC

ENGCAU, CHECK ENGINE LIGHT
 ENGWAR, STOP ENGINE LIGHT
 OTHERS(ACCORDING TO J1939)

DSC TO NHL

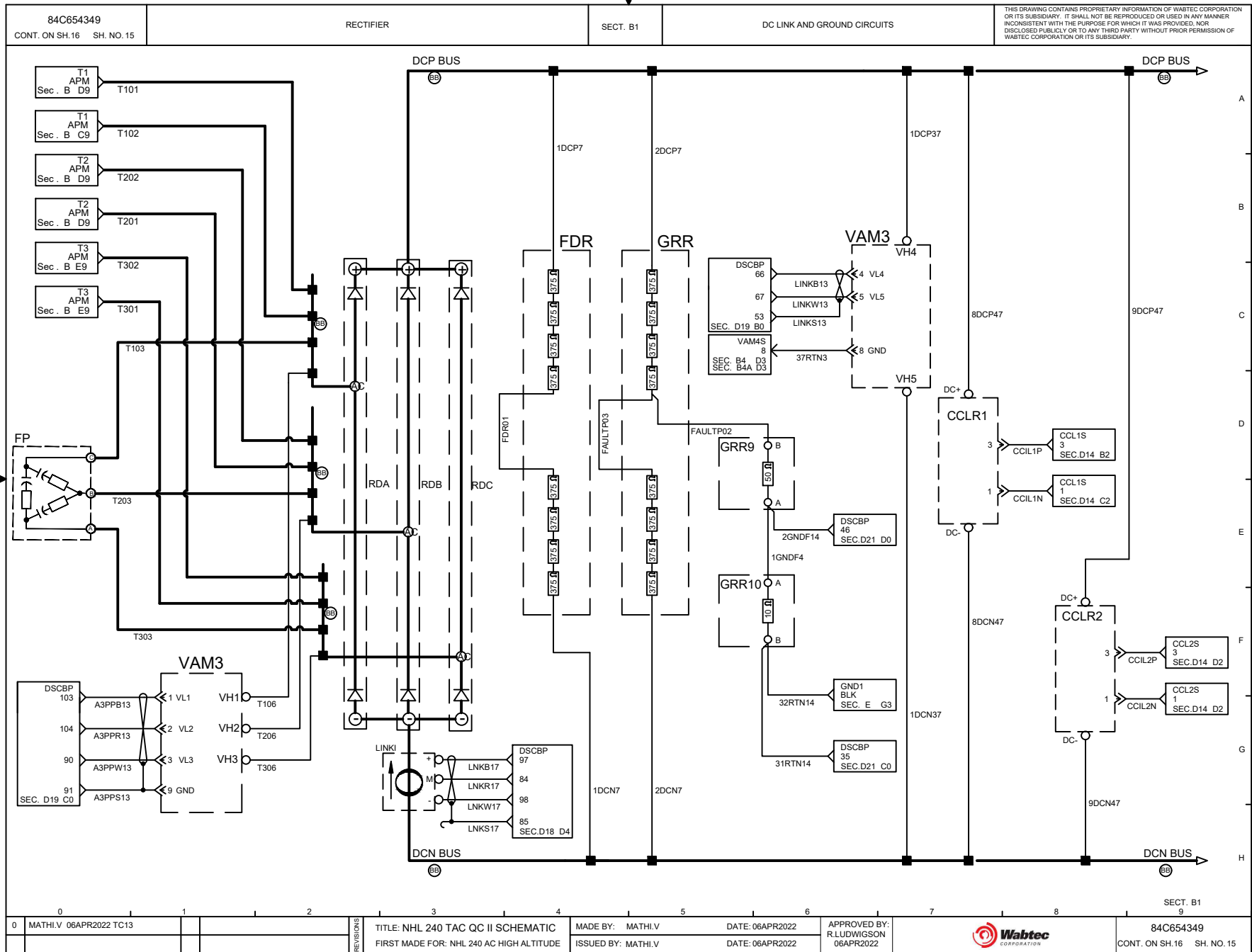
SET SPEED 1
 SET SPEED 2
 RETARD LIGHT
 RETARD LIGHT FOR OEM SYSTEM
 REVERSE LIGHT
 PROPEL LIGHT
 SPEED LIMIT OR SYSTEM FAULT LIGHT
 NO PROPULSION LIGHT
 NO RETARD LIGHT
 RETARD CONTINOUS LIGHT
 TRUCK SPEED
 ENGINE SPEED
 SEE OTHER DATA IN 41A328105AM OEM
 CAN ICD

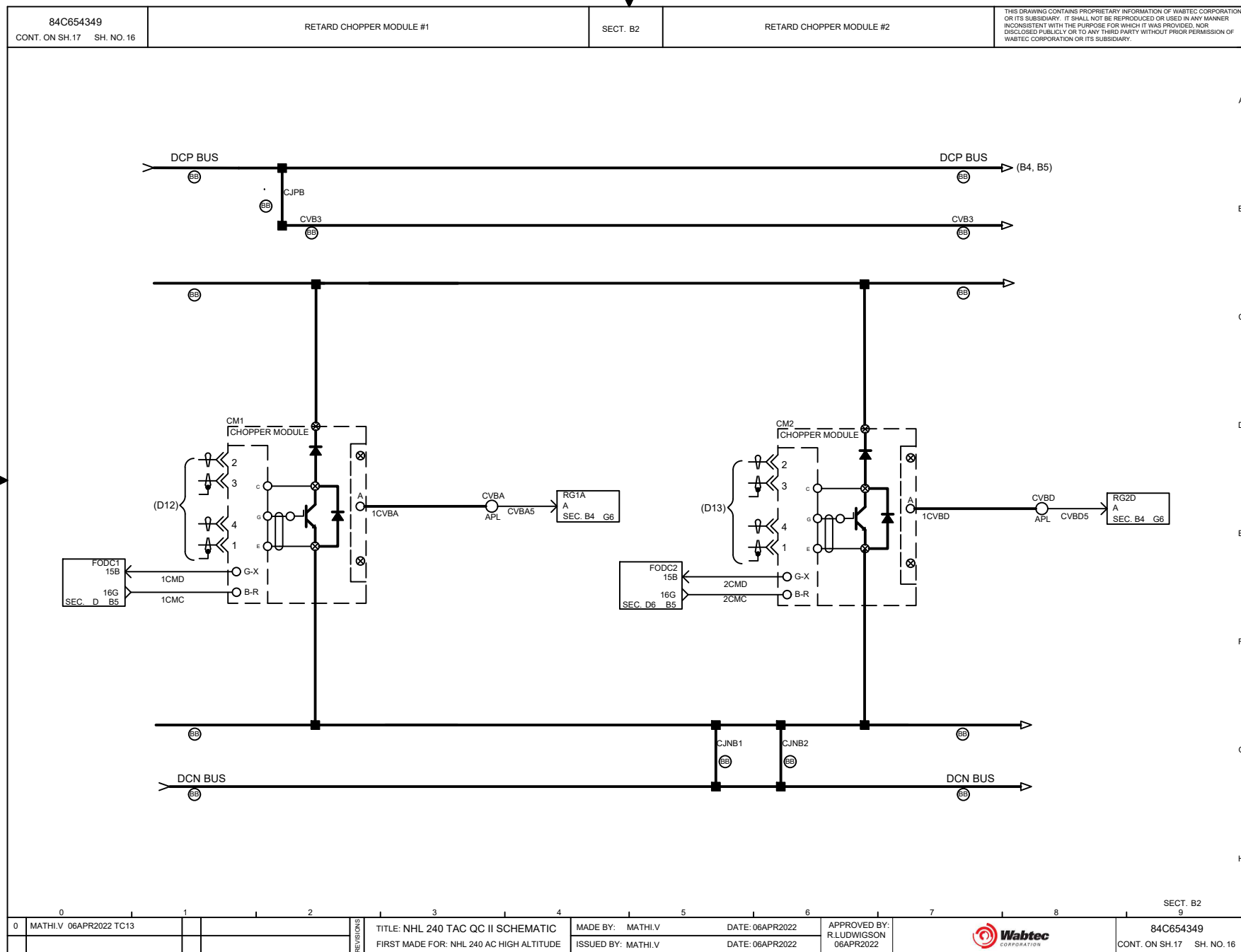
DSC TO ENGINE

TSC1/ENGINE RPM COMMAND

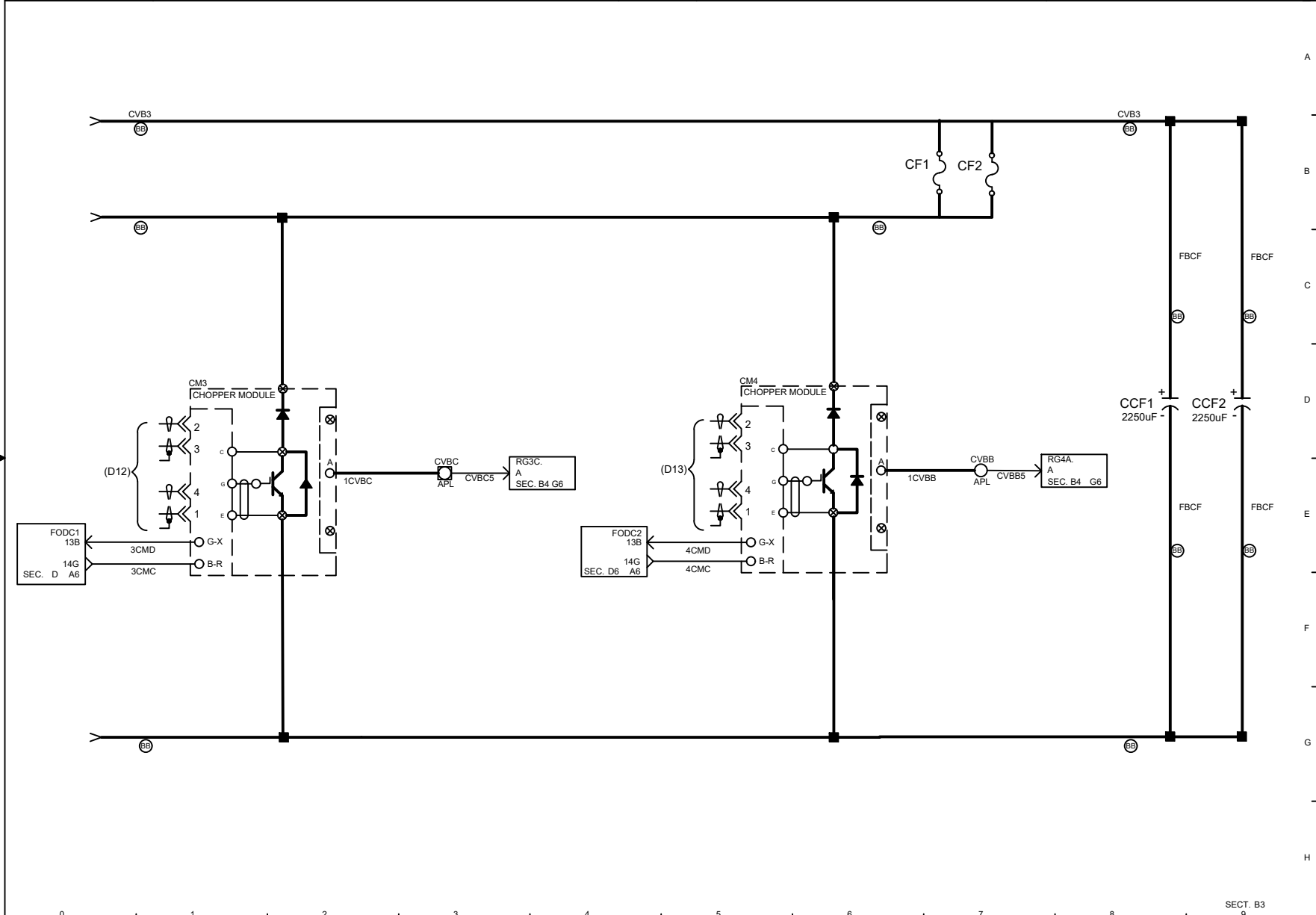
0	1	2	3	4	5	6	7	8	9
A	SARIKA A 24APR2023 TC13			TITLE: NHL 240 TAC QC II SCHEMATIC	MADE BY: MATHI.V	DATE: 06APR2022	APPROVED BY: R.LUDWIGSON		SECT. A12
0	MATHI.V 06APR2022 TC13			FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	ISSUED BY: MATHI.V	DATE: 06APR2022	06APR2022	84C654349	CONT. ON SH.14 SH. NO. 13







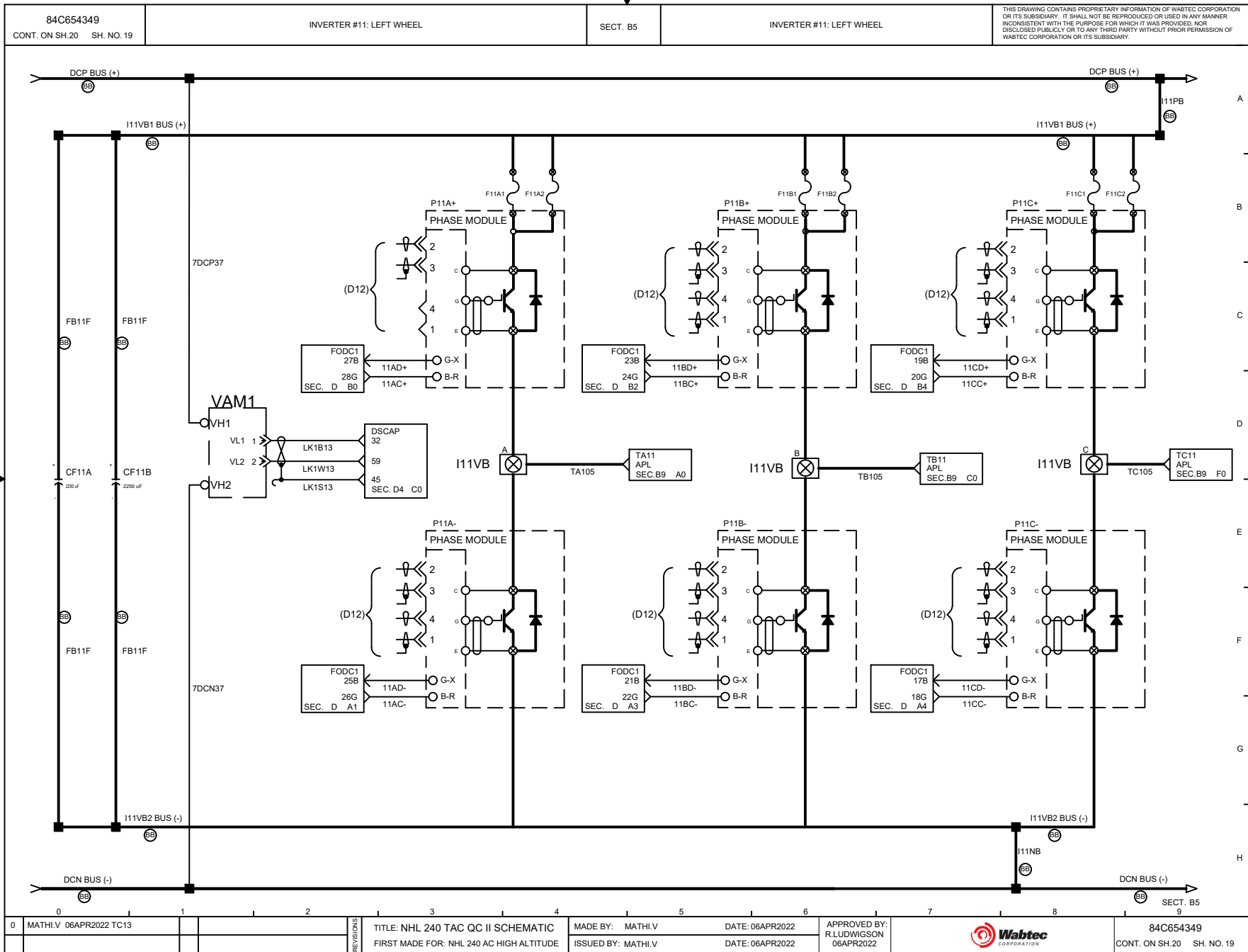
84C654349 CONT. ON SH.18 SH. NO.17	RETARD CHOPPER MODULE #3	SECT. B3	RETARD CHOPPER MODULE #4	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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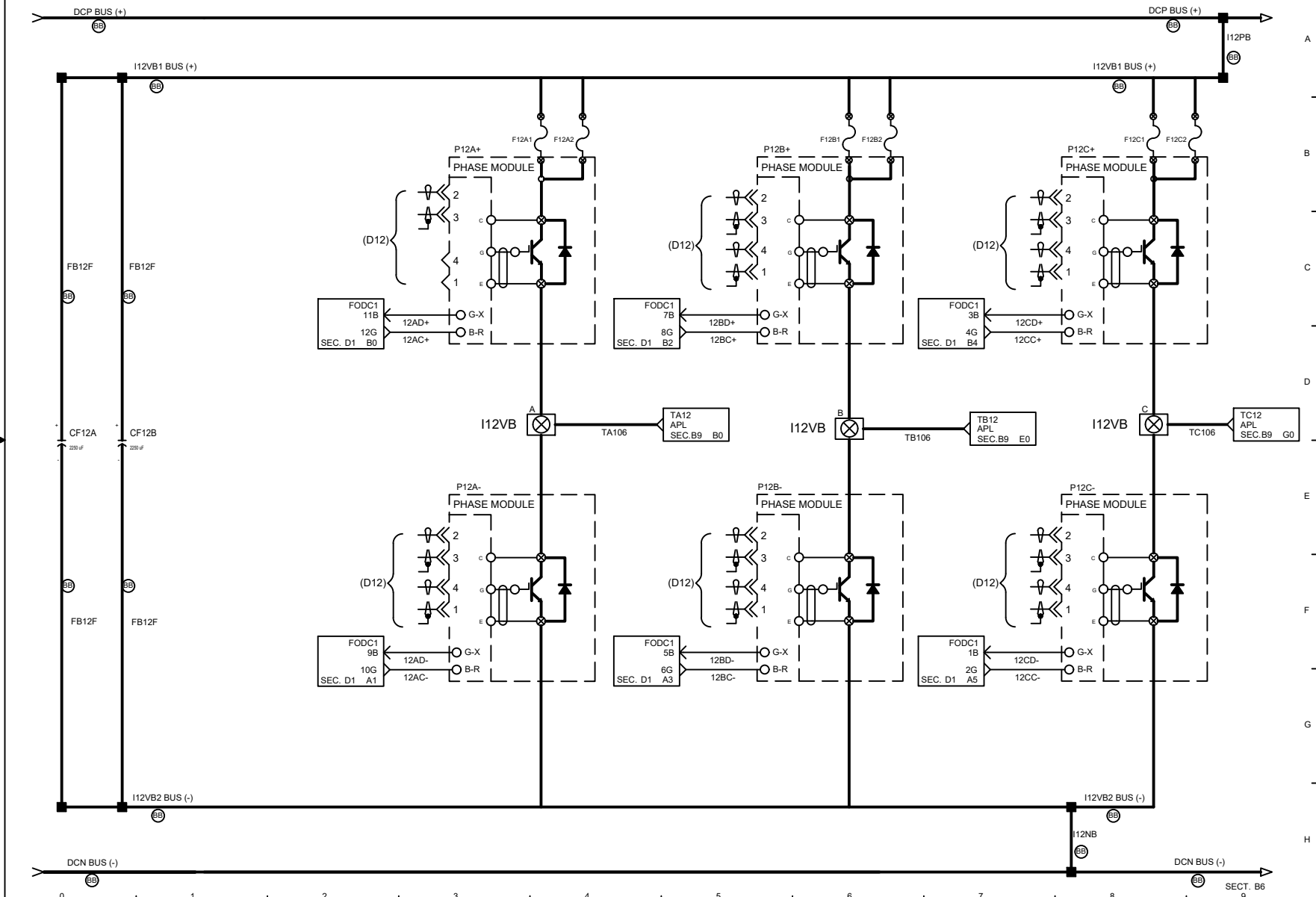


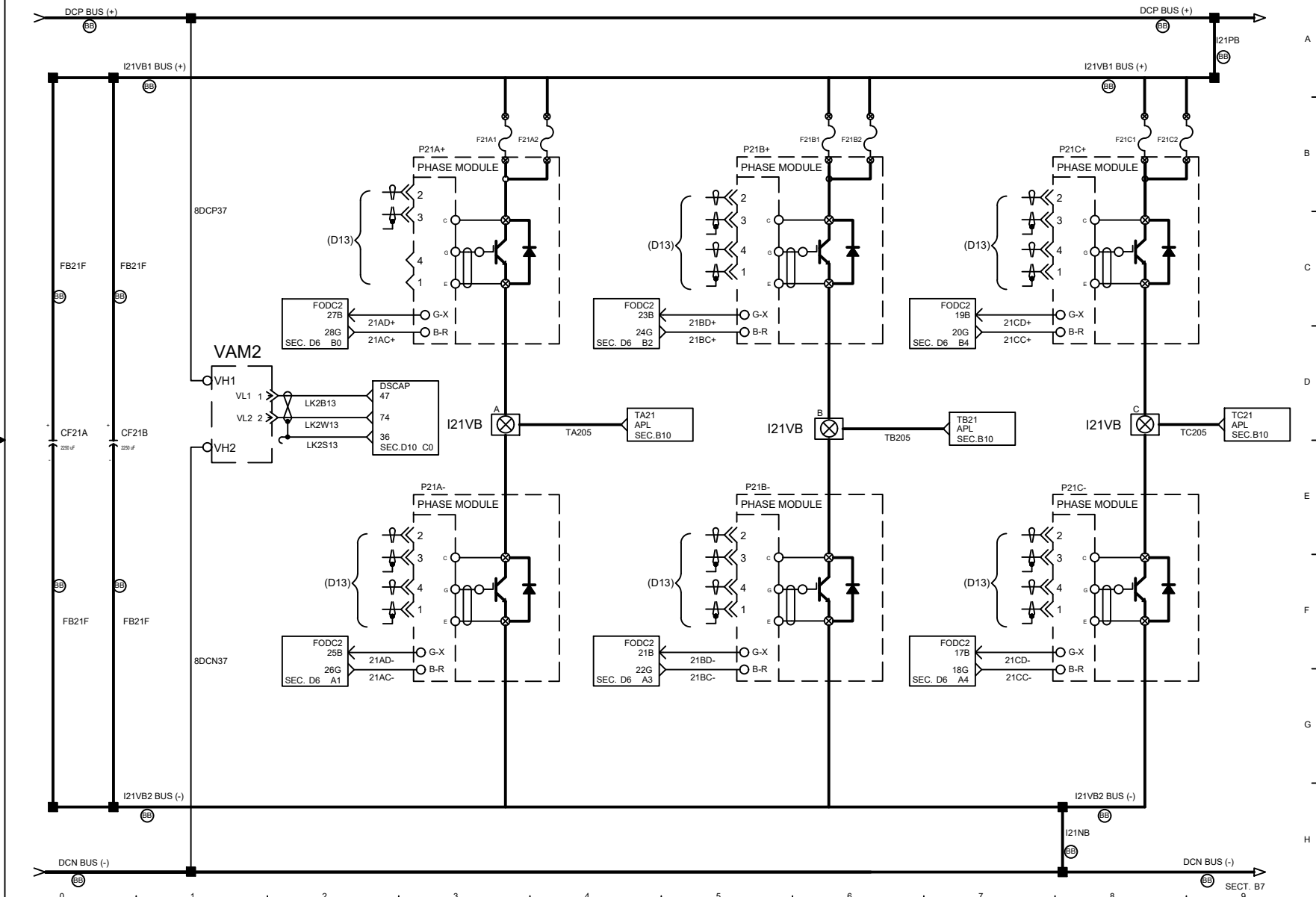
0 MATHI.V 06APR2022 TC13	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATHI.V ISSUED BY: MATHI.V	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: R.LUDWIGSON 06APR2022		SECT. B3 84C654349 CONT. ON SH.18 SH. NO. 17
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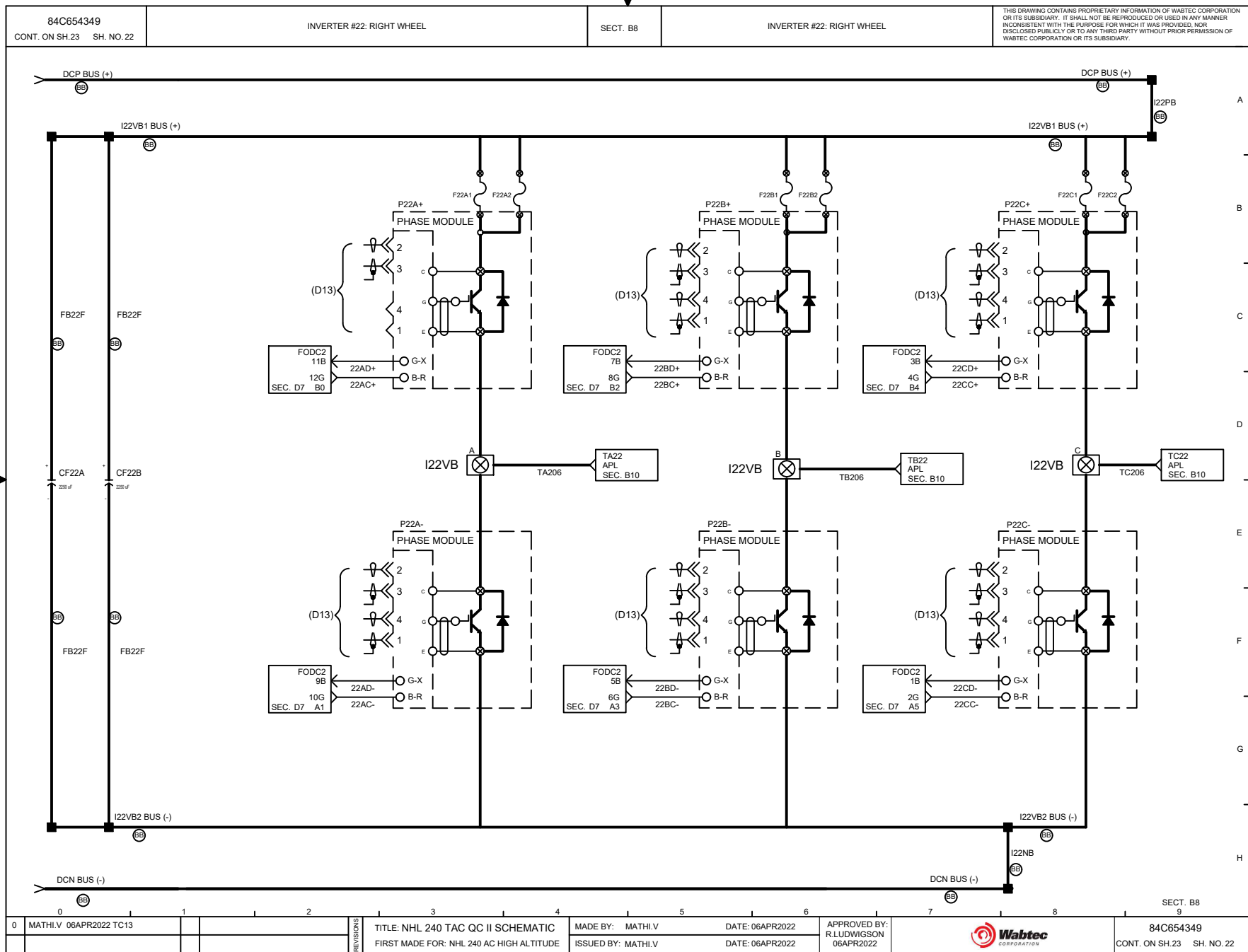


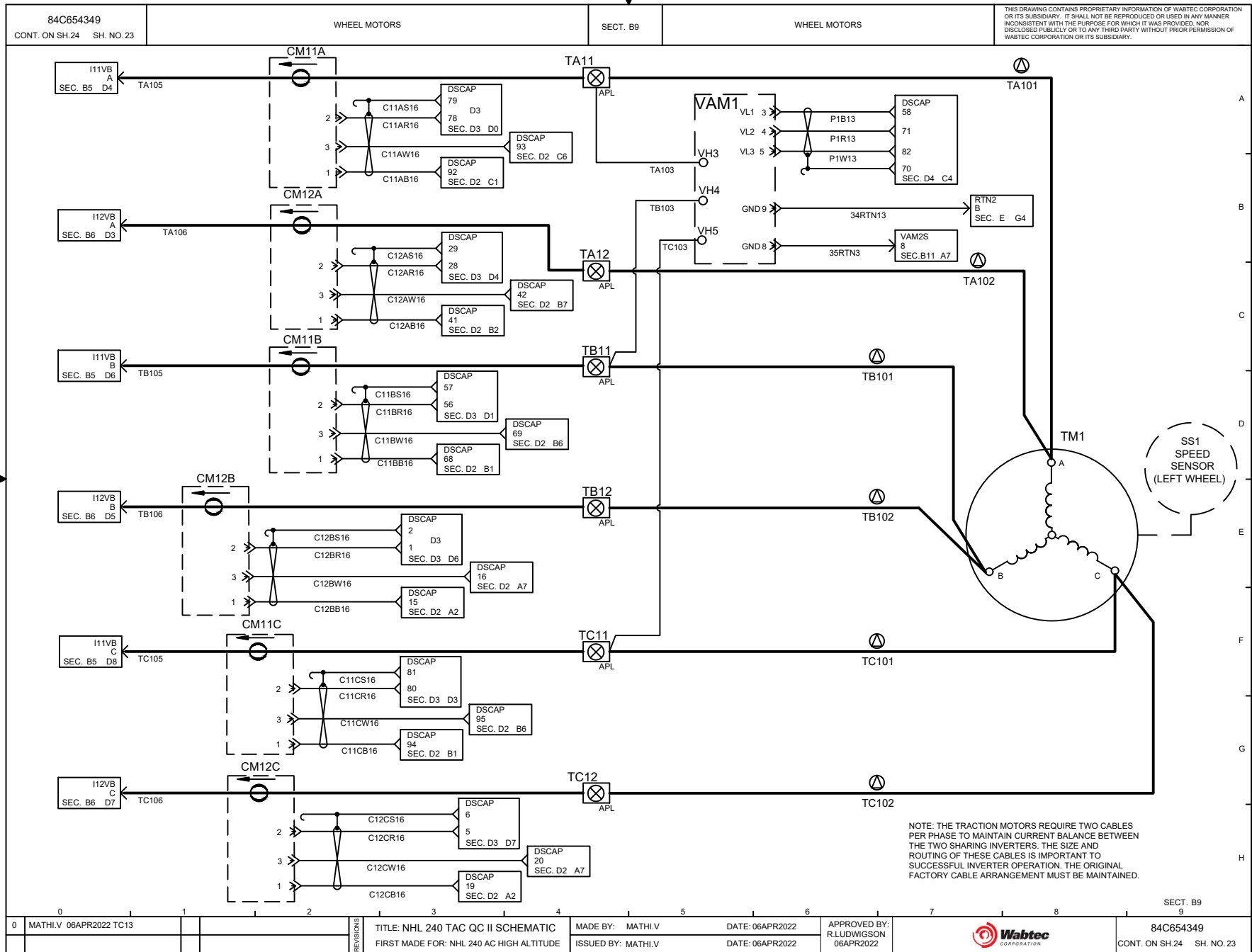
ACES

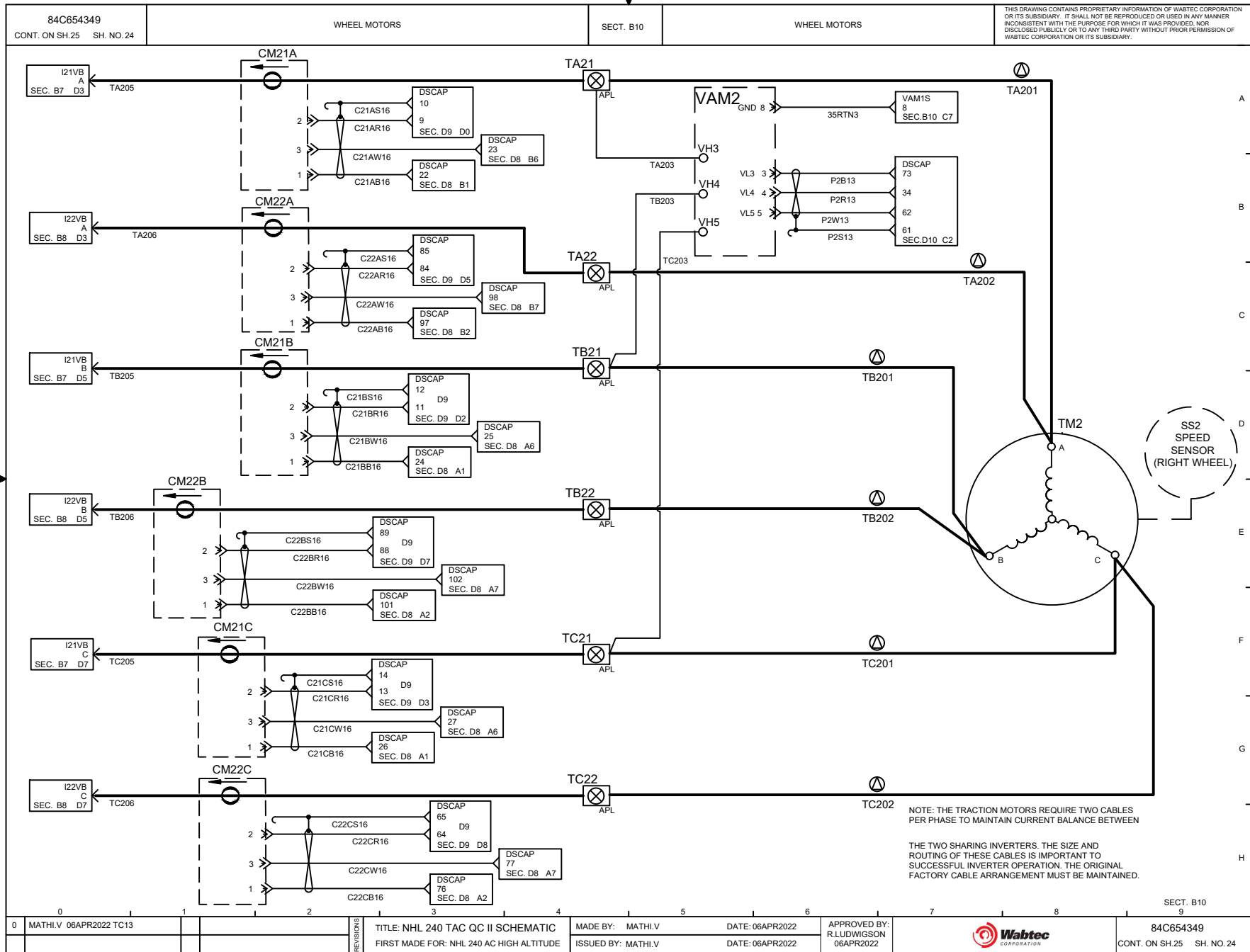


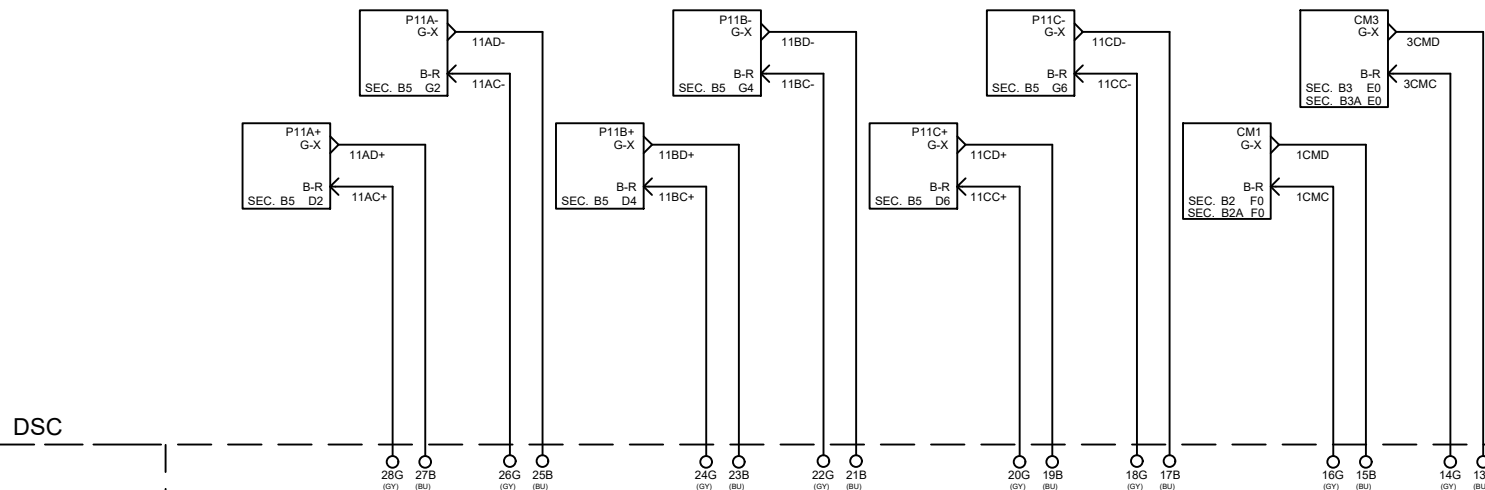




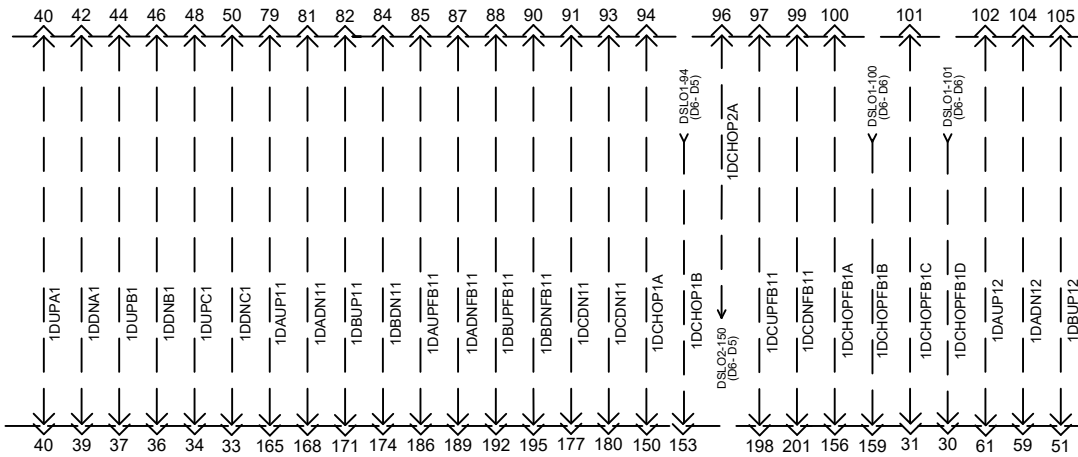




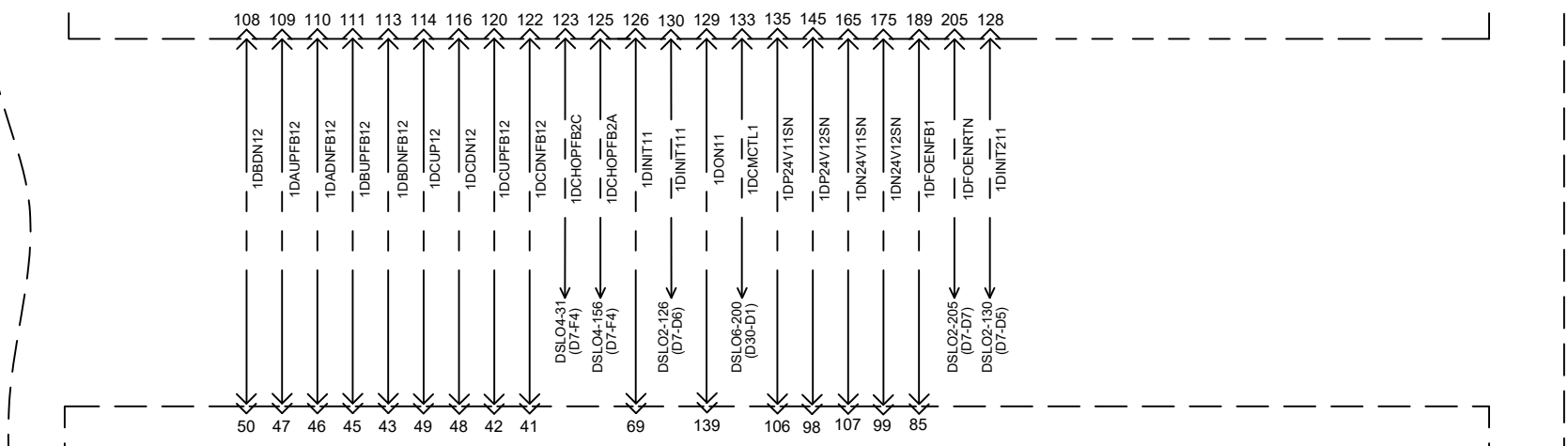
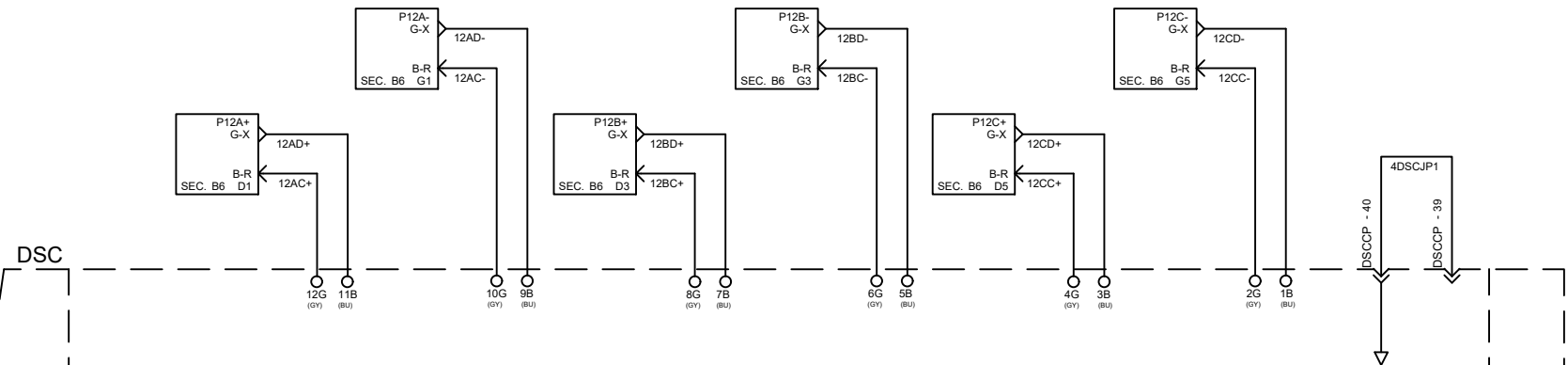


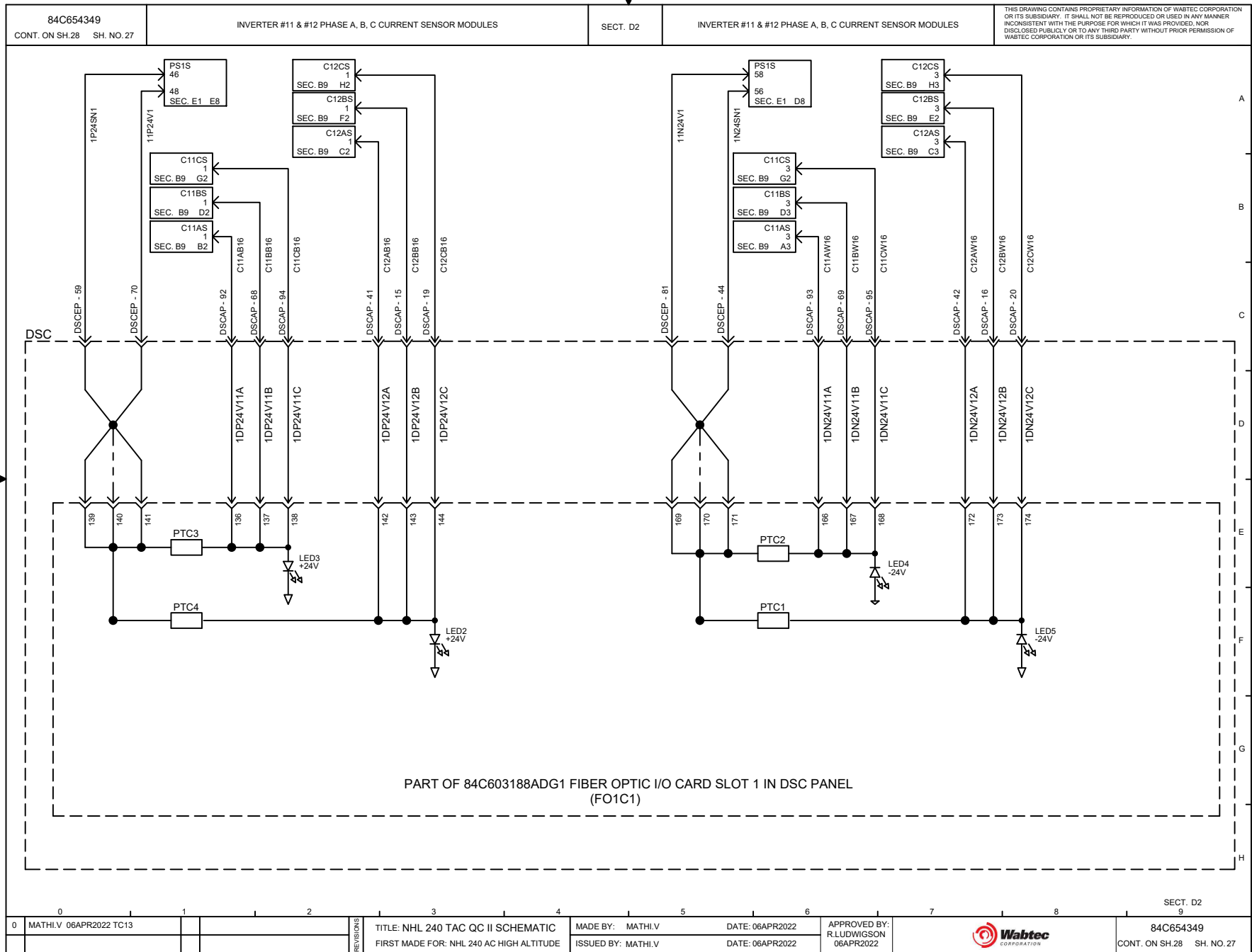


PART OF 84C603188ADG1 FIBER OPTIC I/O CARD SLOT 1 IN DSC PANEL (FODC1)

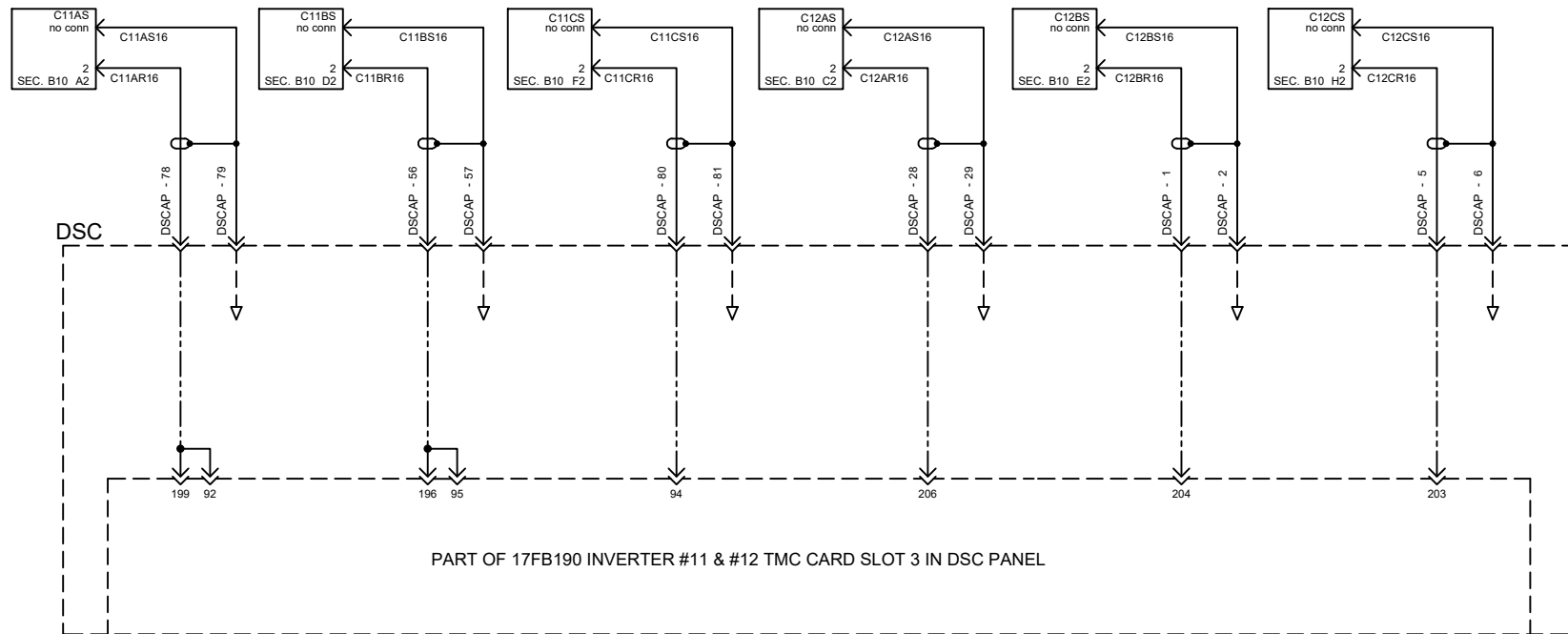


PART OF 17FB190 INVERTER #11 & #12 TMC CARD SLOT 3 IN DSC PANEL





84C654349 CONT. ON SH.29 SH. NO.28	INVERTER #11 & #12 TMC: PHASE CURRENT I/O	SECT. D3	INVERTER #11 & #12 TMC: PHASE CURRENT I/O	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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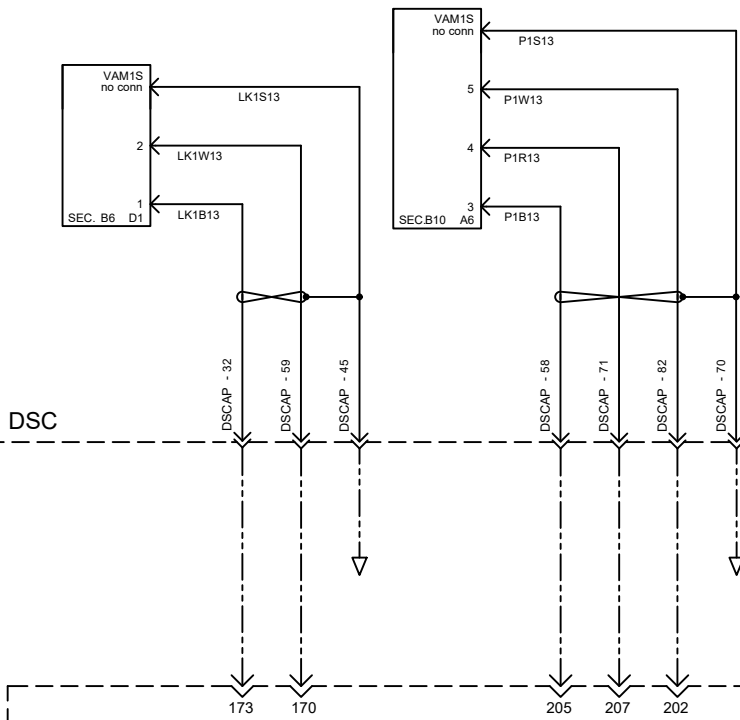


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ACES

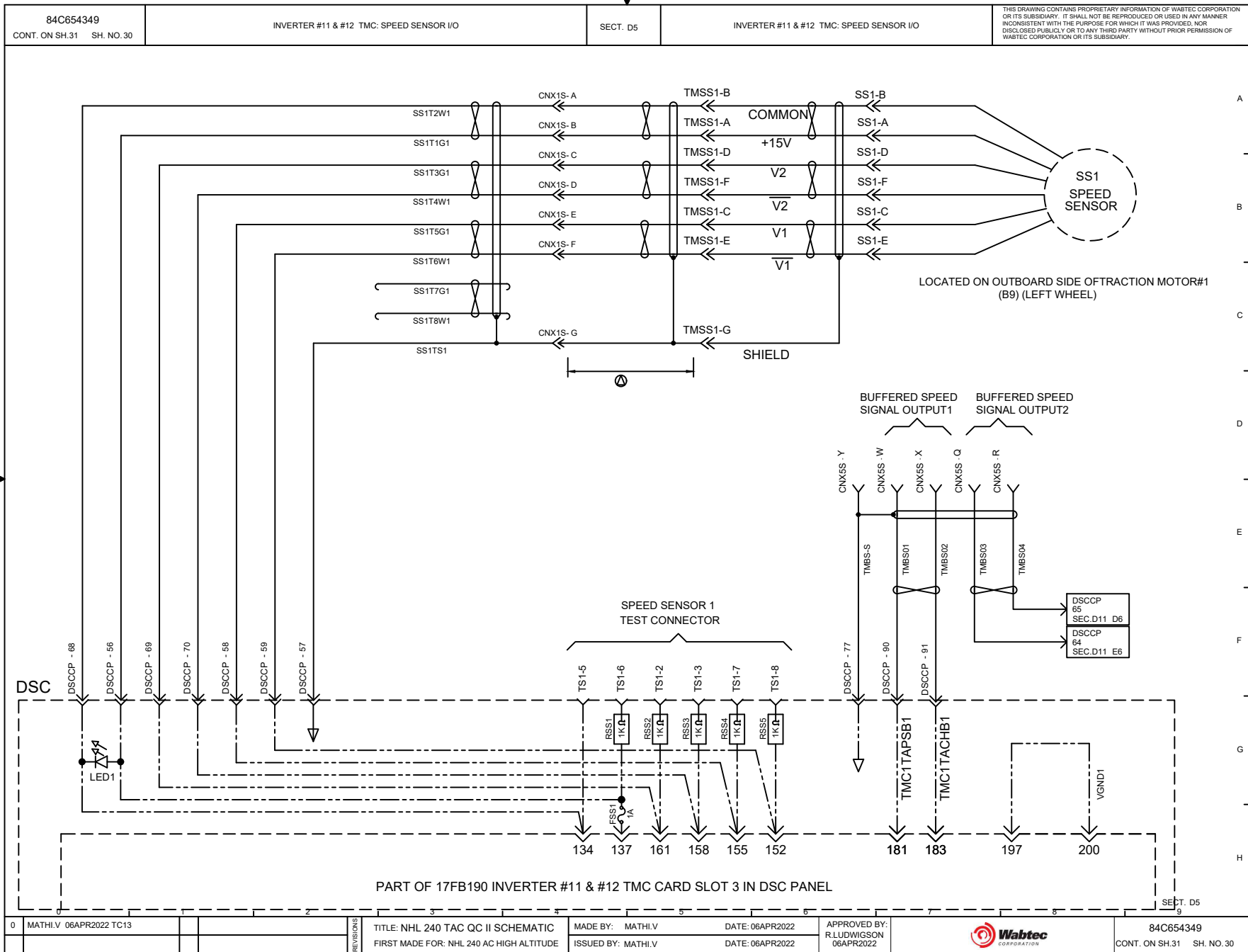
84C654349 CONT. ON SH.30 SH. NO.29	INVERTER #11 LINK AND INVERTER #11 & #12 PHASE VOLTAGE I/O	SECT. D4	BLANK	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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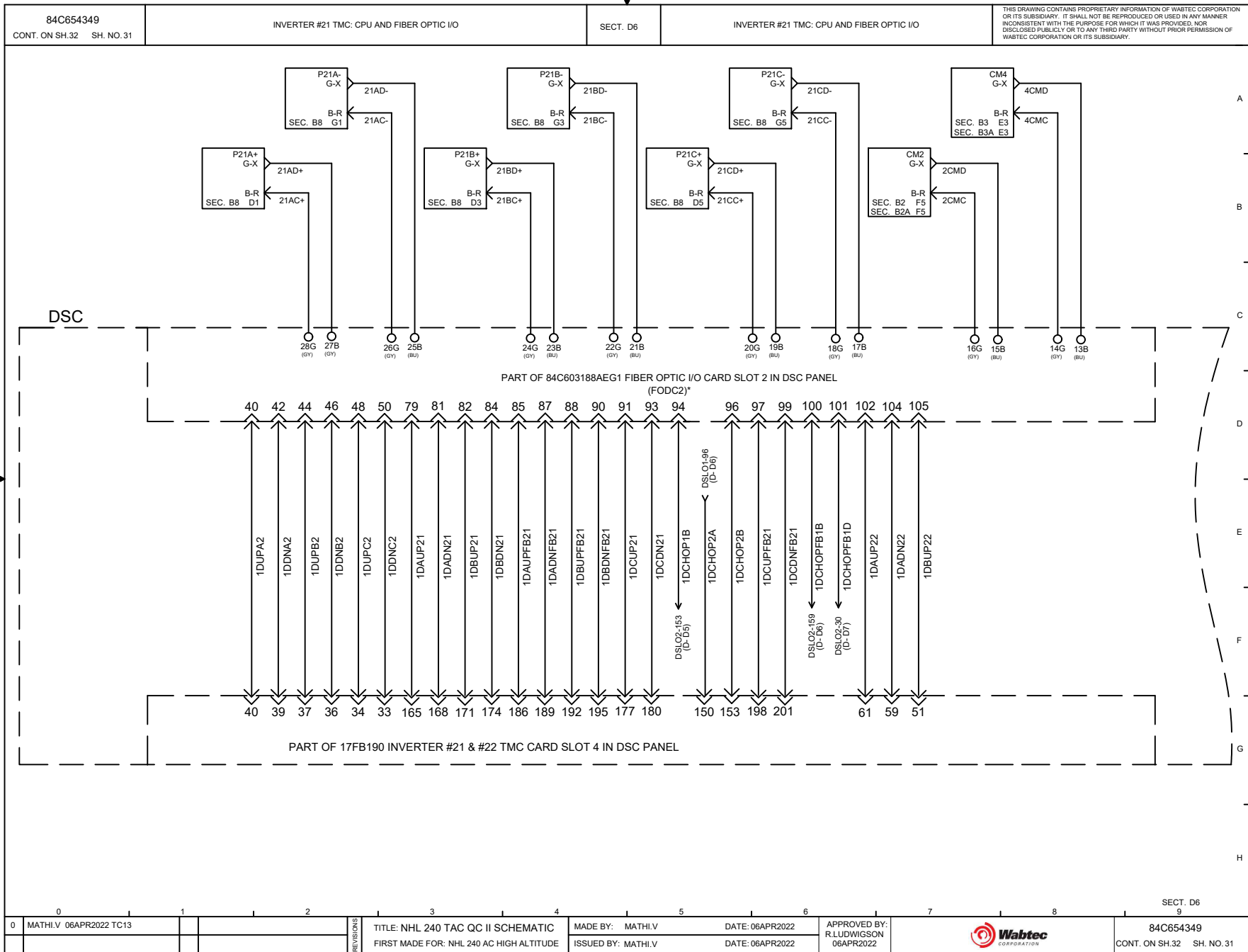


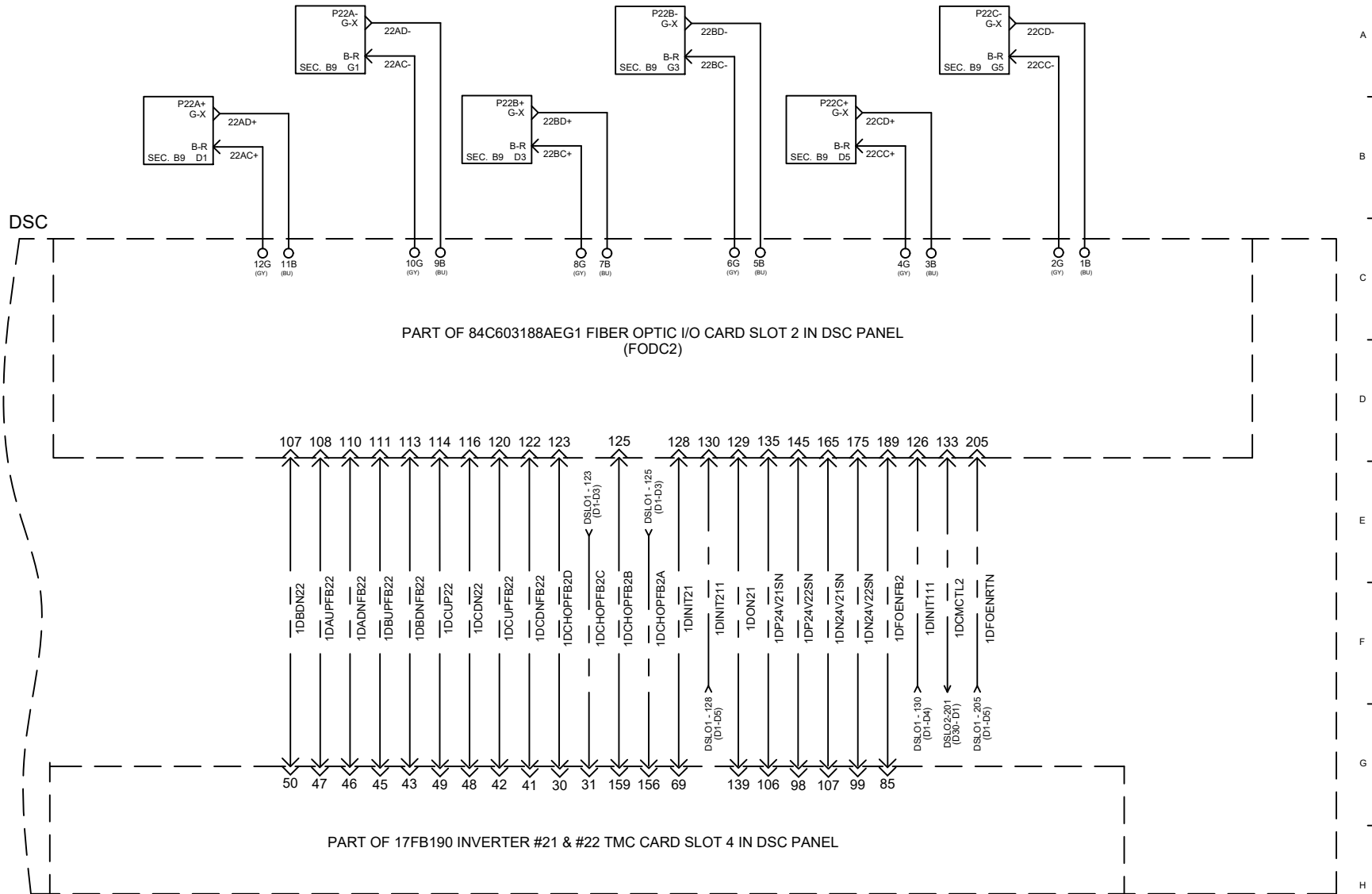
PART OF 17FB190 INVERTER #11 & #12 TMC CARD SLOT 3 IN DSC PANEL

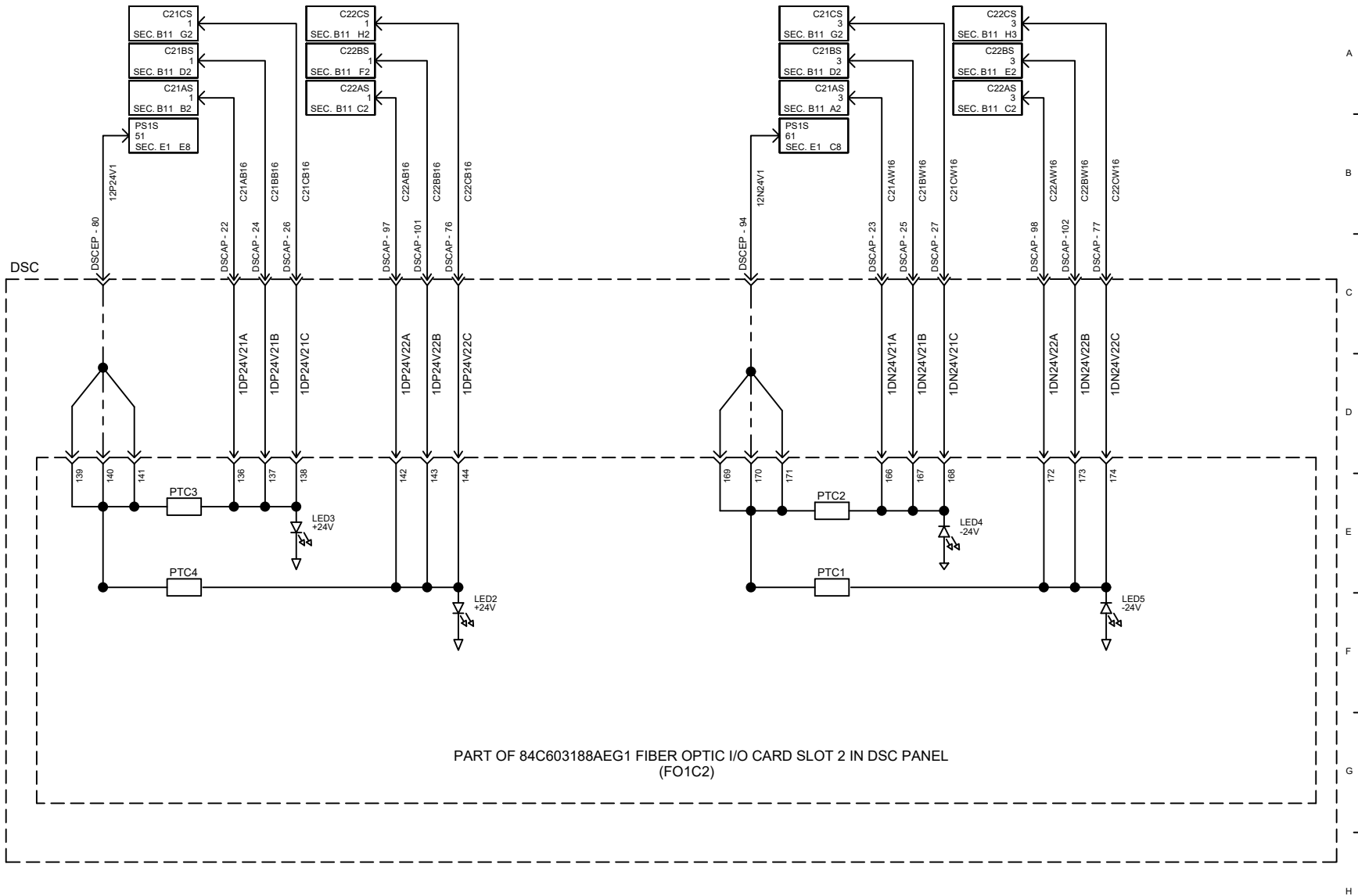
0 MATHI.V 06APR2022 TC13	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATHI.V ISSUED BY: MATHI.V	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: R.LUDWIGSON 06APR2022		84C654349 CONT. ON SH.30 SH. NO.29
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ACES

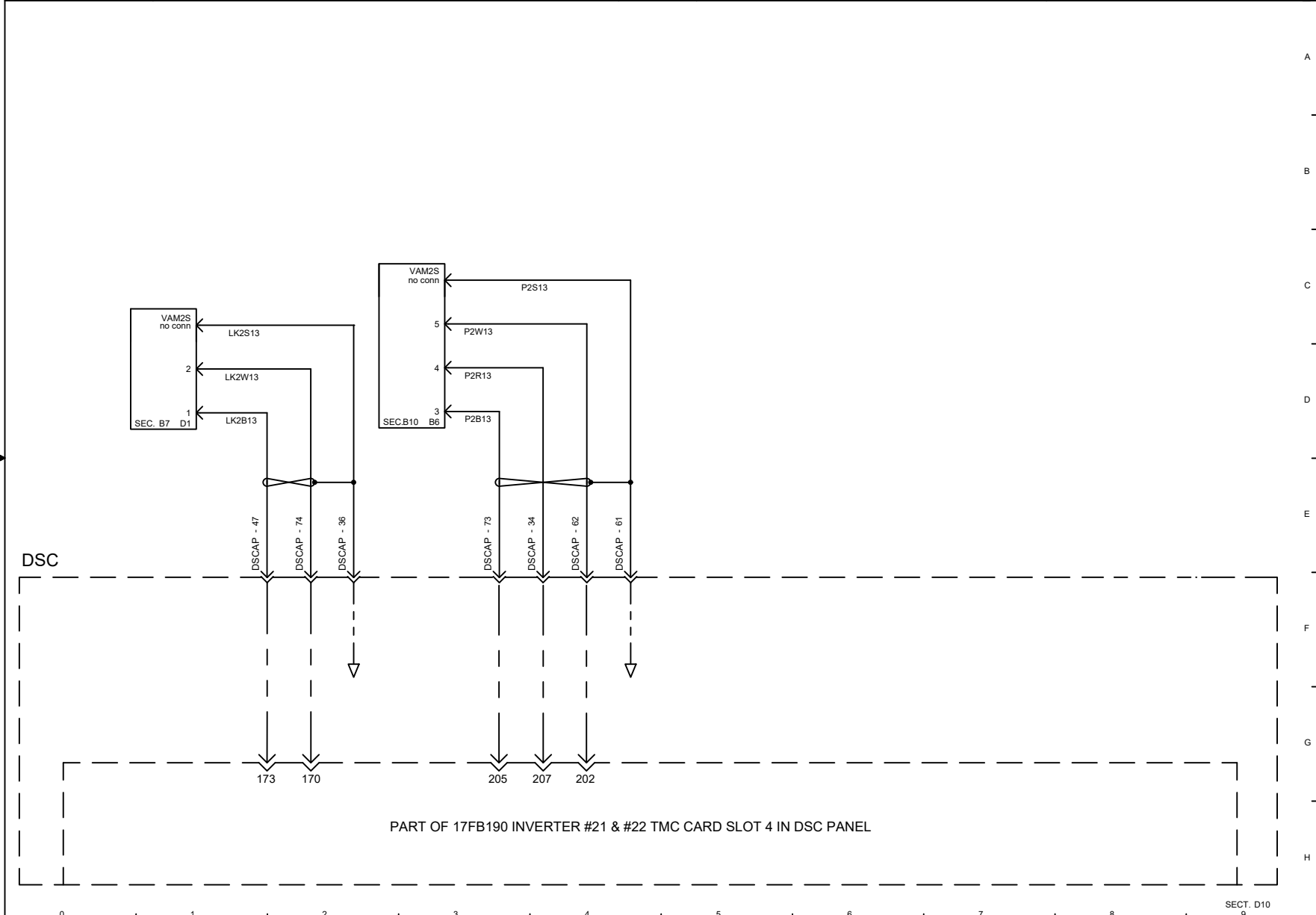






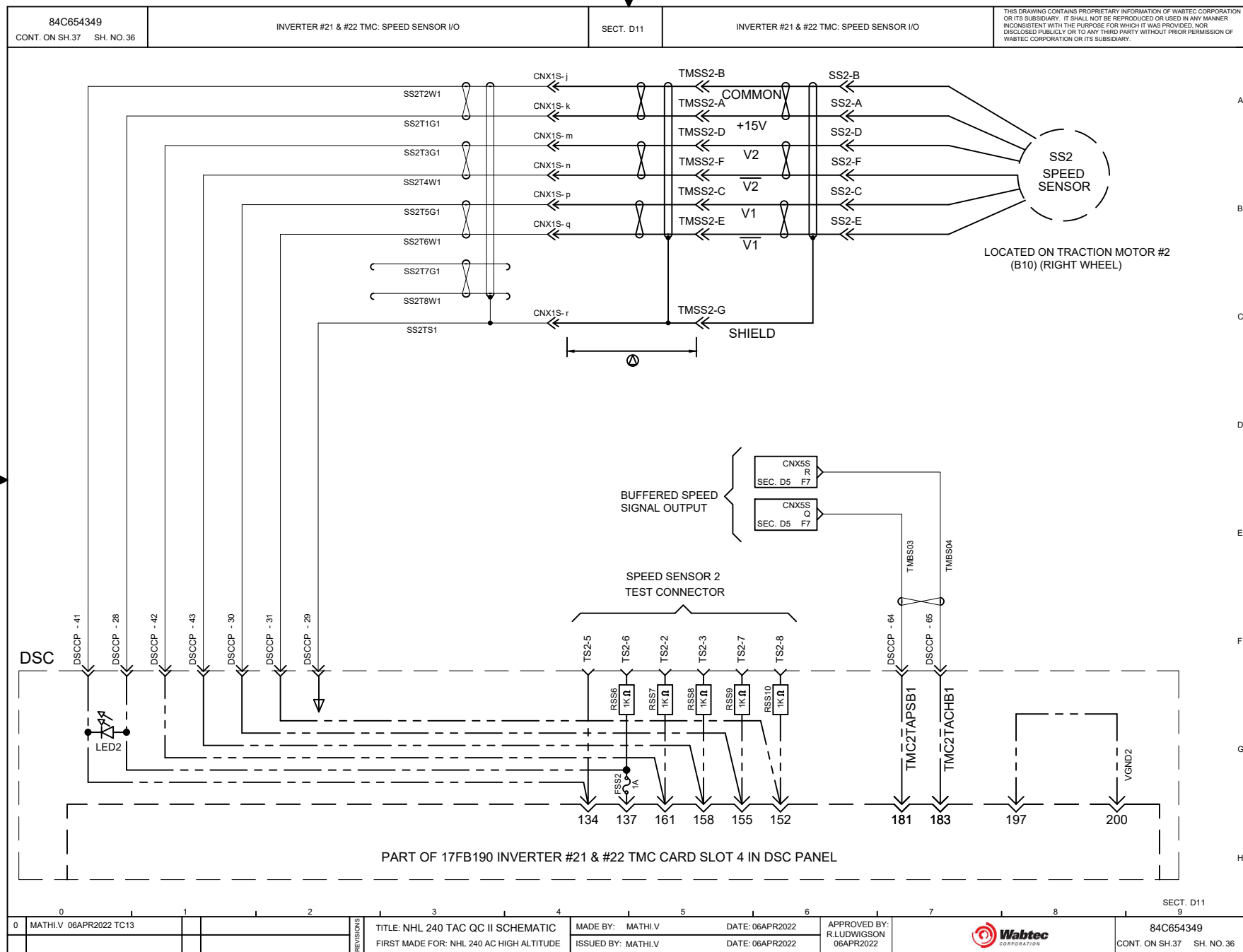


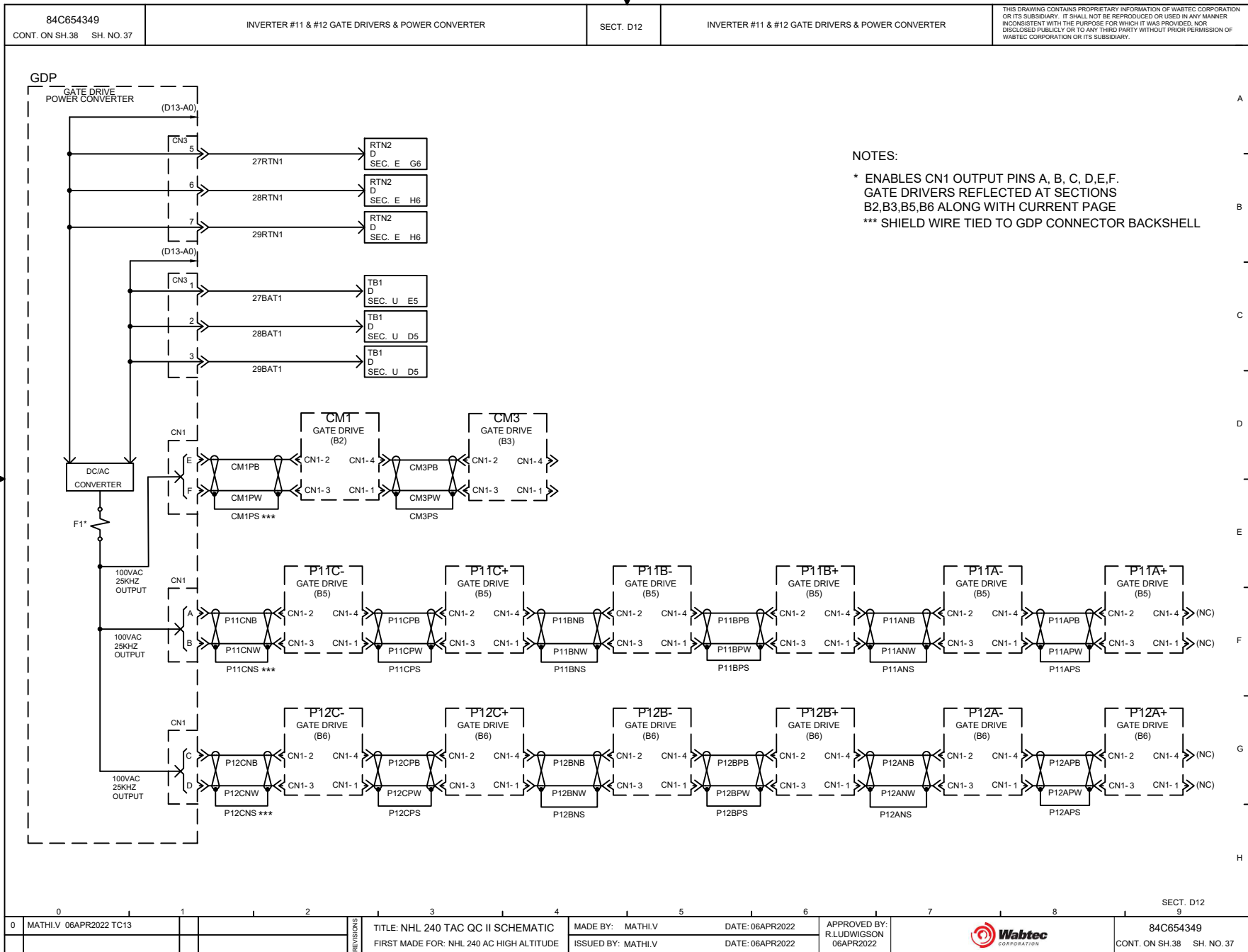
84C654349 CONT. ON SH.36 SH. NO.35	INVERTER #21 LINK AND INVERTER #21 & #22 PHASE VOLTAGE I/O	SECT. D10	BLANK	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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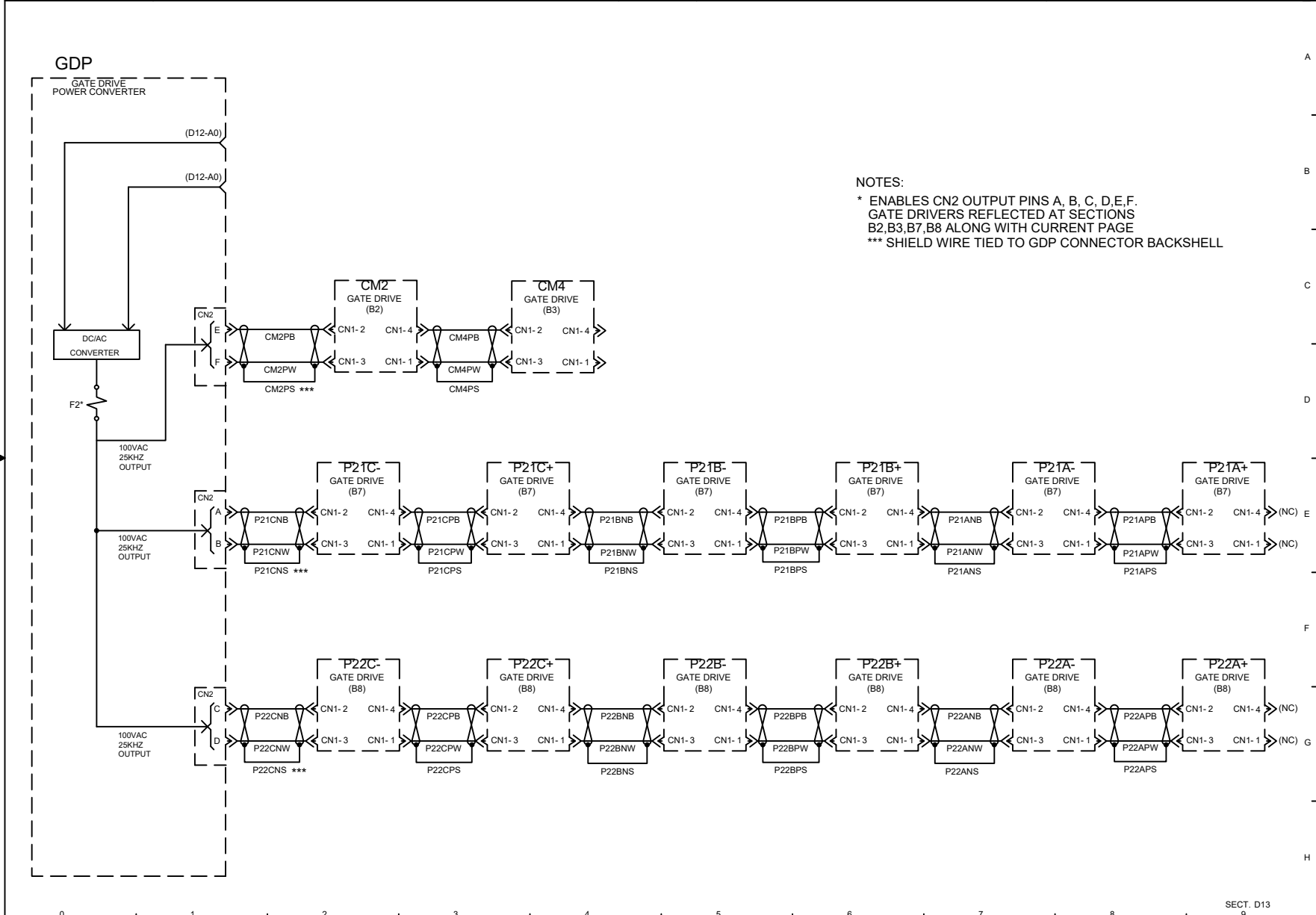


0 MATHI.V 06APR2022 TC13	1	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATHI.V ISSUED BY: MATHI.V	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: R.LUDWIGSON 06APR2022		84C654349 CONT. ON SH.36 SH. NO.35
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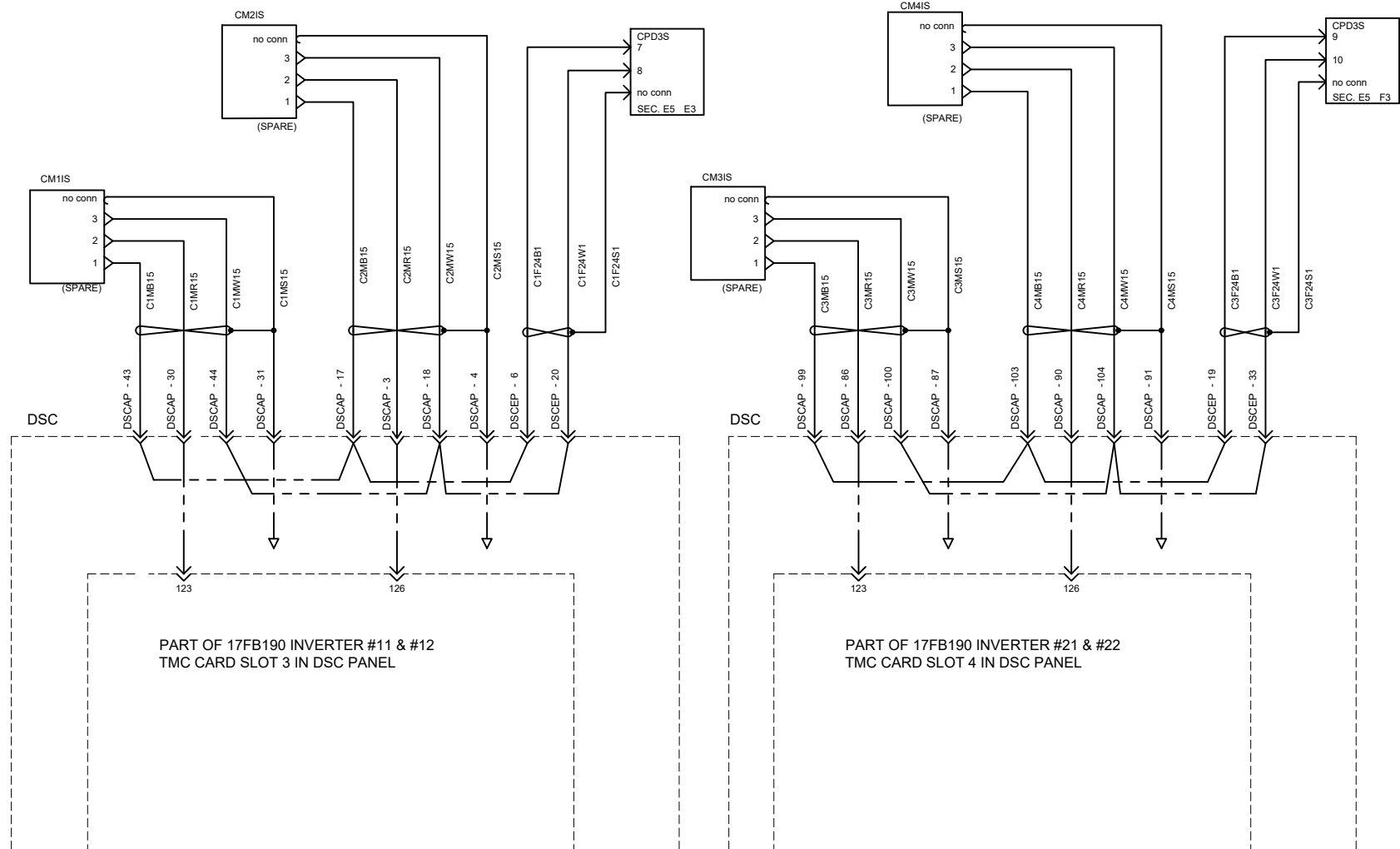
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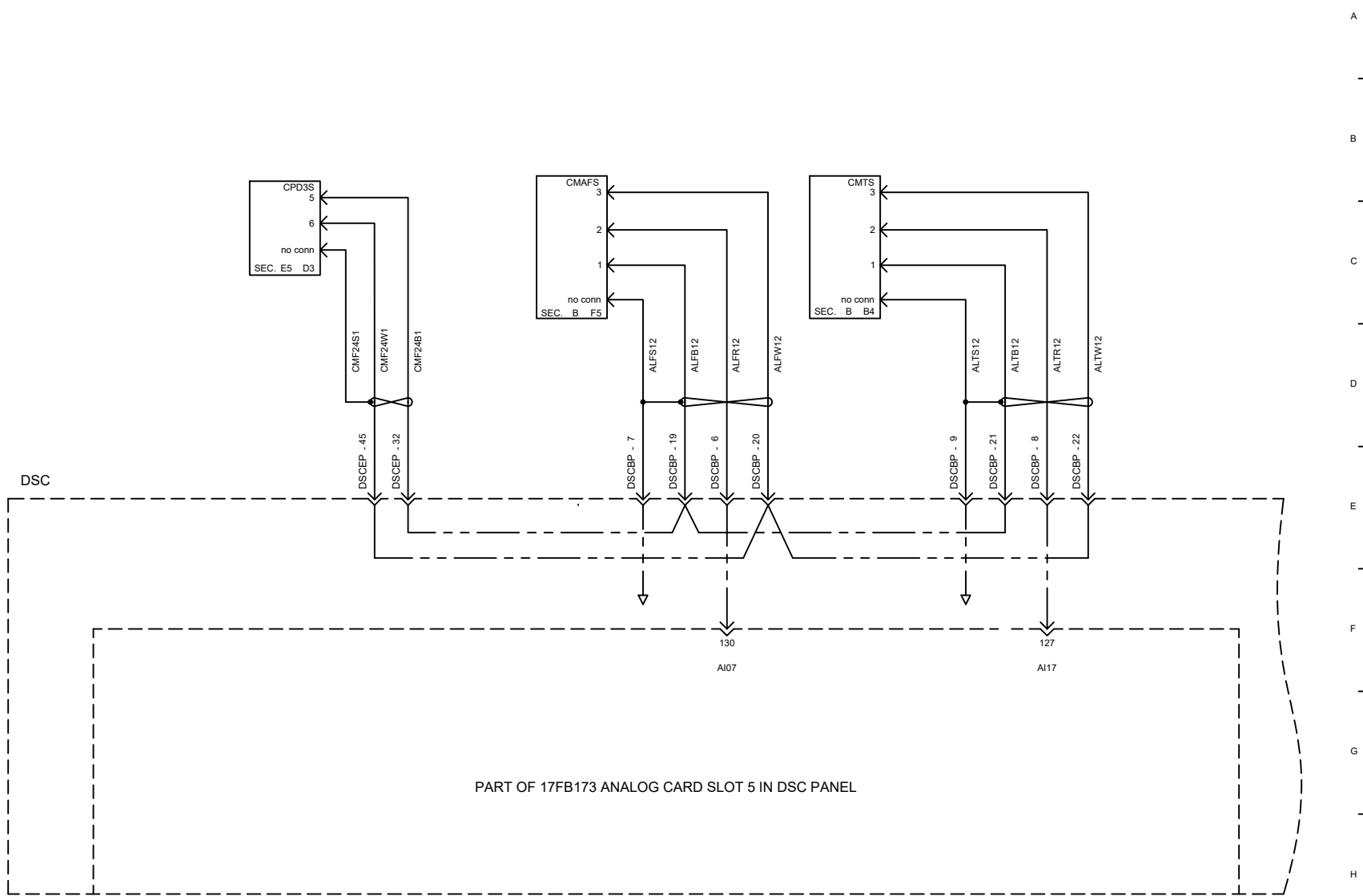




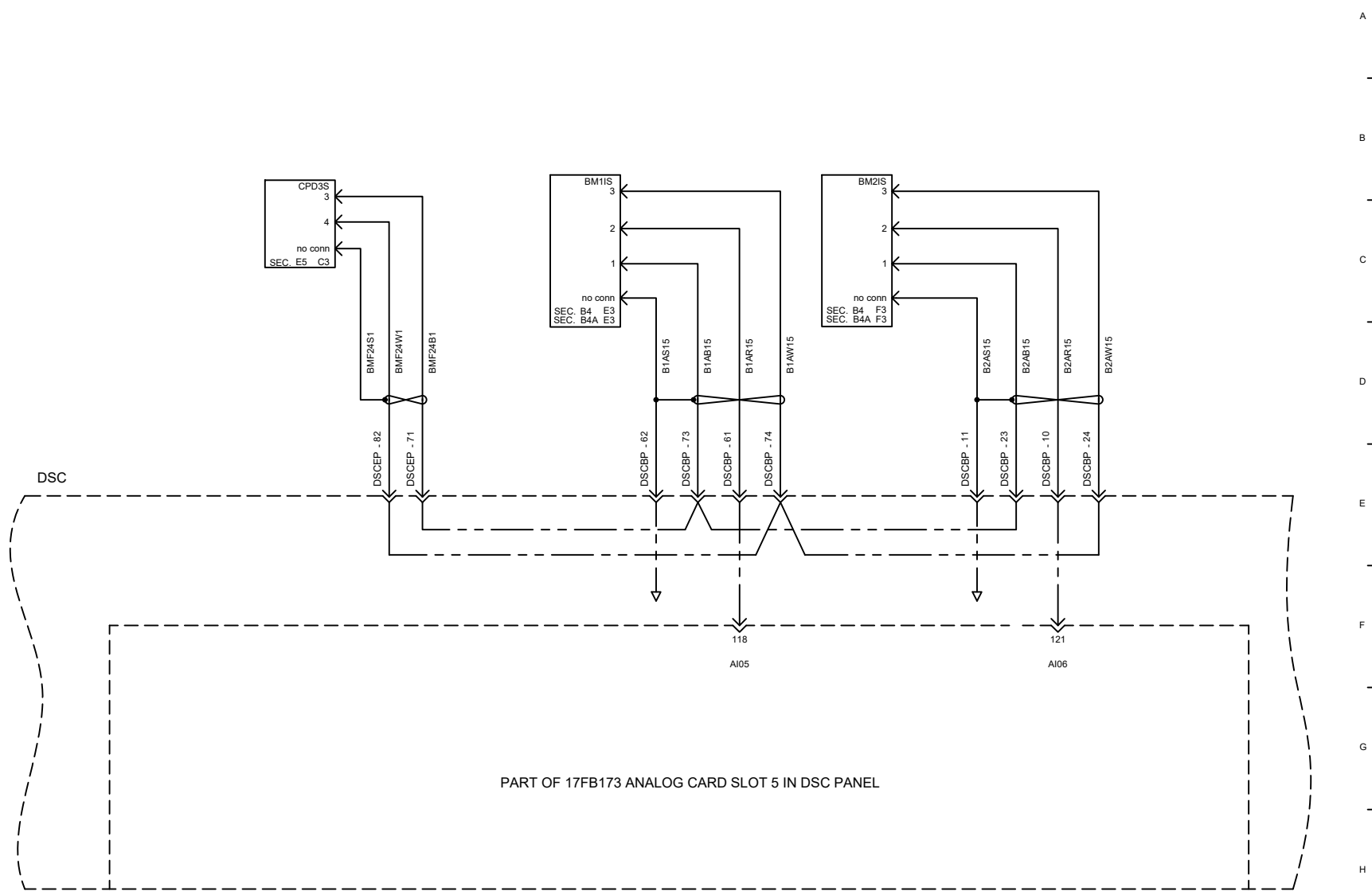


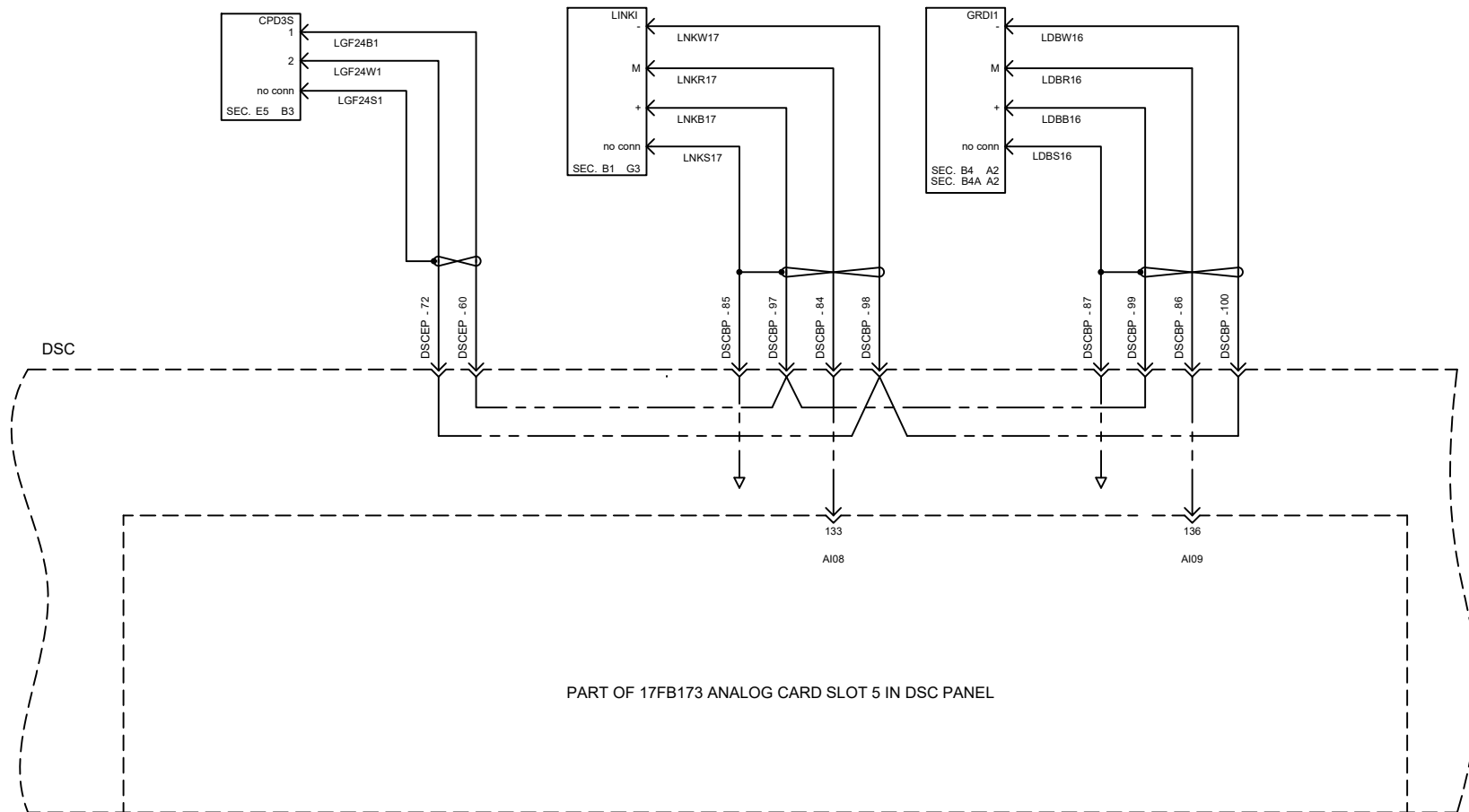
* CONNECTORS WERE CONNECTED TO CHOPPER MODULE CURRENT SENSORS
 FOR 17KH560D1 CM CURRENT SENSORS ARE NOT APPLICABLE
 CONNECTORS USED AS SPARE* FOR THIS CONTROL GROUP

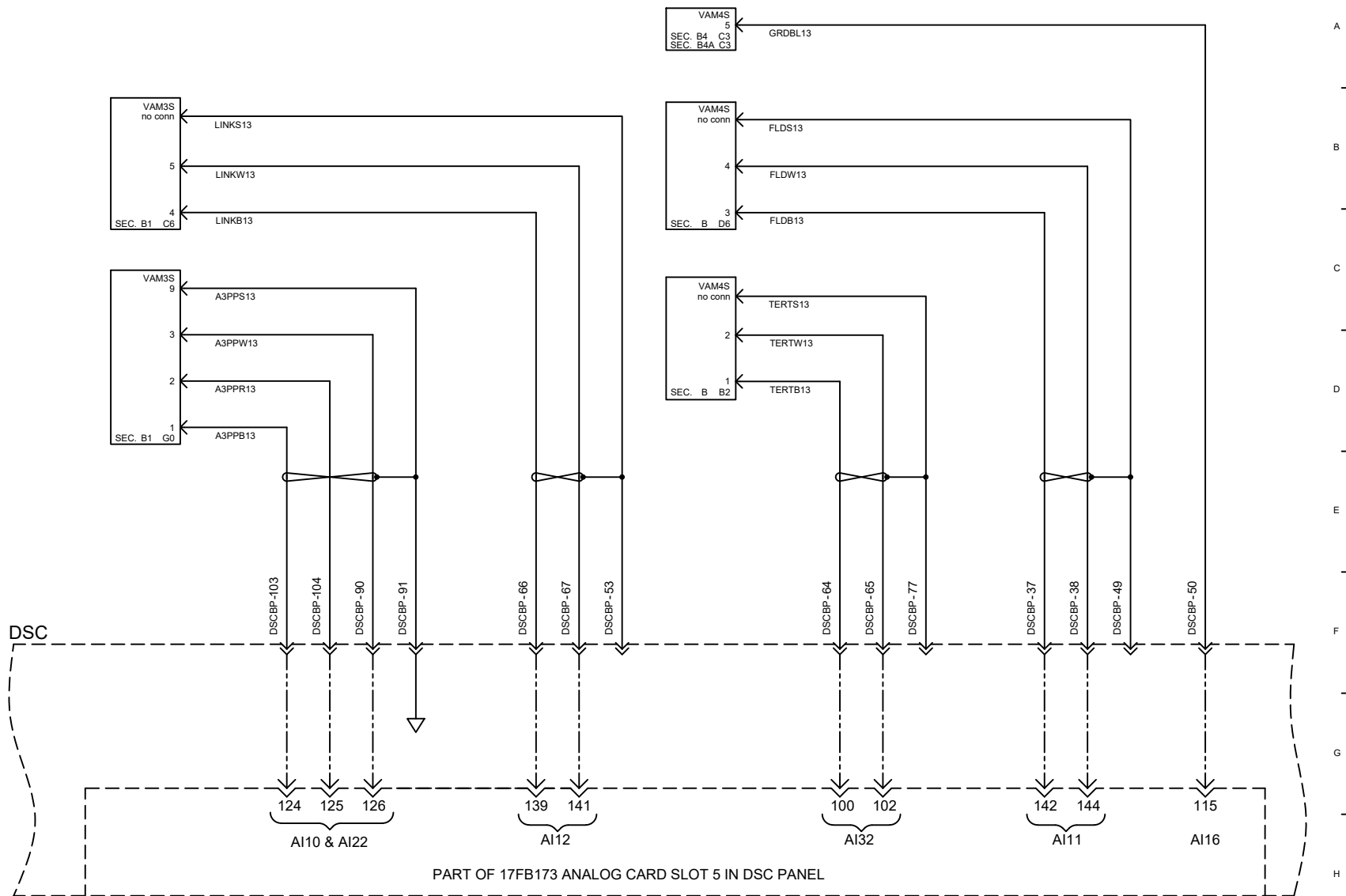




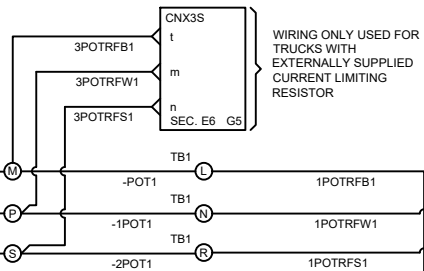
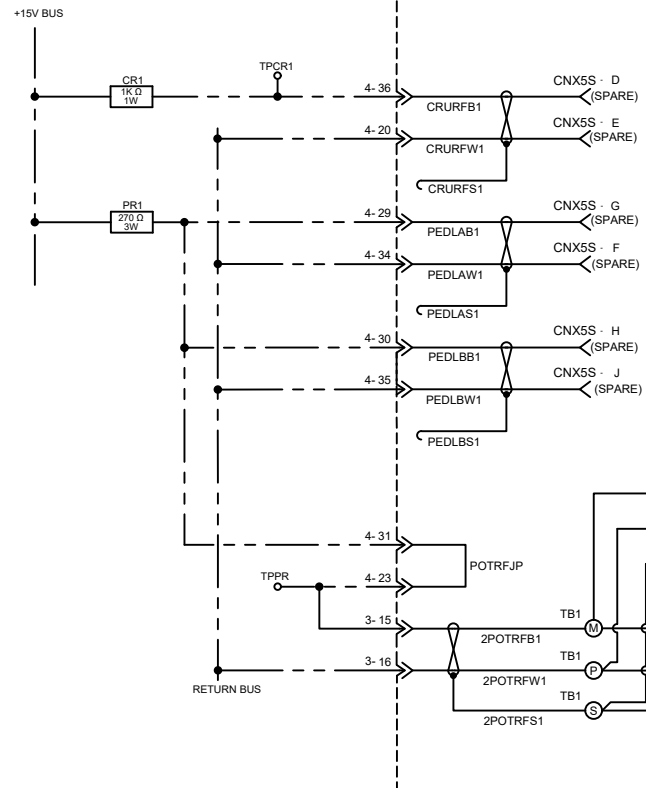
PART OF 17FB173 ANALOG CARD SLOT 5 IN DSC PANEL



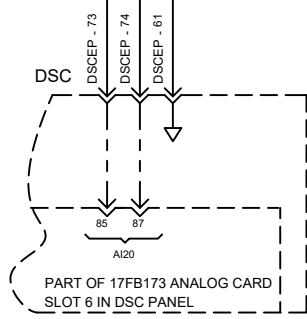




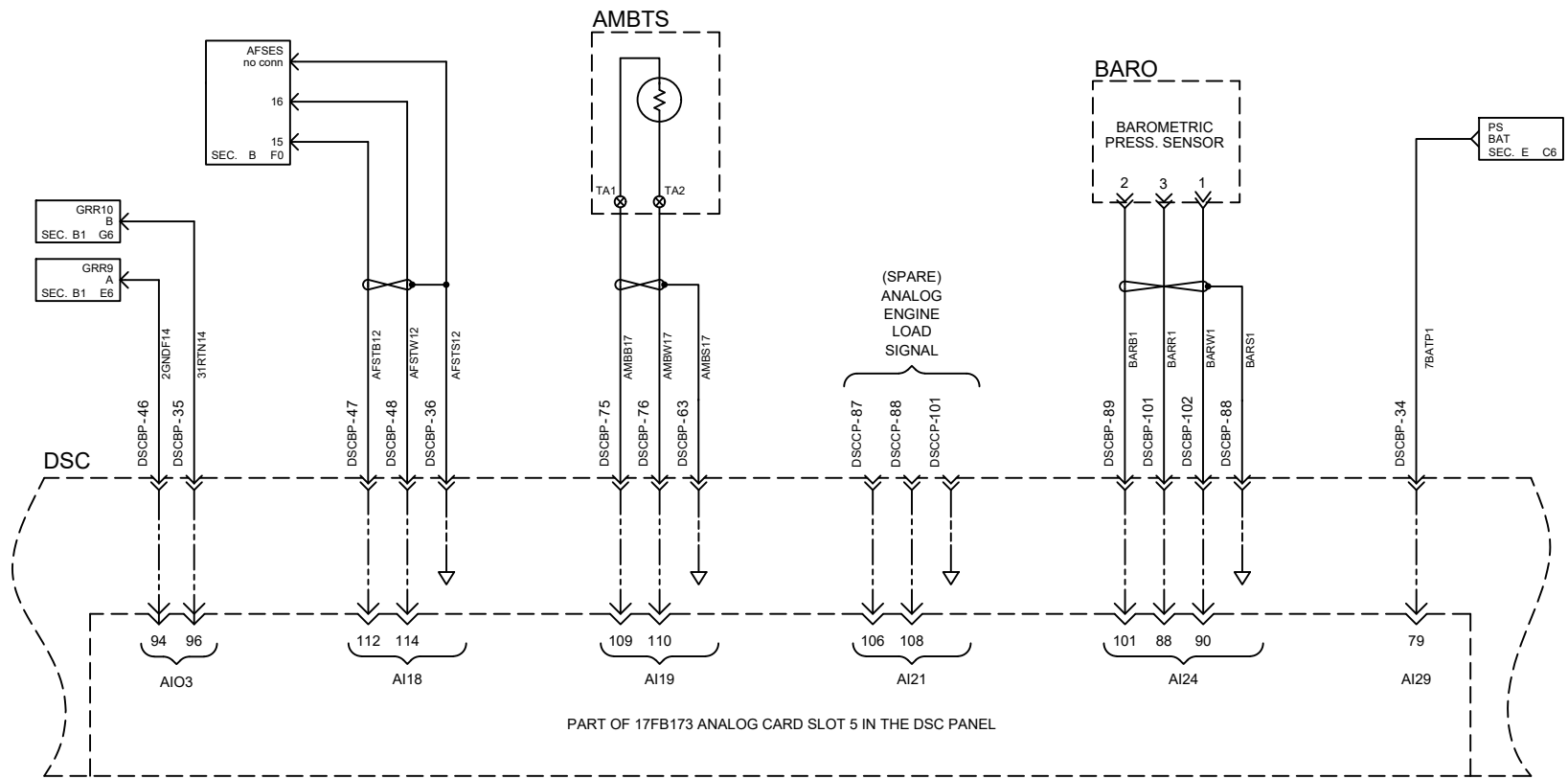
CPD

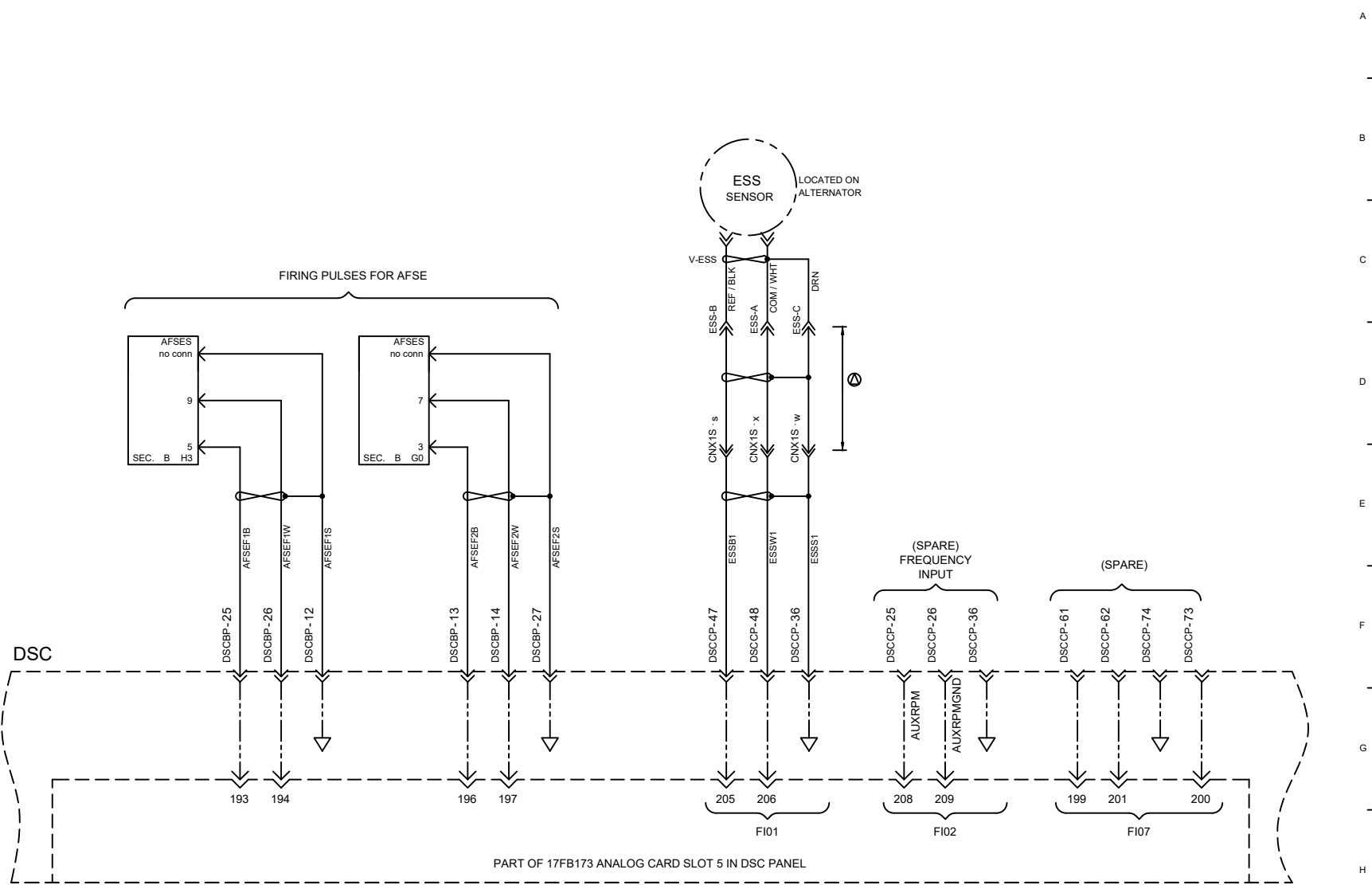


WIRING ONLY USED FOR TRUCKS WITH EXTERNALLY SUPPLIED CURRENT LIMITING RESISTOR

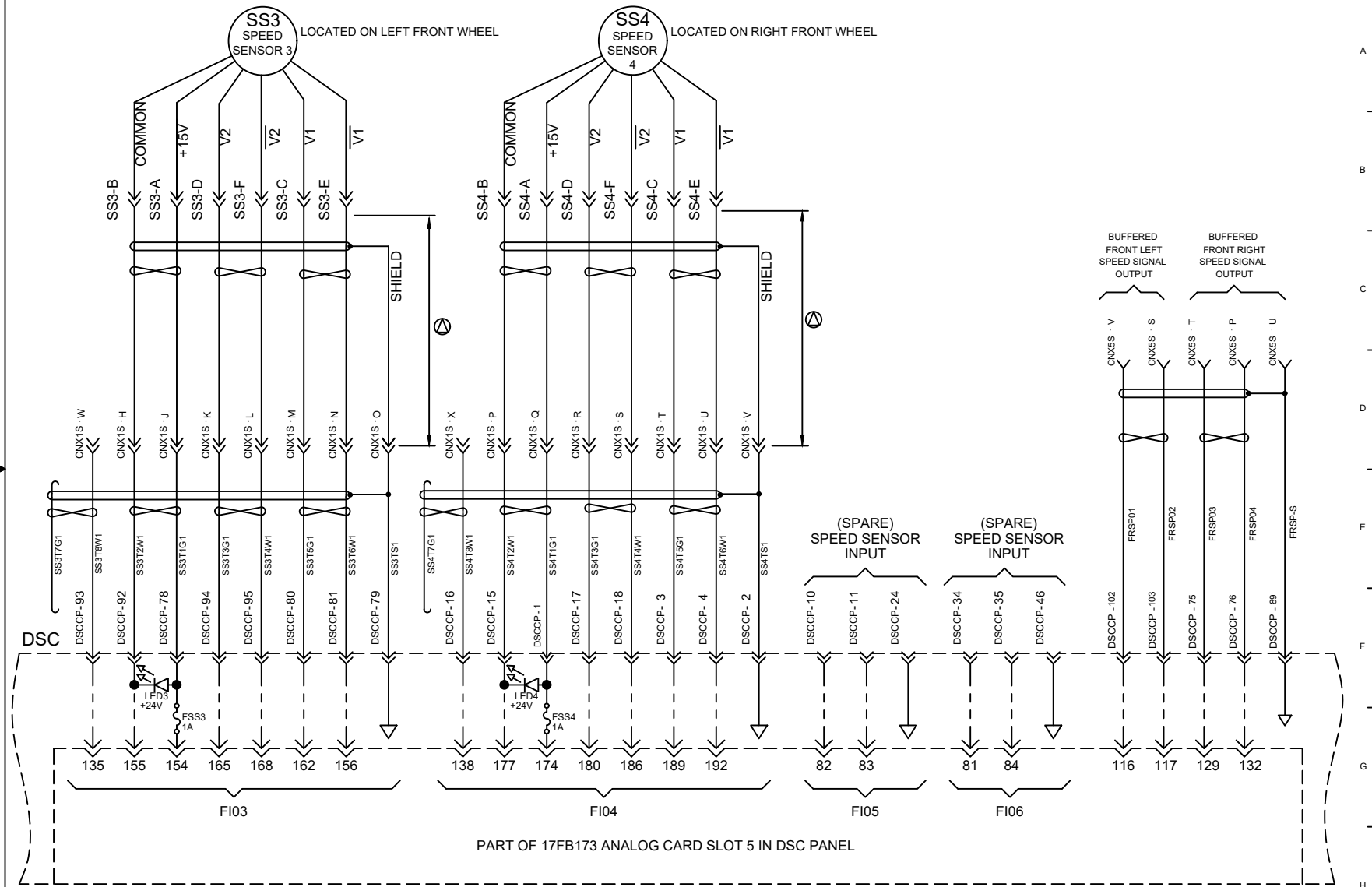


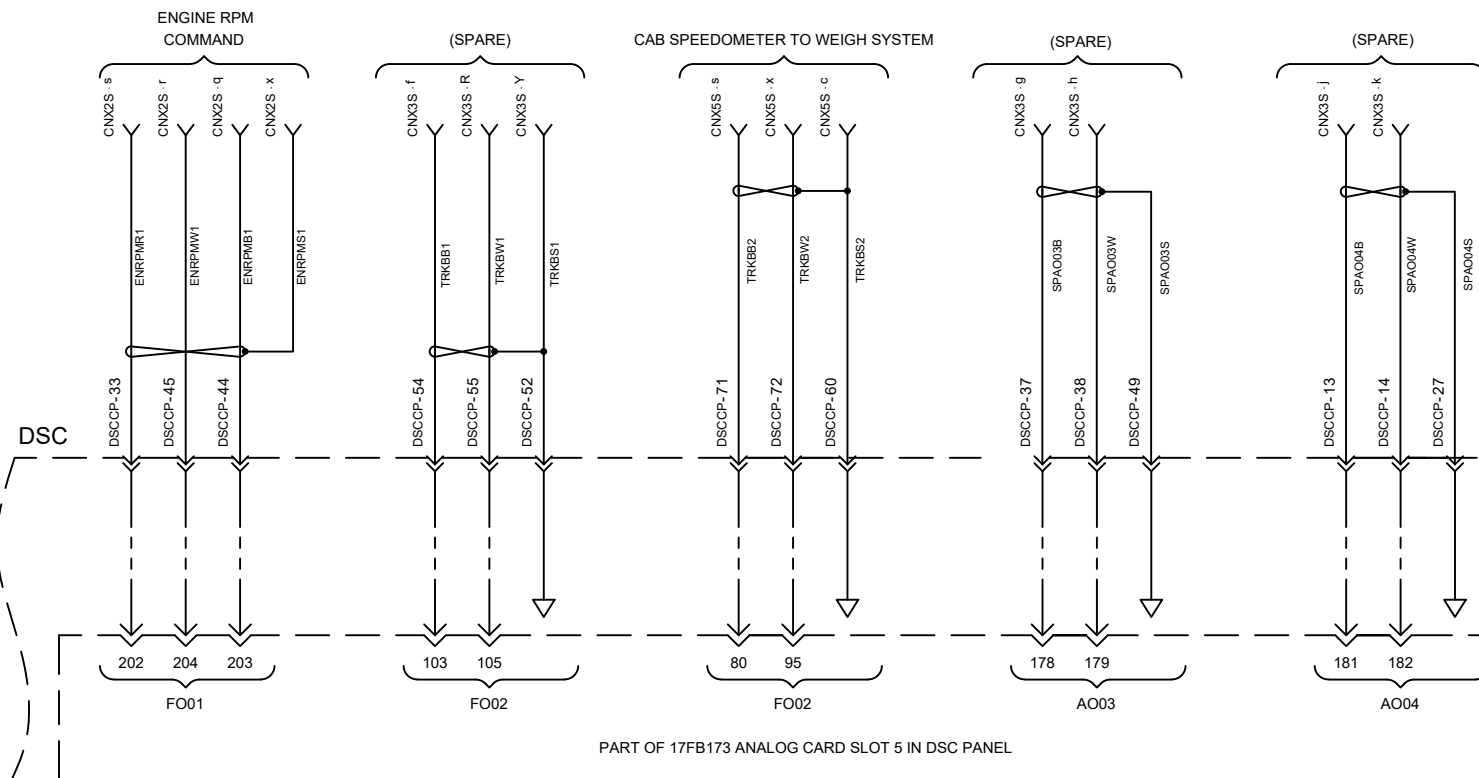
PART OF 17FB173 ANALOG CARD SLOT 6 IN DSC PANEL

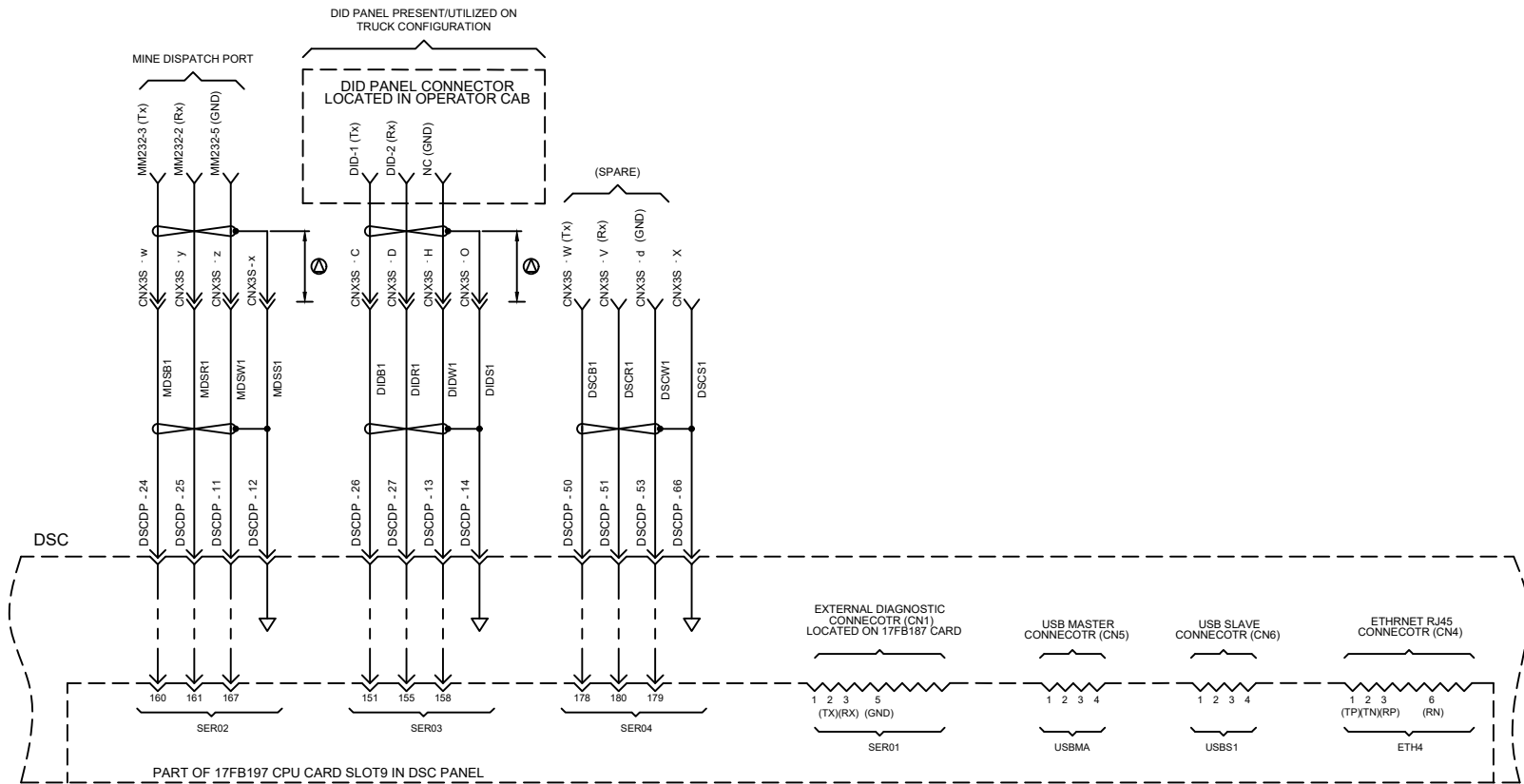


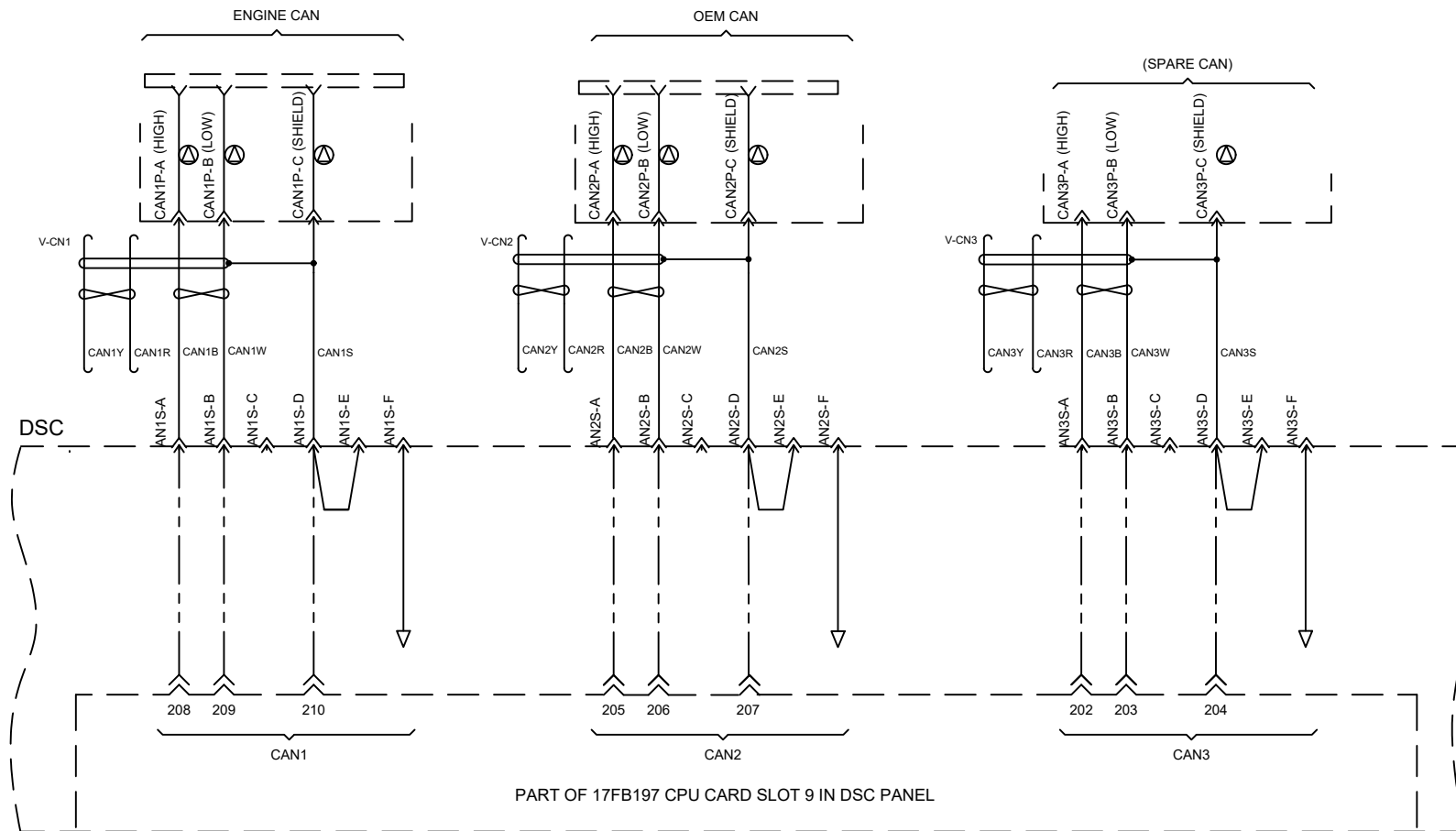


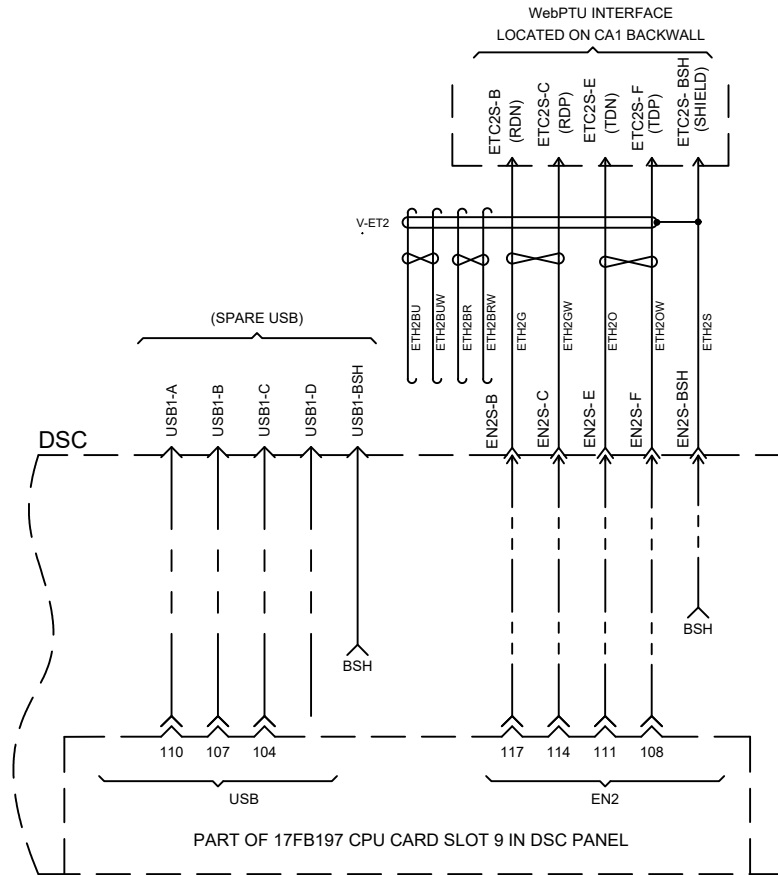
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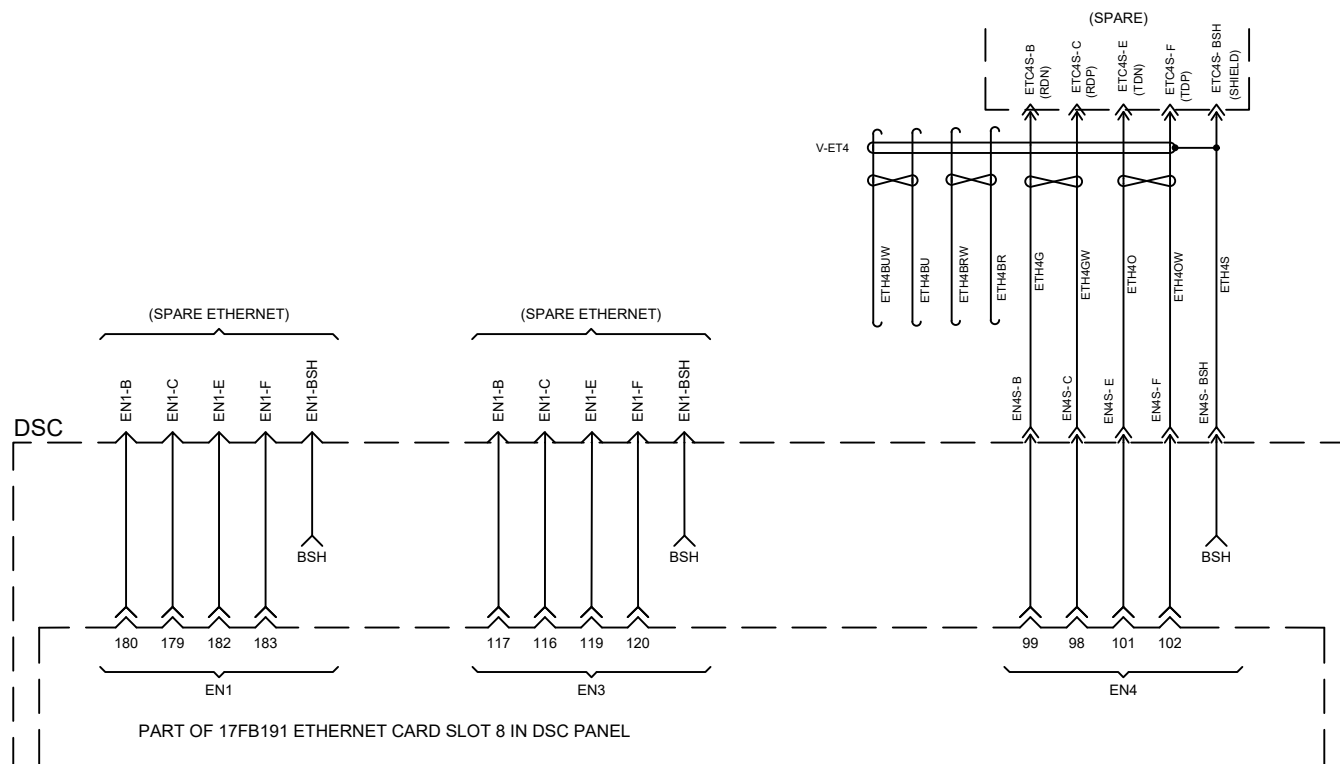


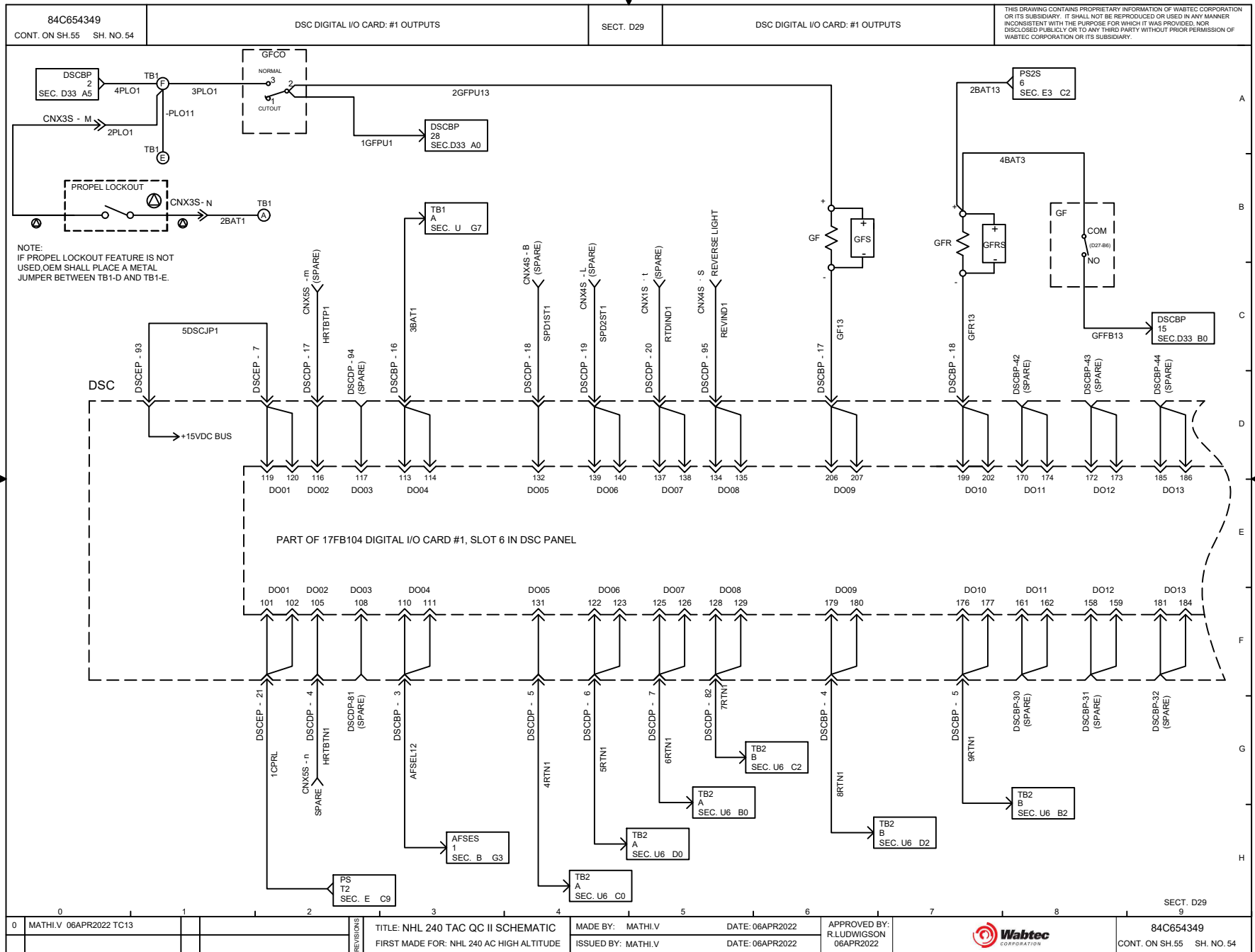


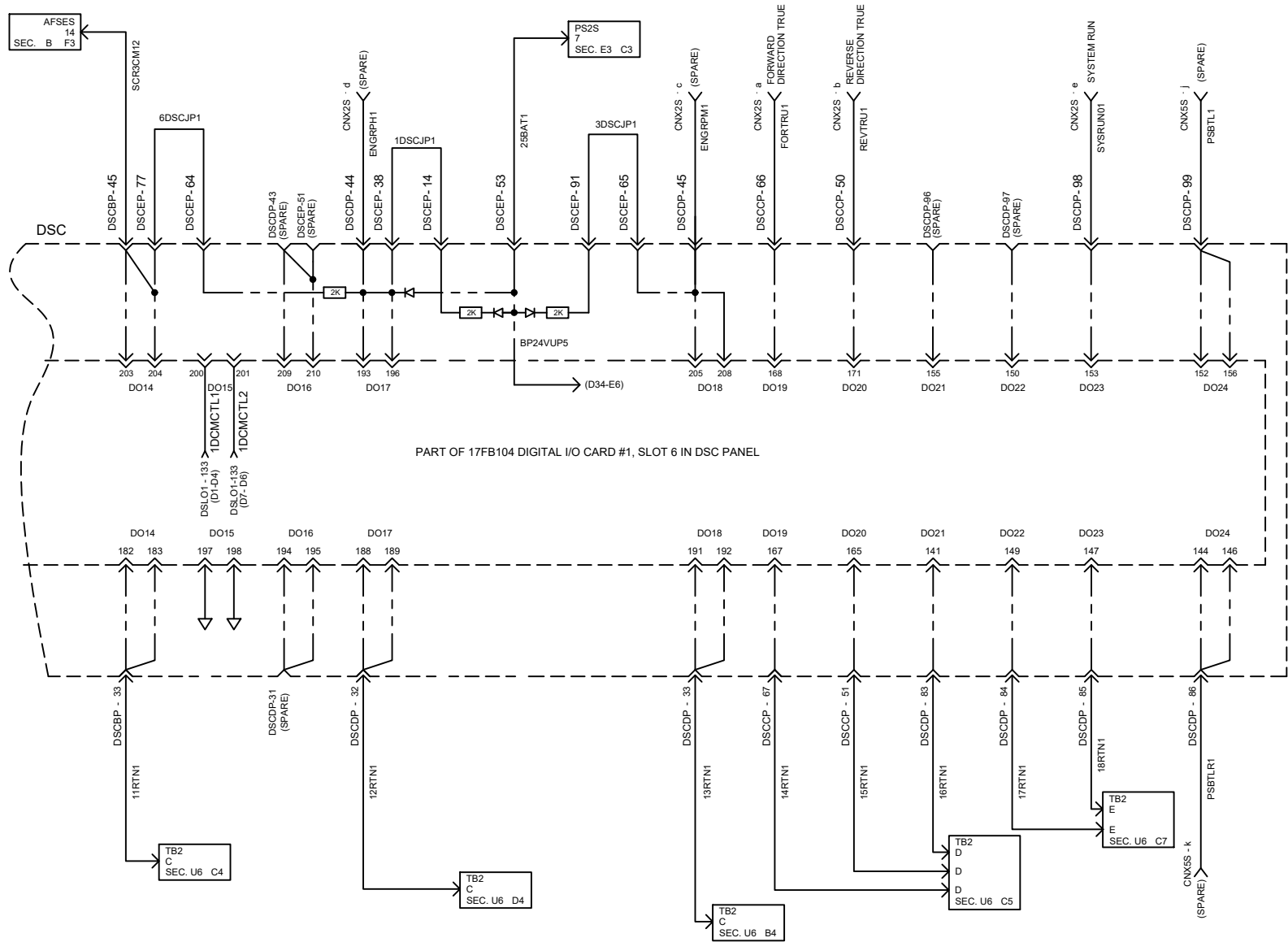




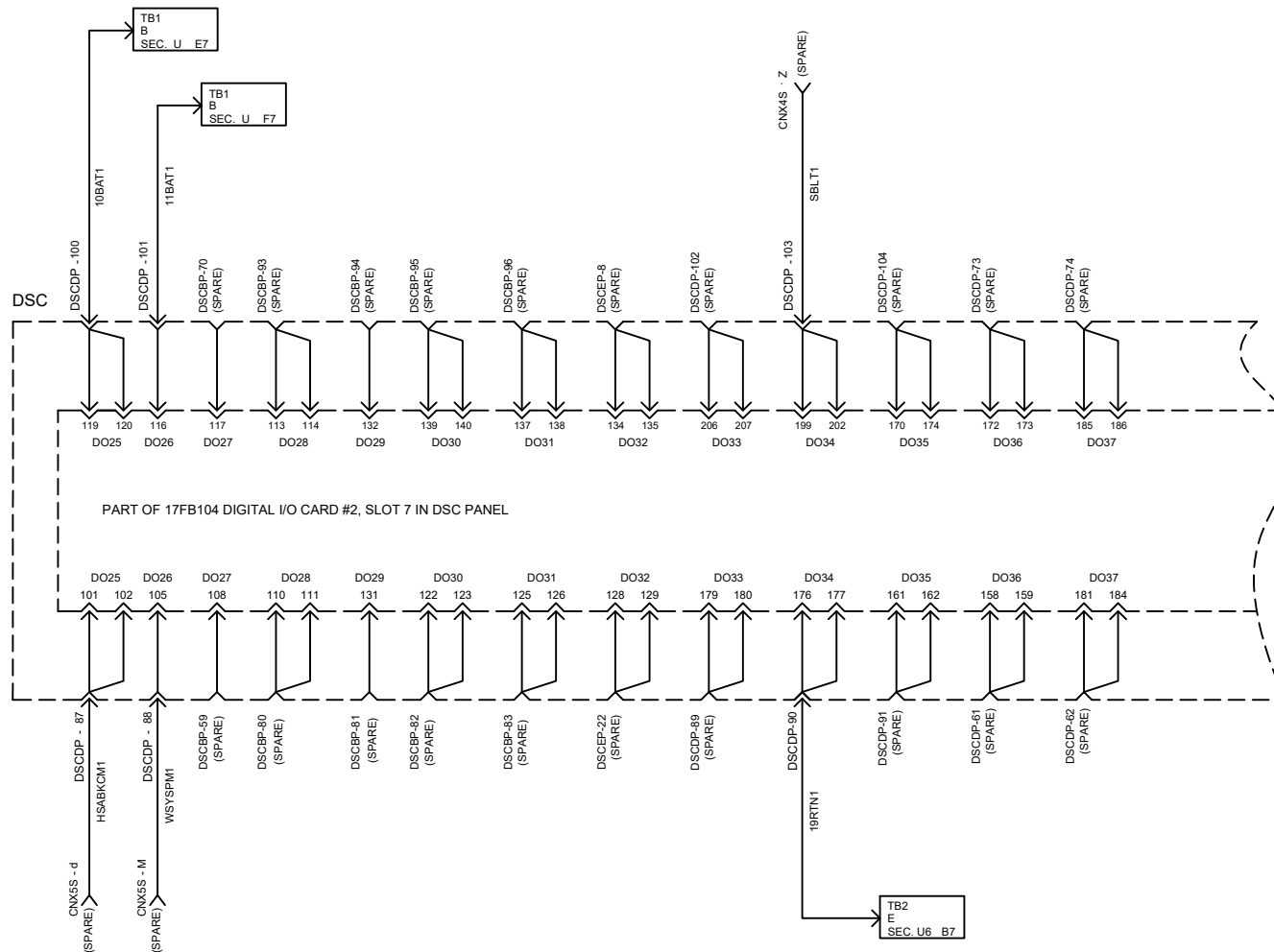


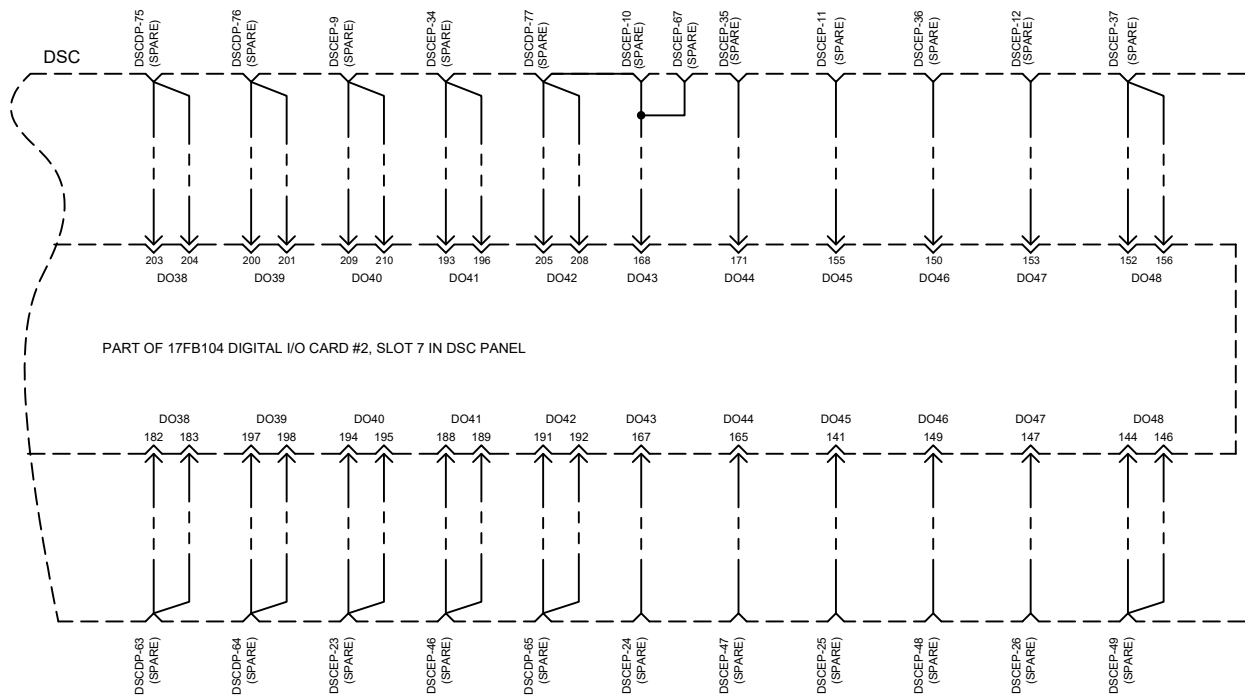


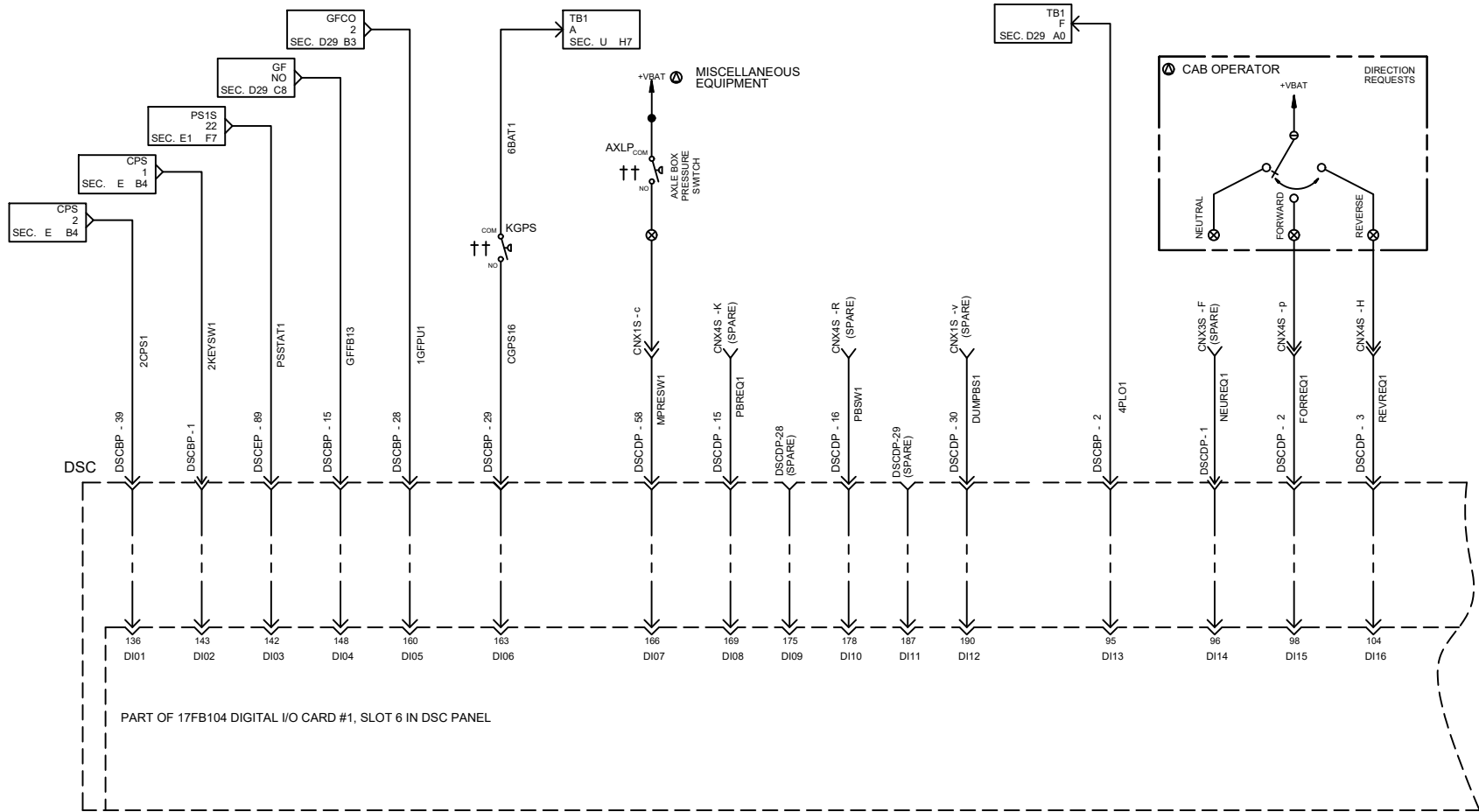




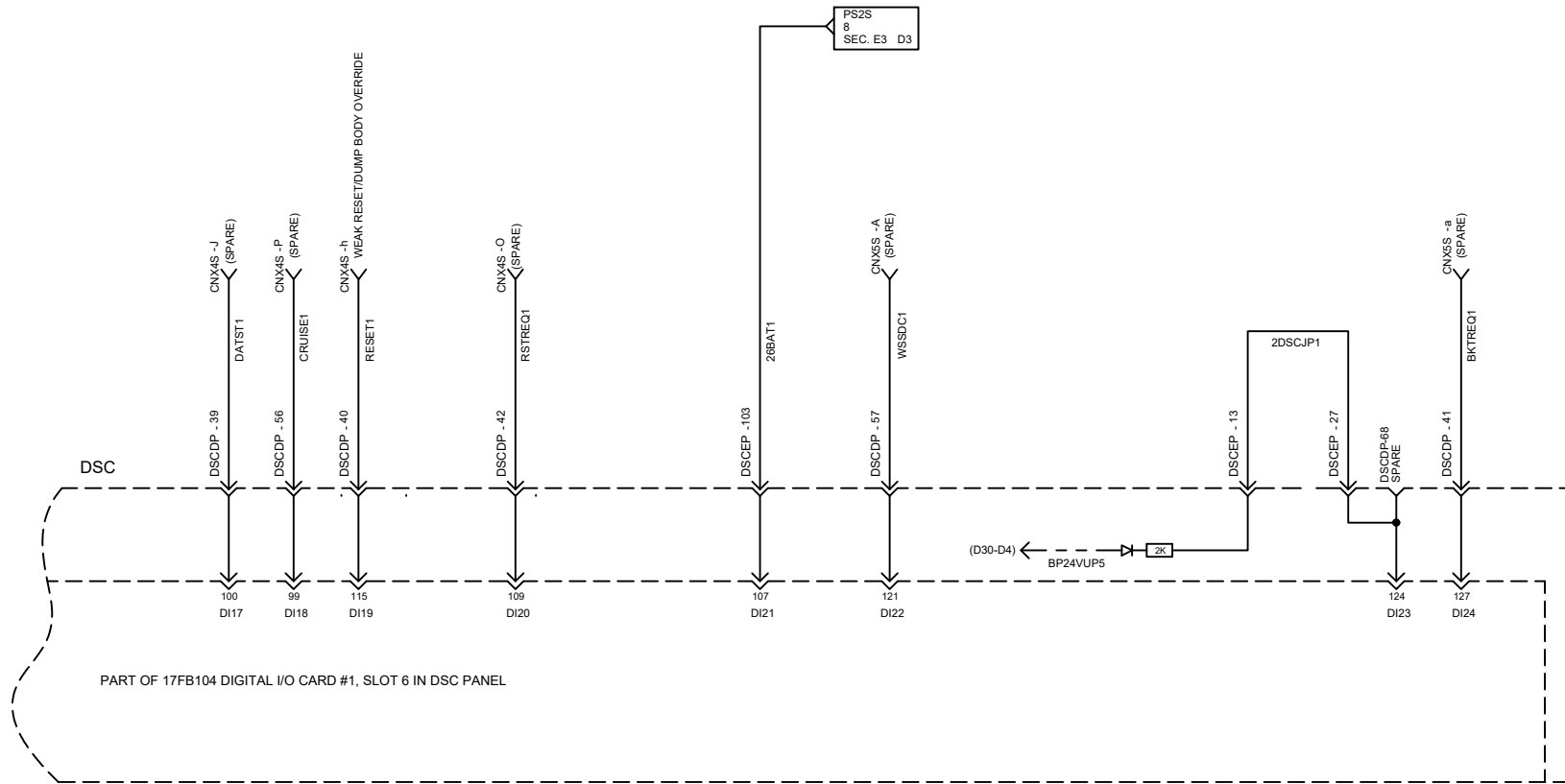
PART OF 17FB104 DIGITAL I/O CARD #1, SLOT 6 IN DSC PANEL

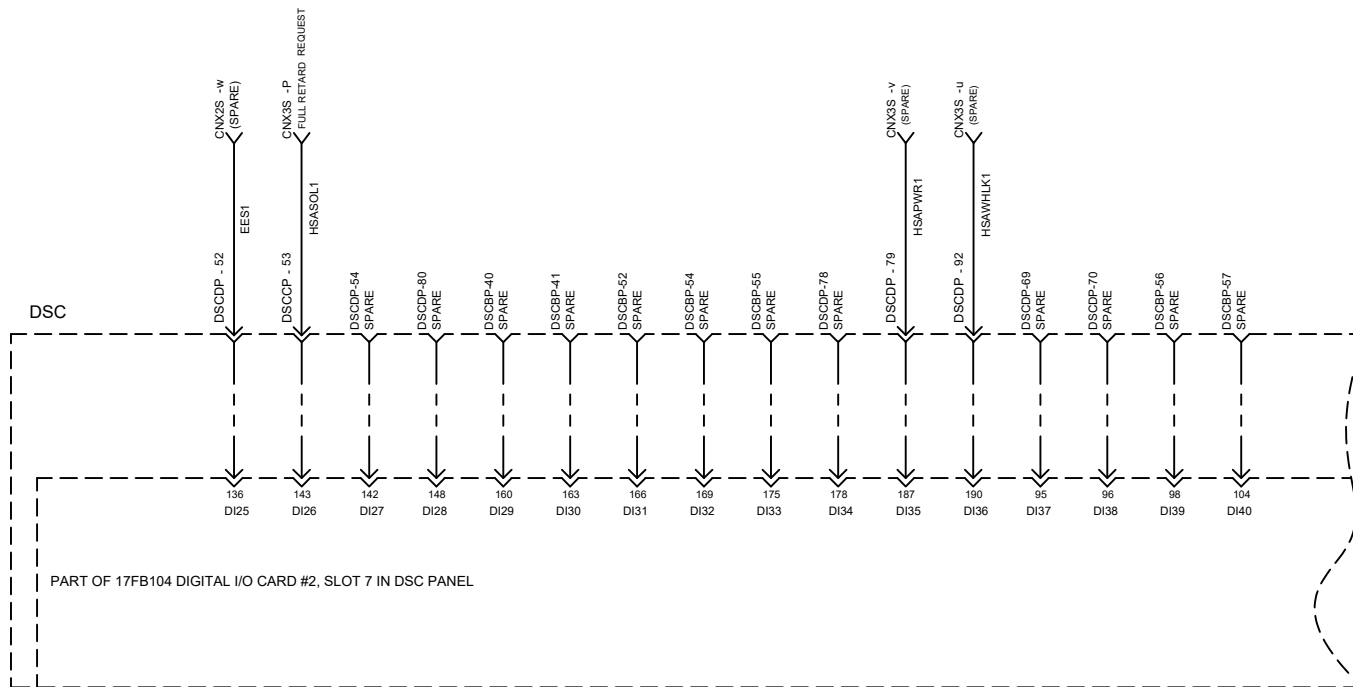


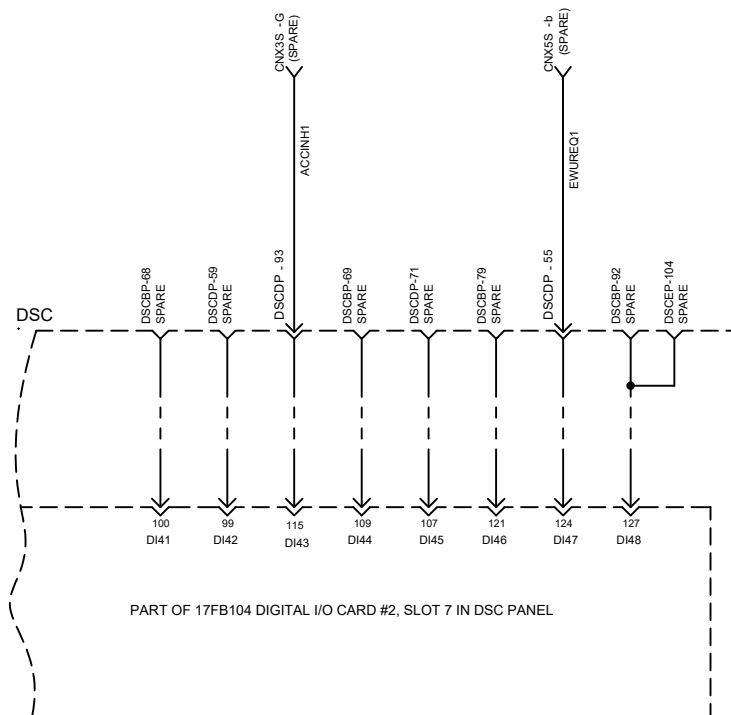


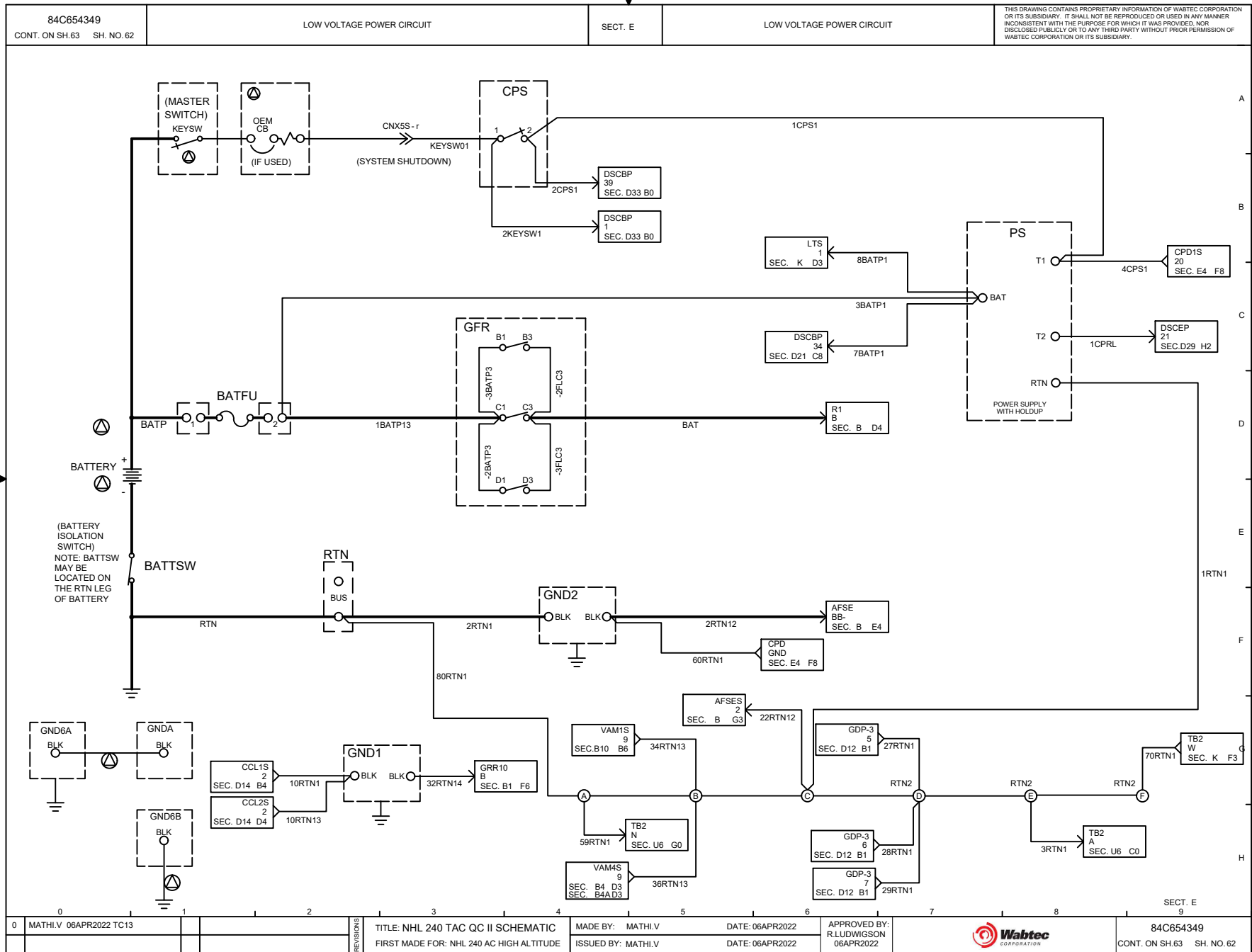


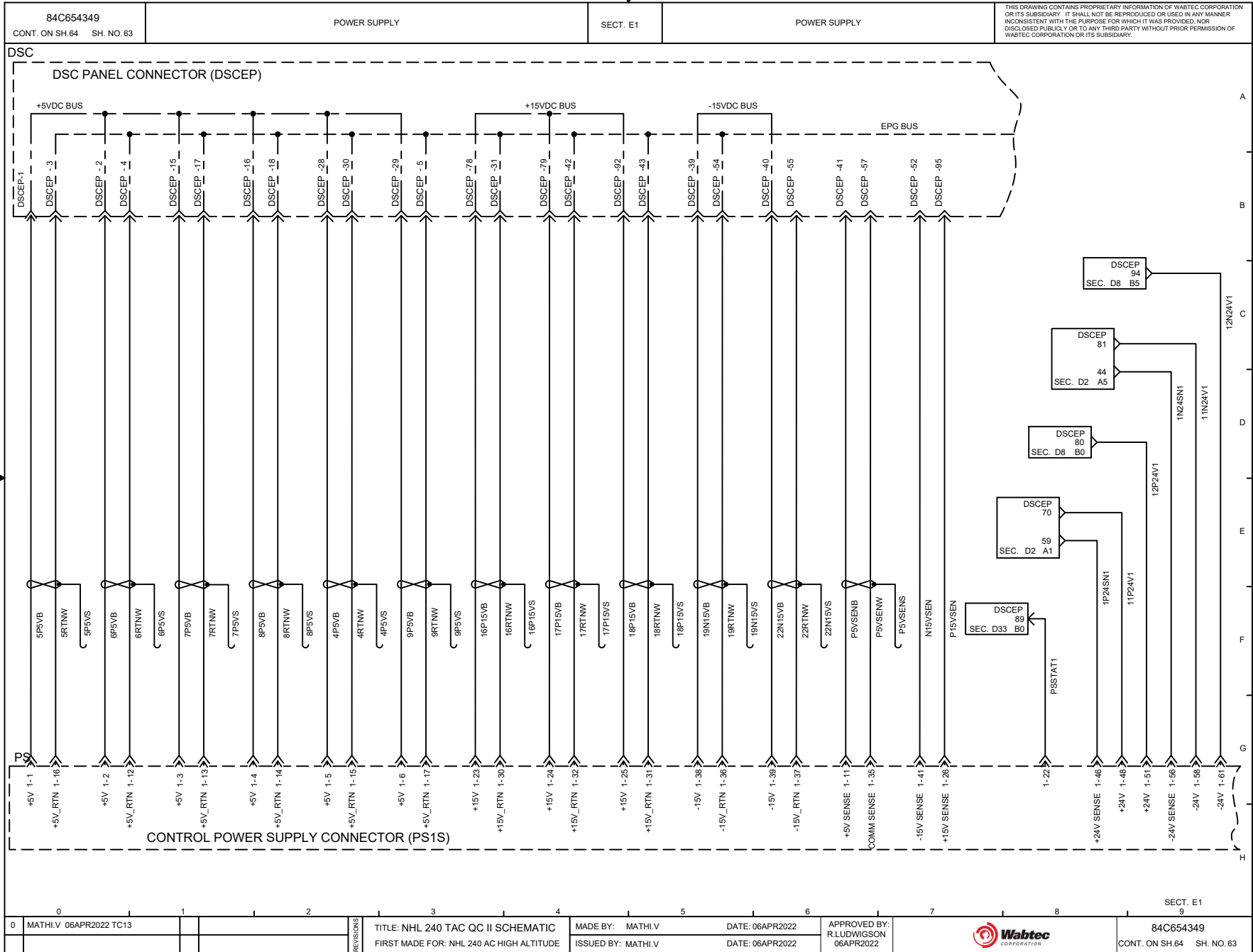
NOTES:
 †† CIRCUIT SHOWN IN LOW AIR POSITION.

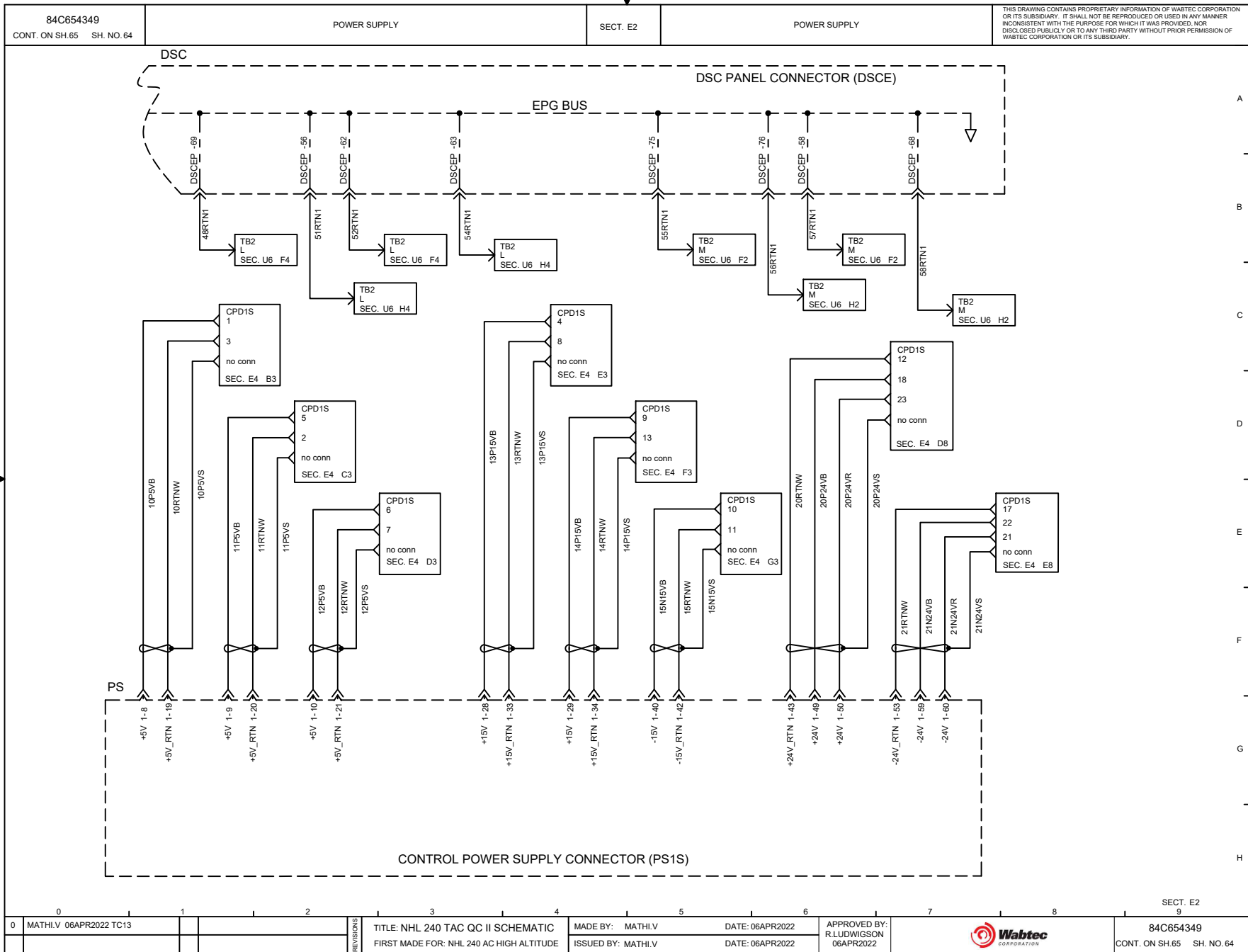




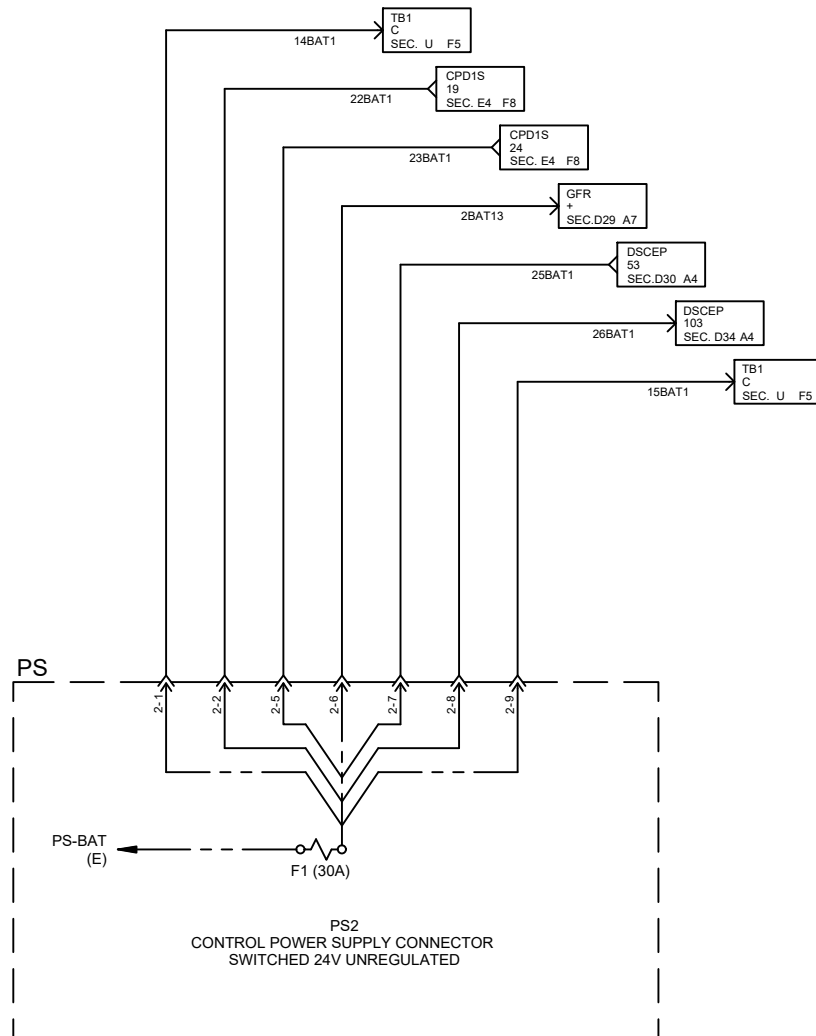






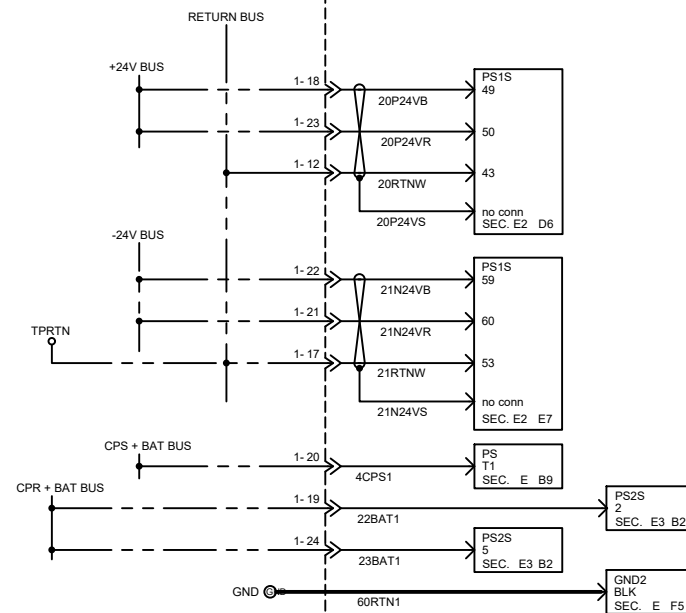


84C654349 CONT. ON SH.66 SH. NO.65	CONTROL POWER SUPPLY CONNECTOR (PS2)	SECT. E3	BLANK	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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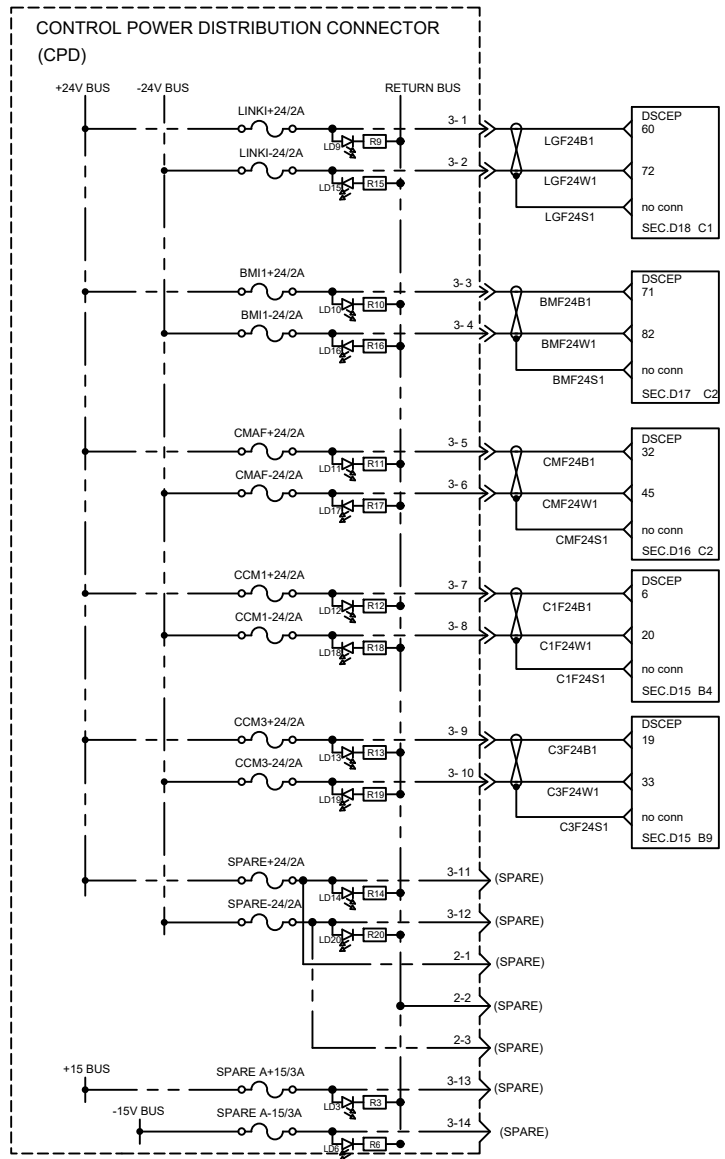
0 MATH.I.V 06APR2022 TC13	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATH.I.V ISSUED BY: MATH.I.V	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: R.LUDWIGSON 06APR2022		SECT. E3 9 84C654349 CONT. ON SH.66 SH. NO.65
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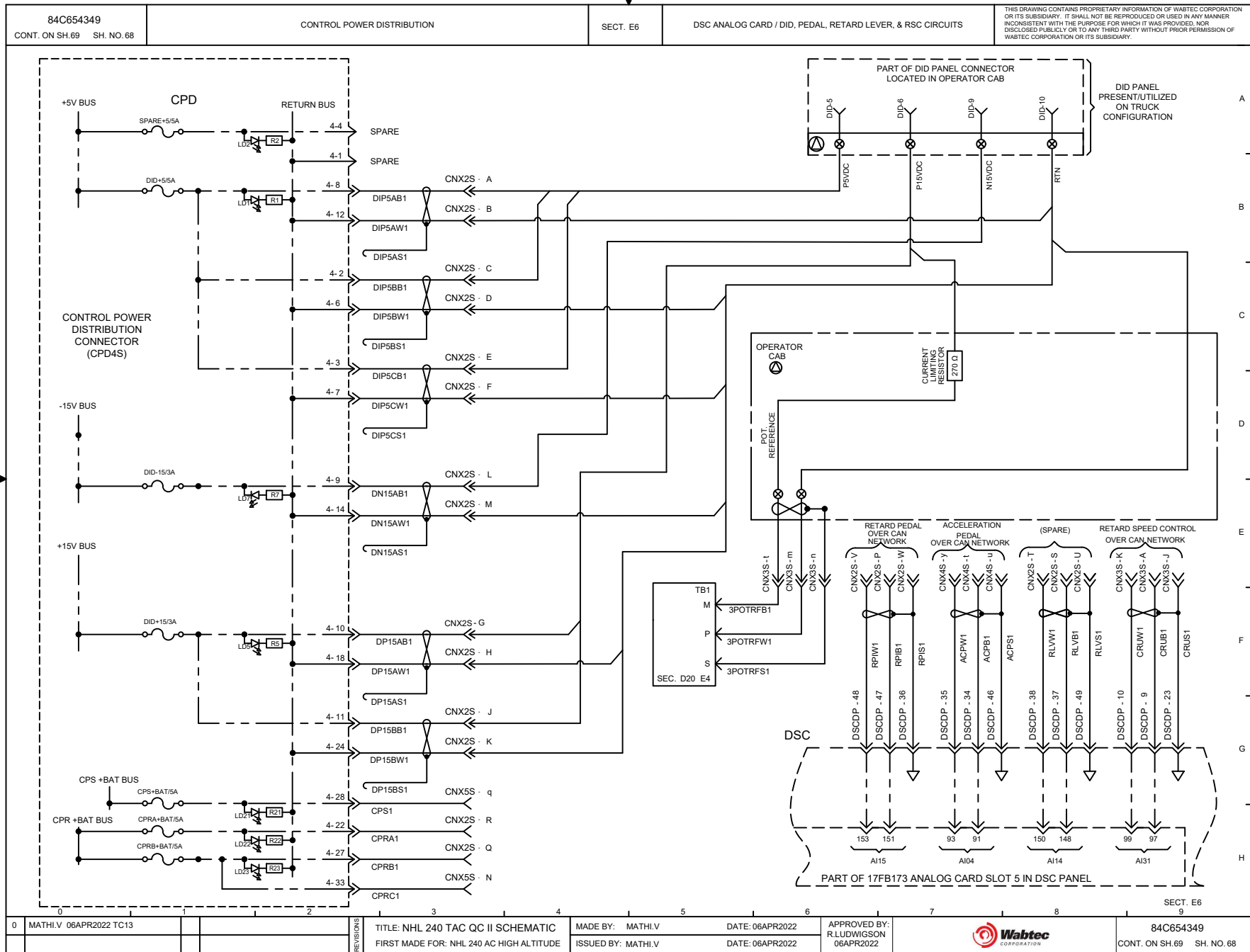
ACES

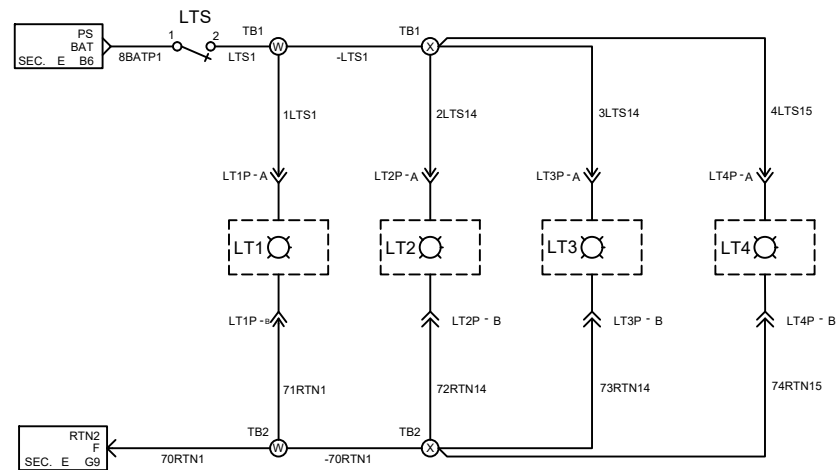


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CORPORATION

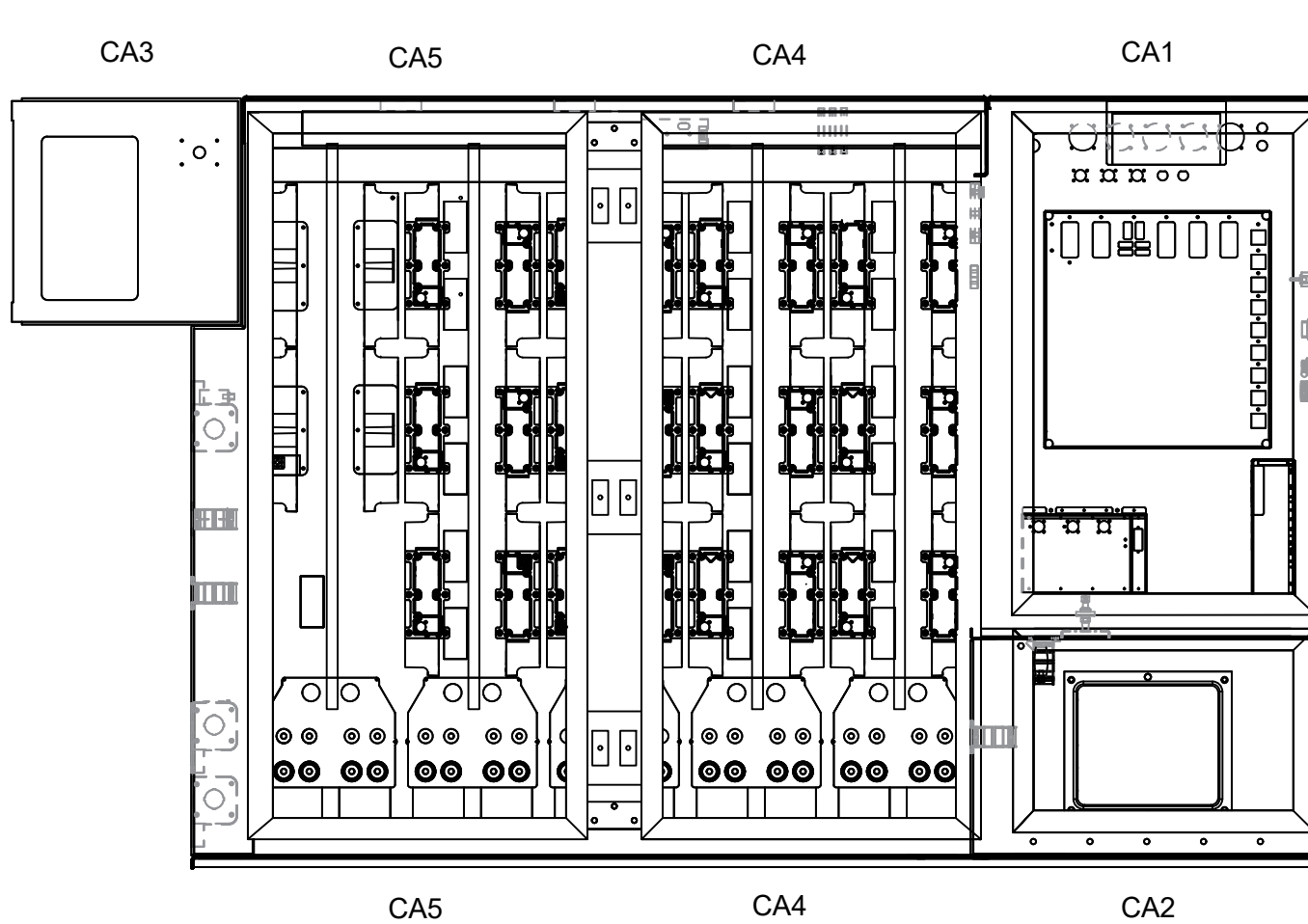
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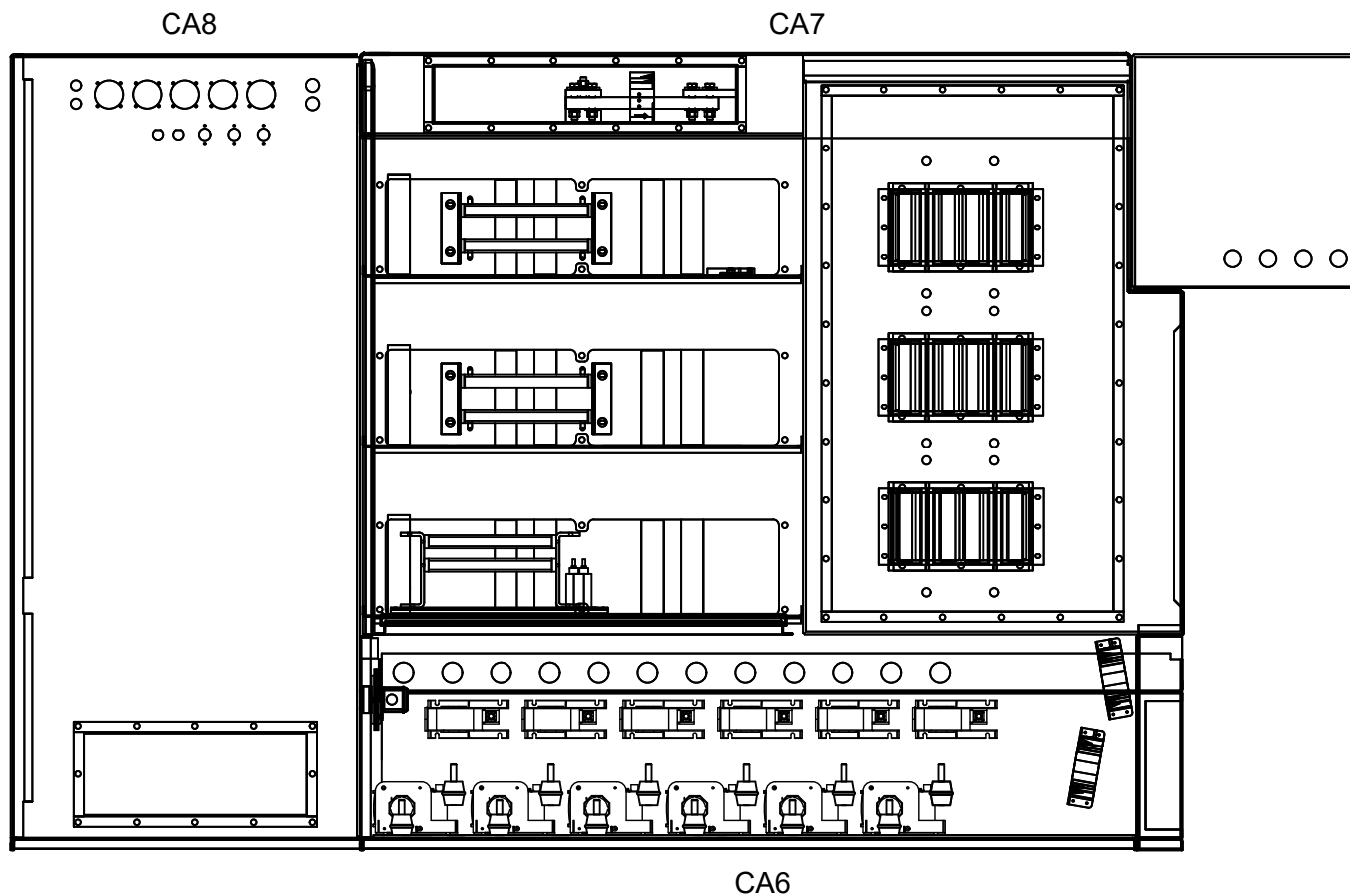
84C654349 CONT. ON SH.71 SH. NO.70	CONTROL GROUP AREA LOCATIONS: FRONT VIEW	SECT. P	CONTROL GROUP AREA LOCATIONS: FRONT VIEW	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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FRONT VIEW

0 MATHLV 06APR2022 TC13 	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATHLV ISSUED BY: MATHLV	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: RLUDWIGSON 06APR2022	 ACES	SECT. P 9 84C654349 CONT. ON SH.71 SH. NO.70
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84C654349 CONT. ON SH.73 SH. NO.72	CONTROL GROUP AREA LOCATIONS: BACK VIEW	SECT. P2	CONTROL GROUP AREA LOCATIONS: BACK VIEW	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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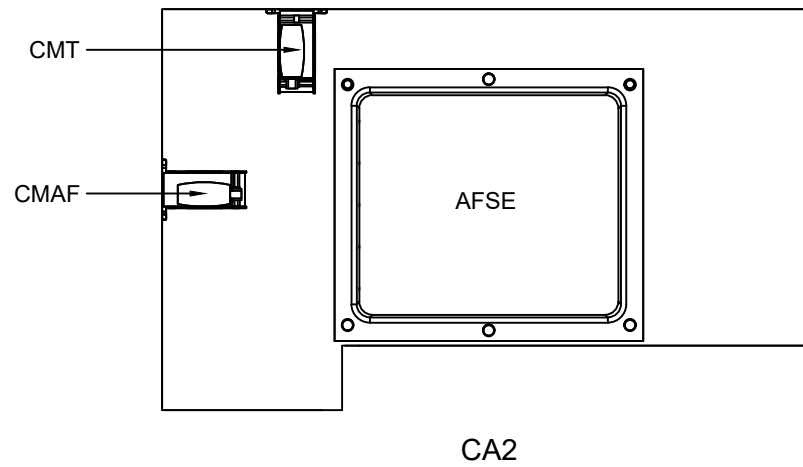


BACK VIEW

0 MATHLV 06APR2022 TC13	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATHLV ISSUED BY: MATHLV	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: RLUDWIGSON 06APR2022		SECT. P2 9 84C654349 CONT. ON SH.73 SH. NO.72
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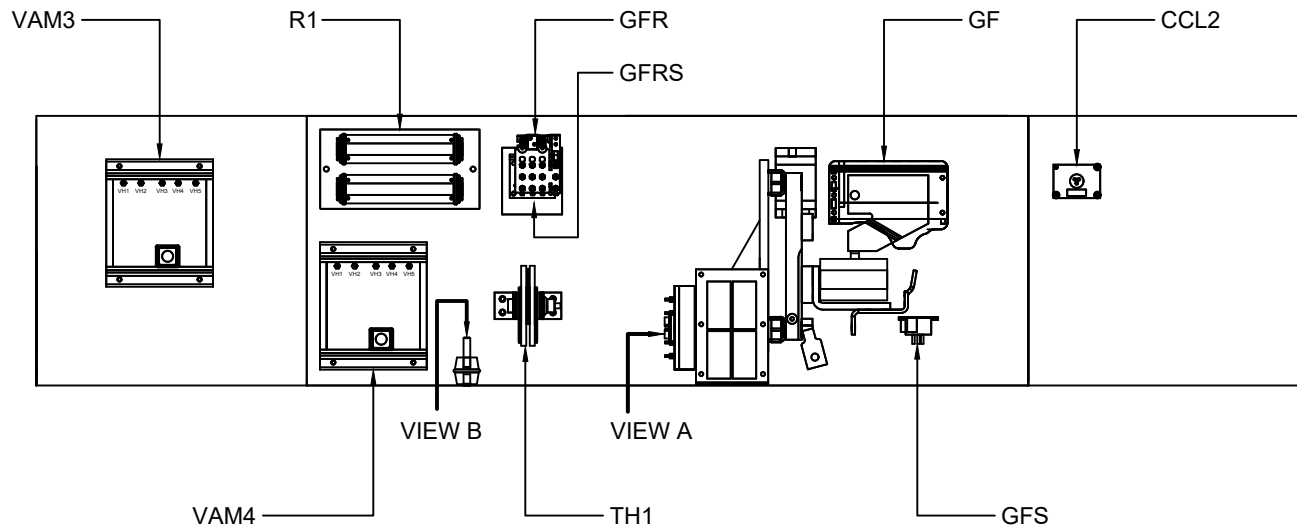
ACES

84C654349 CONT. ON SH.75 SH. NO.74	CONTROL GROUP AREA LOCATIONS: CA2	SECT. P4	CONTROL GROUP AREA LOCATIONS: CA2	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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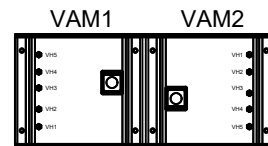


										SECT. P4	
										9	
0	MATHLV 06APR2022 TC13										

84C654349 CONT. ON SH.76 SH. NO.75	CONTROL GROUP AREA LOCATIONS: CA3	SECT. P5	CONTROL GROUP AREA LOCATIONS: CA3	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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CA3



VIEW A

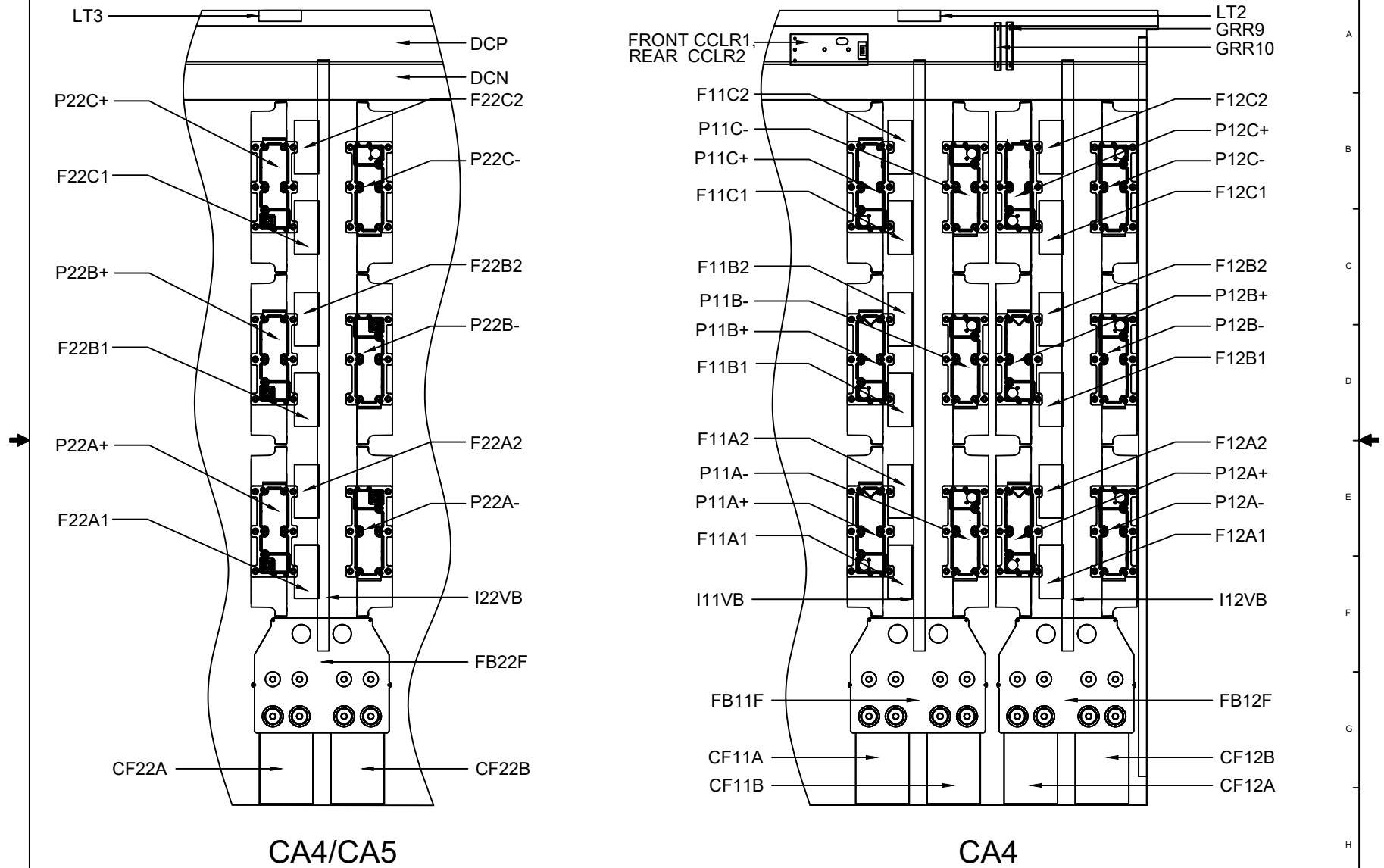
- T19
- T20
- F1
- F2

VIEW B

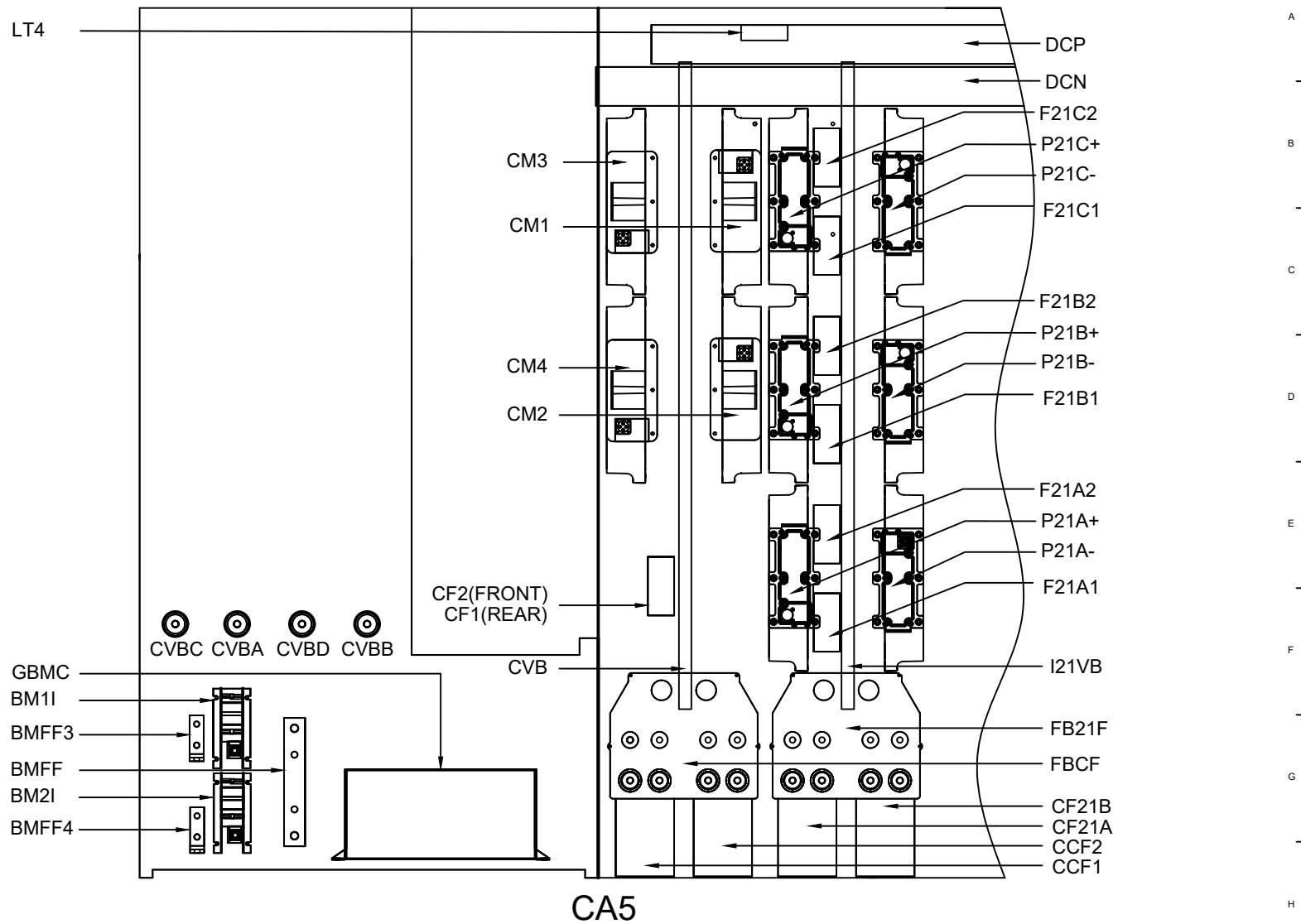
									SECT. P5										
0	1		2		3		4		5		6		7		8		9		
MATHLV 06APR2022 TC13					REVISIONS	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE				MADE BY: MATHLV ISSUED BY: MATHLV		DATE: 06APR2022 DATE: 06APR2022		APPROVED BY: R.LUDWIGSON 06APR2022				84C654349 CONT. ON SH.76 SH. NO.75	

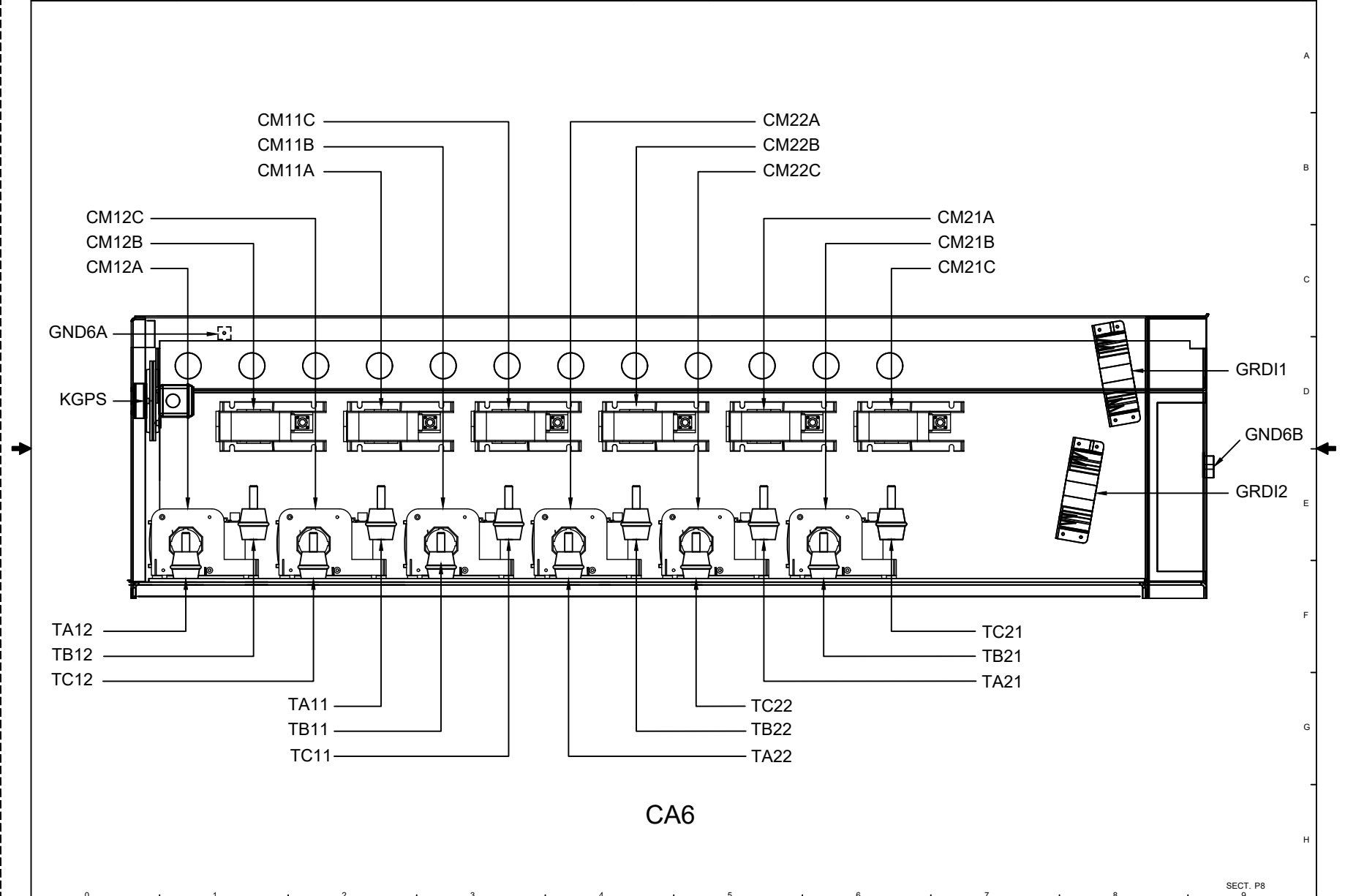
ACES

84C654349 CONT. ON SH.77 SH. NO. 76	CONTROL GROUP AREA LOCATIONS: CA4	SECT. P6	CONTROL GROUP AREA LOCATIONS: CA4	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
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0	1	2	3	4	5	6	7	8	9
0	MATH.IV 06APR2022 TC13	TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	MADE BY: MATH.IV ISSUED BY: MATH.IV	DATE: 06APR2022 DATE: 06APR2022	APPROVED BY: R.LUDWIGSON 06APR2022	 ACES		84C654349	CONT. ON SH.77 SH. NO. 76



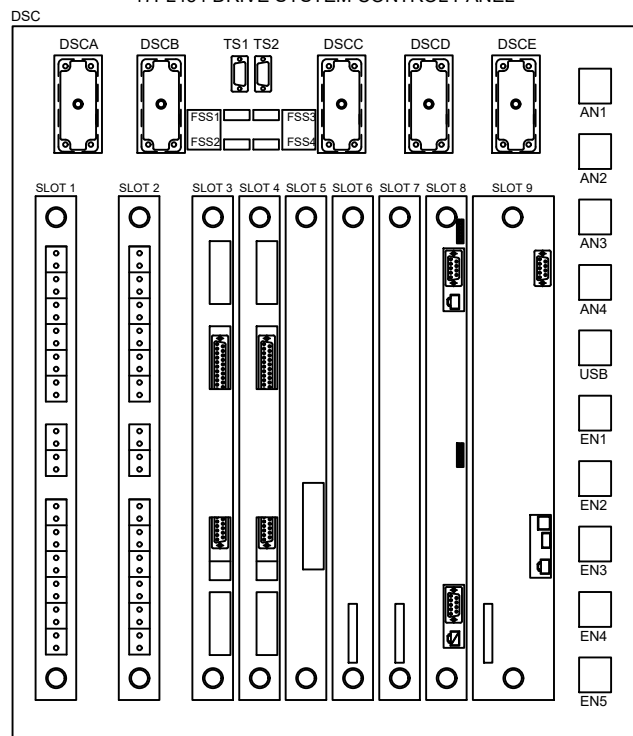


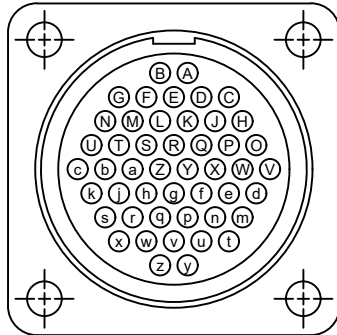


										SECT. P9									
0		1		2		3		4		5		6		7		8		9	
0	MATH.I.V 06APR2022 TC13					REVISIONS	TITLE: NHL 240 TAC QC II SCHEMATIC		MADE BY: MATH.I.V		DATE: 06APR2022		APPROVED BY: R.LUDWIGSSON 06APR2022			84C654349			
					FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE		ISSUED BY: MATH.I.V		DATE: 06APR2022		CONT. ON SH.80 SH. NO. 79								

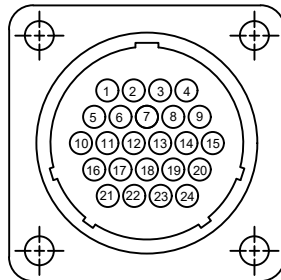
SLOT	TAG	DESCRIPTION	FACE PLATE CONNECTORS
1	DSC1: FODC1	INVERTER #1 FIBER OPTIC FODC1 (84C603188ADG1)	NONE
2	DSC2: FODC2	INVERTER #2 FIBER OPTIC FODC2 (84C603188AEG1)	NONE
3	DSC3: TMC	INVERTER #1 TMC (17FB190)	(1) RJ45, USB-A, USB-B, 16 PIN HOLDER
4	DSC4: TMC	INVERTER #2 TMC (17FB190)	(1) RJ45, USB-A, USB-B, 16 PIN HOLDER
5	DSC5: ANALG	ANALOG (17FB173)	(1) DB25
6	DSC6: I/O	DIGITAL I/O (17FB104) CARD #1	NONE
7	DSC7: I/O	DIGITAL I/O (17FB104) CARD #2	NONE
8	DSC8: ETH	ETHERNET SWITCH CARD (17FB191)	(2) DB9, (1) RJ45
9	DSC9: CPU	SYSTEM CPU (17FB197)	(1) DB9, RJ45, USB-A, USB-B

17FL454 DRIVE SYSTEM CONTROL PANEL

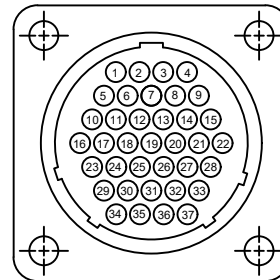




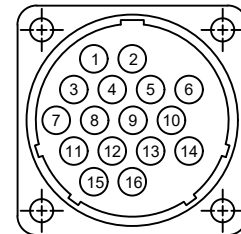
FLANGE RECEPTACLE (SOCKETS)
(CNX1, CNX2, CNX3, CNX4, CNX5)



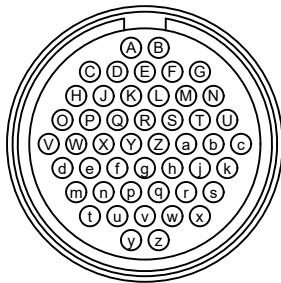
FLANGE RECEPTACLE (PINS)
(CPD1S)



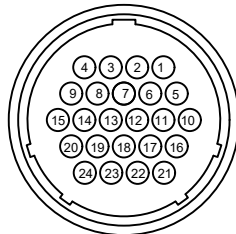
FLANGE RECEPTACLE (PINS)
(CPD4S)



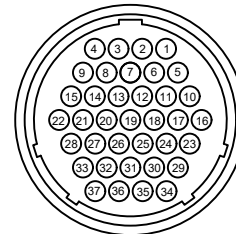
FLANGE RECEPTACLE (PINS)
(CPD3S, AFSES)



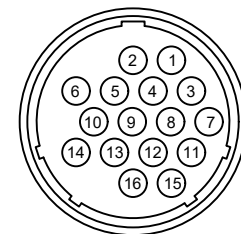
CABLE PLUG (PINS)
(CNX1, CNX2, CNX3, CNX4, CNX5)



CABLE PLUG (SOCKETS)
(CPD1S)

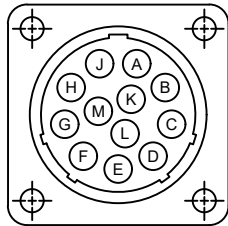


CABLE PLUG (SOCKETS)
(CPD4S)

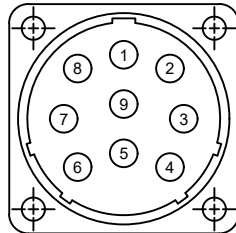


CABLE PLUG (SOCKETS)
(CPD3S, AFSES)

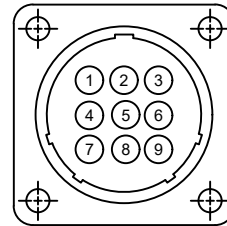
ACES



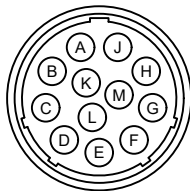
FLANGE RECEPTACLE (PINS)
(GDP-1,GDP-2)



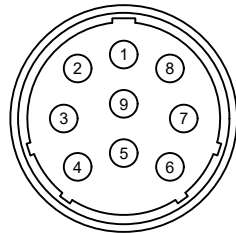
FLANGE RECEPTACLE (PINS)
(GDP-3)



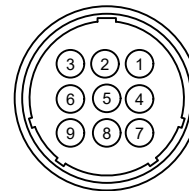
FLANGE RECEPTACLE (PINS)
(PS2S,VAM1S,VAM2S,VAM3S,VAM4S)



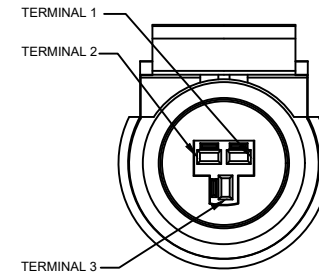
CABLE PLUG (SOCKETS)
(GDP-1,GDP-2)



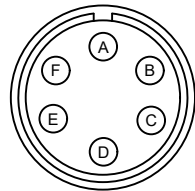
CABLE PLUG (SOCKETS)
(GDP-3)



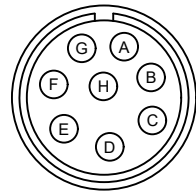
CABLE PLUG (SOCKETS)
(PS2S,VAM1S,VAM2S,VAM3S,VAM4S)



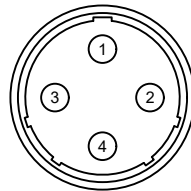
CABLE PLUG (SOCKETS)
(BAROS)



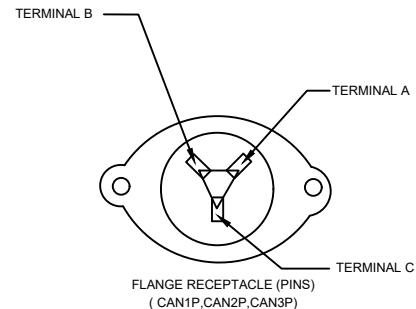
CABLE PLUG (SOCKETS)
(AN1S,AN2S,AN3S)



CABLE PLUG (SOCKETS)
(EN2S,EN3S,EN4S,ETC2S)



CABLE PLUG (SOCKETS)
(CM1IS,CM1-S,CM2IS,CM2-S,CM3IS,
CM3-S,CM4IS,CM4-S,CMAFS,CMTS,
BM1IS,BM2IS,
11A+S,11A-S,11B+S,11B-S,
11C+S,11C-S,12A+S,12A-S,
12B+S,12B-S,12C+S,12C-S,
21A+S,21A-S,21B+S,21B-S,
21C+S,21C-S,22A+S,22A-S,
22B+S,22B-S,22C+S,22C-S)



FLANGE RECEPTACLE (PINS)
(CAN1P,CAN2P,CAN3P)



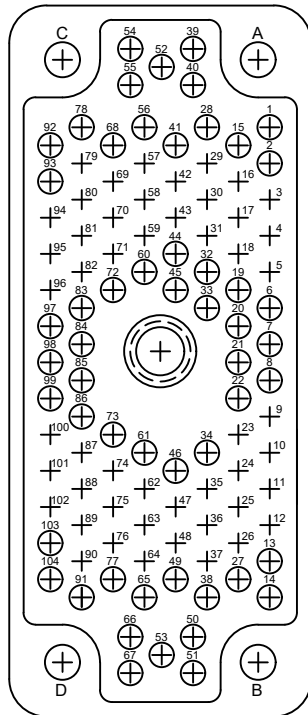
FLANGE RECEPTACLE (SOCKETS)
(CCL1S,CCL2S)
(CLR1S,CLR2S)



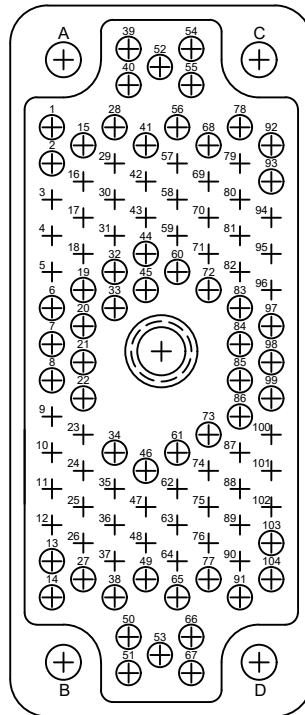
FLANGE RECEPTACLE (PINS)
(LT1P-LT4P)



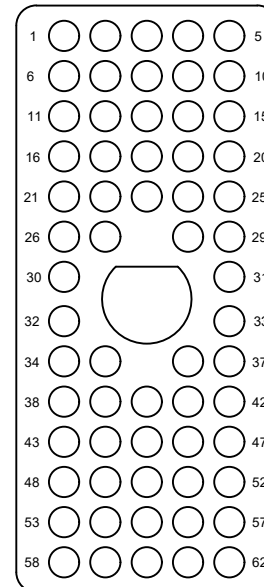
FLANGE RECEPTACLE (SOCKETS)
(DID-S)



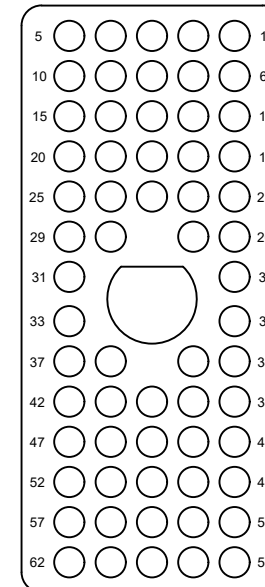
FLANGE RECEPTACLE (SOCKETS)
(DSCAP, DSCBP, DSCCP, DSCDP, DSCEP)



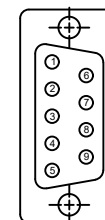
CABLE PLUG (PINS)
(DSCAP, DSCBP, DSCCP, DSCDP, DSCEP)



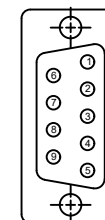
FLANGE RECEPTACLE (PINS)
(PS1S)



CABLE PLUG (SOCKETS)
(PS1S)

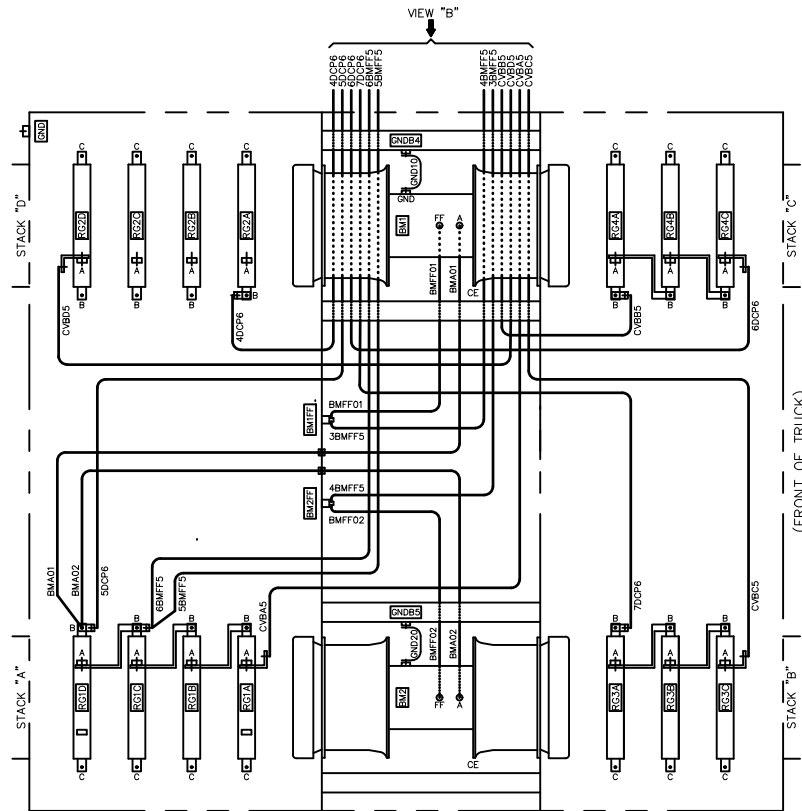
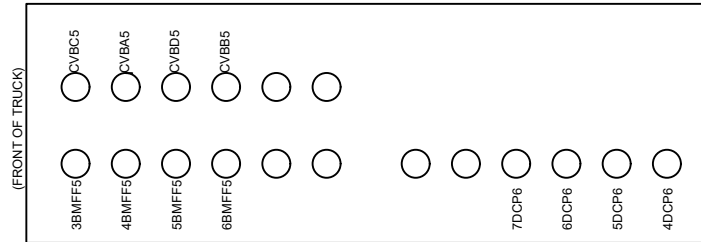


FLANGE
(SOCKETS)
TS1, TS2



CABLE PLUG
(PINS)
TS1, TS2

VIEW B
(TOP)



G2

ACES

84C654349 CONT. ON SH.88 SH. NO. 87	LAMINATED BUS BARS	SECT. R5	BLANK	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
	LONGITUDINAL CROSS SECTION OF LAMINATED INVERTER VERTICAL BUS BARS (IVB11, IVB12, IVB21, & IVB22) INVERTER VERTICAL BUS BARS IVB11, IVB12, IVB21, AND IVB22 ARE LOW INDUCTANCE LAMINATED BUS BARS THAT CONNECT THE PHASE MODULES TO THE DC LINK. EACH BUS BAR CONTAINS 3 LAYERS: "+", "-", AND "PHASE" REFER TO THE INDIVIDUALLY ELECTRICALLY ISOLATED LAYERS OF EACH BUS BAR. + - PHASE A B C + - PHASE			A B C
	LONGITUDINAL CROSS SECTION OF LAMINATED CHOPPER VERTICAL BUS BAR (CVB) CHOPPER VERTICAL BUS BAR CVB IS A LOW INDUCTANCE LAMINATED BUS BAR THAT CONNECTS THE CHOPPER MODULES TO THE DC LINK. THE BUS BAR CONTAINS 3 LAYERS: "+", "-", AND "PHASE" REFER TO THE INDIVIDUALLY ELECTRICALLY ISOLATED LAYERS OF THE BUS BAR. - + 3 2 1 LAYERS FUSE			D E
	LONGITUDINAL CROSS SECTION OF LAMINATED INVERTER VERTICAL BUS BARS (FB11F, FB12F, FB21F, FB22F, FBCF) CAPACITOR BUS BARS FB11F, FB12F, FB21F, & FB22F ARE LOW INDUCTANCE LAMINATED BUS BARS THAT CONNECT THE FILTER CAPACITORS TO THE VERTICAL BUS BARS. EACH BUS BAR CONTAINS 2 LAYERS: "+", "-", AND "PHASE" REFER TO THE INDIVIDUALLY ELECTRICALLY ISOLATED LAYERS OF EACH BUS BAR. + - 1 2 LAYERS			F G H
0 MATH.I.V 06APR2022 TC13	1 2 3 4 5 6 7 8 9 SECT. R5 0 1 2 3 4 5 6 7 8 9 TITLE: NHL 240 TAC QC II SCHEMATIC FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE MADE BY: MATH.I.V ISSUED BY: MATH.I.V DATE: 06APR2022 DATE: 06APR2022 APPROVED BY: R.LUDWIGSON 06APR2022 84C654349 CONT. ON SH.88 SH. NO. 87			Wabtec CORPORATION



Wabtec
CORPORATION

ACES

84C654349	INTERFACE CABLE LIST & WIRING GUIDELINES	SECT. R6	INTERFACE CABLE LIST & WIRING GUIDELINES	THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.
CONT. ON SH.89 SH. NO.88				

NHL 240T HIGH ALTITUDE TRUCK CONFIGURATION							
CABLE NAME	17KG560		EXTERNAL		NOTES	CABLE SIZE	SCHEM. SECTION
	DEVICE	TERMINAL	DEVICE	TERMINAL			
					HIGH VOLTAGE CABLES		
T101	RDA	AC	ALT	T1	CABLE PROVIDED WITH EXTERNAL DEVICE	925/24	B, B1
T102	RDA	AC	ALT	T1	CABLE PROVIDED WITH EXTERNAL DEVICE	925/24	B, B1
T201	RDB	AC	ALT	T2	CABLE PROVIDED WITH EXTERNAL DEVICE	925/24	B, B1
T202	RDB	AC	ALT	T2	CABLE PROVIDED WITH EXTERNAL DEVICE	925/24	B, B1
T301	RDC	AC	ALT	T3	CABLE PROVIDED WITH EXTERNAL DEVICE	925/24	B, B1
T302	RDC	AC	ALT	T3	CABLE PROVIDED WITH EXTERNAL DEVICE	925/24	B, B1
T19	T19	APL	ALT	T19	CABLE PROVIDED WITH EXTERNAL DEVICE	225/24	B
T20	T20	APL	ALT	T20	CABLE PROVIDED WITH EXTERNAL DEVICE	225/24	B
F1	F1	APL	ALT	F1	CABLE PROVIDED WITH EXTERNAL DEVICE	225/24	B
F2	F2	APL	ALT	F2	CABLE PROVIDED WITH EXTERNAL DEVICE	225/24	B
TA101	TA11	APL	TM1	TA	⊗	550/24	B9
TA102	TA12	APL	TM1	TA	⊗	550/24	B9
TB101	TB11	APL	TM1	TB	⊗	550/24	B9
TB102	TB12	APL	TM1	TB	⊗	550/24	B9
TC101	TC11	APL	TM1	TC	⊗	550/24	B9
TC102	TC12	APL	TM1	TC	⊗	550/24	B9
TA201	TA21	APL	TM2	TA	⊗	550/24	B10
TA202	TA22	APL	TM2	TA	⊗	550/24	B10
TB201	TB21	APL	TM2	TB	⊗	550/24	B10
TB202	TB22	APL	TM2	TB	⊗	550/24	B10
TC201	TC21	APL	TM2	TC	⊗	550/24	B10
TC202	TC22	APL	TM2	TC	⊗	550/24	B10
3BMFF5	BMFF3	APL	BM1FF	APL	CABLE PROVIDED WITH EXTERNAL DEVICE, THROUGH BM1I CURRENT SENSOR	225/24	B4
4BMFF5	BMFF4	APL	BM2FF	APL	CABLE PROVIDED WITH EXTERNAL DEVICE, THROUGH BM2I CURRENT SENSOR	225/24	B4
5BMFF5	BMFF	BUS	RG1C	B	CABLE PROVIDED WITH EXTERNAL DEVICE	225/24	B4
6BMFF5	BMFF	BUS	RG1C	B	CABLE PROVIDED WITH EXTERNAL DEVICE	225/24	B4
4DCP6	DCP	BUS	RG2A	B	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B4
5DCP6	DCP	BUS	RG1D	B	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B4
6DCP6	DCP	BUS	RG4C	A	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B4
7DCP6	DCP	BUS	RG3A	B	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B4
CVBA5	CVBA	APL	RG1A	A	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B2, B4
CVBB5	CVBB	APL	RG4A	B	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B3, B4
CVBC5	CVBC	APL	RG3C	A	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B3, B4
CVBD5	CVBD	APL	RG2D	A	CABLE PROVIDED WITH EXTERNAL DEVICE	650/24	B2, B4
LOW VOLTAGE CABLES							
BATP1	BATFU	1	BATTSW	M	⊗	275/24	E
RTN1	RTN	BUS	BATTERY	-	⊗	275/24	E

NOTES:

- CABLE SIZE BASED ON 2000 VOLT / 125 DEGREE C INSULATION FOR HIGH VOLTAGE CABLES. CABLE SIZE BASED ON 600 VOLT / 125 DEGREE C INSULATION FOR LOW VOLTAGE CABLES.
- ALL EXTERNAL CABLE BUSHINGS MUST HAVE SEALING RINGS APPLIED UPON INSTALLATION.
- THE TRACTION MOTORS REQUIRE TWO CABLES PER PHASE TO MAINTAIN CURRENT BALANCE BETWEEN THE TWO SHARING INVERTERS. THE SIZE AND ROUTING OF THESE CABLES IS IMPORTANT TO SUCCESSFUL INVERTER OPERATION. THE ORIGINAL FACTORY CABLE ARRANGEMENT MUST BE MAINTAINED.

WIRING/EMC GUIDELINES

- BATTERY LEVEL AND SIGNAL WIRING MUST BE RUN SEPARATELY FROM HIGH VOLTAGE CABLES.
- WHERE SIGNAL LEADS OR LOW VOLTAGE LEADS INTERSECT POWER CABLING, IT SHOULD BE DONE AT RIGHT ANGLES. IF PARALLEL RUNS OF HIGH VOLTAGE AND SIGNAL/LOW VOLTAGE LEADS CANNOT BE AVOIDED, THE LENGTH MUST BE KEPT TO A MINIMUM AND MAXIMUM POSSIBLE SEPARATION SHOULD BE MAINTAINED.
- PROPER PRACTICES FOR LOW LEVEL WIRING MUST BE USED. (EX. TWISTED/SHIELDED CABLE, PROPER GROUNDING, ETC.)

0	MATHI.V 06APR2022 TC13	1	2	3	4	5	6	7	8	9	SECT. R6
0	TITLE: NHL 240 TAC QC II SCHEMATIC		MADE BY: MATHI.V		DATE: 06APR2022		APPROVED BY: R.LUDWIGSON		06APR2022		84C654349
	FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE		ISSUED BY: MATHI.V		DATE: 06APR2022						CONT. ON SH.89 SH. NO.88



ACES

84C654349

CONT. ON SH.91 SH. NO. 90

TERMINAL BOARD INDEX

SECT. S

TERMINAL BOARD INDEX

BMFF3

TERM	LOCATION	WIRES					
APL	SEC.B4-D3	3BMFF					

BMFF4

TERM	LOCATION	WIRES					
APL	SEC.B4-F3	4BMFF					

CVBA

TERM	LOCATION	WIRES					
APL	SEC.B2-D3	1CVBA					

CVBB

TERM	LOCATION	WIRES					
APL	SEC.B3-D6	1CVBB					

CVBC

TERM	LOCATION	WIRES					
APL	SEC.B3-D1	1CVBC					

CVBD

TERM	LOCATION	WIRES					
APL	SEC.B2-D7	1CVBD					

F1

TERM	LOCATION	WIRES					
APL	SEC.B-B5	F103	F104				

F2

TERM	LOCATION	WIRES					
APL	SEC.B-E7	F201					

RTN2

TERM	LOCATION	WIRES					
A	SEC.E-G4	59RTN1	80RTN1				
B	SEC.E-G5	36RTN13	34RTN13				
C	SEC.E-G6	22RTN12	1RTN1				
D	SEC.E-G7	27RTN1	26RTN1	29RTN1			
E	SEC.E-G8	3RTN1					
F	SEC.E-G9	70RTN1					

T19

TERM	LOCATION	WIRES					
APL	SEC.B-A5	T1902	T1901				

T20

TERM	LOCATION	WIRES					
APL	SEC.B-B0	T2001	T2002				

TB1

TERM	LOCATION	WIRES					
A	SEC.U-F8	2BAT1	3BAT1	6BAT1	-BAT1		
B	SEC.U-E8	-BAT1	10BAT1	11BAT1	-1BAT1		

TB1

TERM	LOCATION	WIRES					
C	SEC.U-E6	-1BAT1	14BAT1	15BAT1	-2BAT1		
D	SEC.U-E6	-2BAT1	27BAT1	28BAT1	29BAT1		
E	SEC.D29-B1	-PLO11					
F	SEC.D29-A1	3PLO1	2PLO1	-PLO11	4PLO1		
L	SEC.D20-E4	1POTRFB1	-POT1				
M	SEC.D20-E3	-POT1	3POTRFB1	2POTRFB1			
N	SEC.D20-F4	1POTRFW1	-1POT1				
P	SEC.D20-F3	-1POT1	2POTRFW1	3POTRFW1			
R	SEC.D20-F4	1POTRFS1	-2POT1				
S	SEC.D20-F3	-2POT1	2POTRFS1	3POTRFS1			
W	SEC.K-C4	LTS1	-LTS1	1LTS1			
X	SEC.K-C5	-LTS1	3LTS14	4LTS15	2LTS14		

TB2

TERM	LOCATION	WIRES					
A	SEC.U6-C1	-RTN1	4RTN1	5RTN1	6RTN1	3RTN1	
B	SEC.U6-C3	-1RTN1	7RTN1	8RTN1	9RTN1	-RTN1	
C	SEC.U6-C4	-2RTN1	11RTN1	12RTN1	13RTN1	-1RTN1	
D	SEC.U6-C6	-3RTN1	14RTN1	15RTN1	16RTN1	-2RTN1	
E	SEC.U6-C8	17RTN1	18RTN1	19RTN1	-3RTN1		
L	SEC.U6-F4	48RTN1	51RTN1	52RTN1	54RTN1	-8RTN1	
M	SEC.U6-F3	-6RTN1	55RTN1	56RTN1	57RTN1	58RTN1	-9RTN1
N	SEC.U6-F1	-9RTN1	59RTN1				
W	SEC.K-F4	70RTN1	-70RTN1	71RTN1			
X	SEC.K-F5	72RTN14	73RTN14	74RTN15	-70RTN1		

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MATH.I.V 06APR2022 TC13

ACES

84C654349

CONT. ON SH.92 SH. NO. 91

DEVICE TABLE

SECT. S1

DEVICE TABLE

11A+S

TERM	LOCATION	WIRE NAME
1	D12-	
2	D12-E7	P11APB
3	D12-F7	P11APW
4		

11A-S

TERM	LOCATION	WIRE NAME
1	D12-F7	P11APW
2	D12-E5	P11ANB
3	D12-F5	P11ANW
4	D12-E7	P11APB

11B+S

TERM	LOCATION	WIRE NAME
1	D12-F5	P11ANW
2	D12-E4	P11BPB
3	D12-F4	P11BPW
4	D12-E5	P11ANB

11B-S

TERM	LOCATION	WIRE NAME
1	D12-F4	P11BPW
2	D12-E2	P11BNB
3	D12-F2	P11BNW
4	D12-E4	P11BPB

11C+S

TERM	LOCATION	WIRE NAME
1	D12-F2	P11BNW
2	D12-E1	P11CPB
3	D12-F1	P11CPW
4	D12-E2	P11BNB

11C-S

TERM	LOCATION	WIRE NAME
1	D12-F1	P11CPW
2	D12-E0	P11CNB
3	D12-F0	P11CNW
4	D12-E1	P11CPB

12A+S

TERM	LOCATION	WIRE NAME
1	D12-	
2	D12-F7	P12APB
3	D12-G7	P12APW
4		

12A-S

TERM	LOCATION	WIRE NAME
1	D12-G7	P12APW
2	D12-F5	P12ANB
3	D12-G5	P12ANW
4	D12-F7	P12APB

12B+S

TERM	LOCATION	WIRE NAME
1	D12-G5	P12ANW
2	D12-F4	P12BPB
3	D12-G4	P12BPW
4	D12-F5	P12ANB

12B-S

TERM	LOCATION	WIRE NAME
1	D12-G4	P12BPW
2	D12-F2	P12BNB
3	D12-G2	P12BNW
4	D12-F4	P12BPB

12C+S

TERM	LOCATION	WIRE NAME
1	D12-G2	P12BNW
2	D12-F1	P12CPB
3	D12-G1	P12CPW
4	D12-F2	P12BNB

12C-S

TERM	LOCATION	WIRE NAME
1	D12-G1	P12CPW
2	D12-F0	P12CNB
3	D12-G0	P12CNW
4	D12-F1	P12CPB

21A+S

TERM	LOCATION	WIRE NAME
1	D13-	
2	D13-E8	P21APB
3	D13-E8	P21APW
4		

21A-S

TERM	LOCATION	WIRE NAME
1	D13-E8	P21APW
2	D13-E7	P21ANB
3	D13-E7	P21ANW
4	D13-E8	P21APB

21B+S

TERM	LOCATION	WIRE NAME
1	D13-E7	P21ANW
2	D13-E5	P21BPB
3	D13-E5	P21BPW
4	D13-E7	P21ANB

21B-S

TERM	LOCATION	WIRE NAME
1	D13-E5	P21BPW
2	D13-E4	P21BNB
3	D13-E4	P21BNW
4	D13-E5	P21BPB

21C+S

TERM	LOCATION	WIRE NAME
1	D13-E4	P21BNW
2	D13-E3	P21CPB
3	D13-E3	P21CPW
4	D13-E4	P21BNB

21C-S

TERM	LOCATION	WIRE NAME
1	D13-E3	P21CPW
2	D13-E1	P21CNB
3	D13-E1	P21CNW
4	D13-E3	P21CPB

22A+S

TERM	LOCATION	WIRE NAME
1	D13-	
2	D13-F8	P22APB
3	D13-F8	P22APW
4		

22A-S

TERM	LOCATION	WIRE NAME
1	D13-F8	P22APW
2	D13-F7	P22ANB
3	D13-F7	P22ANW
4	D13-F8	P22APB

22B+S

TERM	LOCATION	WIRE NAME
1	D13-F7	P22ANW
2	D13-F5	P22BPB
3	D13-F5	P22BPW
4	D13-F7	P22ANB

22B-S

TERM	LOCATION	WIRE NAME
1	D13-F5	P22BPW
2	D13-F4	P22BNB
3	D13-F4	P22BNW
4	D13-F5	P22BPB

22C+S

TERM	LOCATION	WIRE NAME
1	D13-F4	P22BNW
2	D13-F3	P22CPB
3	D13-F3	P22CPW
4	D13-F4	P22BNB

22C-S

TERM	LOCATION	WIRE NAME
1	D13-F3	P22CPW
2	D13-F1	P22CNB
3	D13-F1	P22CNW
4	D13-F3	P22CPB

AFSES

TERM	LOCATION	WIRE NAME
1	B-F3	AFSEL12
2	B-F2	22RTN12
3	B-F1	AFSEF2B
4		
5	B-F2	AFSEF1B
6		
7	B-F1	AFSEF2W
8		
9	B-F2	AFSEF1W
10		
11		
12		
13		
14	B-F3	SCR3CM12
15	B-F0	AFSTB12
16	B-F1	AFSTW12

BAROS

TERM	LOCATION	WIRE NAME
1	D21-C6	BARW1
2	D21-C6	BARB1
3	D21-C6	BARR1

BM1IS

TERM	LOCATION	WIRE NAME
1	B4-E5	B1AB15
2	B4-E5	B1AR15
3	B4-E2	B1AW15
4		

BM2IS

TERM	LOCATION	WIRE NAME
1	B4-F2	B2AB15
2	B4-E7	B2AR15
3	B4-F2	B2AW15
4		

C11AS

TERM	LOCATION	WIRE NAME
1	B10-C1	C11AB16
2	B10-E0	C11AR16
3	B10-C6	C11AW16
4		

C11BS

TERM	LOCATION	WIRE NAME
1	B10-C1	C11BB16
2	B10-E2	C11BR16
3	B10-C6	C11BW16
4		

C11CS

TERM	LOCATION	WIRE NAME
1	B10-C2	C11CB16
2	B10-E3	C11CR16
3	B10-C7	C11CW16
4		

C12AS

TERM	LOCATION	WIRE NAME
1	B10-C2	C12AB16
2	B10-E5	C12AR16
3	B10-C7	C12AW16
4		

C12BS

TERM	LOCATION	WIRE NAME
1	B10-C3	C12BB16
2	B10-E6	C12BR16
3	B10-C8	C12BW16
4		

C12CS

TERM	LOCATION	WIRE NAME
1	B10-C3	C12CB16
2	B10-E8	C12CR16
3	B10-C8	C12CW16
4		

C21AS

TERM	LOCATION	WIRE NAME
1	B11-C1	C21AB16
2	B11-E1	C21AR16
3	B11-C6	C21AW16
4		

C21BS

TERM	LOCATION	WIRE NAME
1	B11-C1	C21BB16
2	B11-E3	C21BR16
3	B11-C6	C21BW16
4		

C21CS

TERM	LOCATION	WIRE NAME
1	B11-C2	C21CB16
2	B11-E4	C21CR16
3	B11-C7	C21CW16
4		

C22AS

TERM	LOCATION	WIRE NAME
1	B11-C2	C22AB16
2	B11-E6	C22AR16
3	B11-C7	C22AW16
4		

C22BS

TERM	LOCATION	WIRE NAME
1	B11-C3	C22BB16
2	B11-E8	C22BR16
3	B11-C8	C22BW16
4		

C22CS

TERM	LOCATION	WIRE NAME
1	B11-C3	C22CB16
2	B11-E9	C22CR16
3	B11-C8	C22CW16
4		

CCL1S

TERM	LOCATION	WIRE NAME
1	D14-B4	CCIL1N
2	D14-B4	10RTN1
3	D14-B4	CCIL1P

CCL2S

TERM	LOCATION	WIRE NAME
1	D14-D4	CCIL2N
2	D14-D4	10RTN13
3	D14-D4	CCIL2P

CLR1S

TERM	LOCATION	WIRE NAME
1	B1-B4	CCIL1N
2		
3	B1-B4	CCIL1P

CLR2S

TERM	LOCATION	WIRE NAME
1	B1-D4	CCIL2N
2		
3	B1-D4	CCIL2P

CM1IS

TERM	LOCATION	WIRE NAME
1	D15-D0	C1MB15
2	D15-C0	C1MR15
3	D15-C0	C1MW15
4		

CM1S

TERM	LOCATION	WIRE NAME
1	D12-D1	CM3PW
2	D12-D0	CM1PB
3	D12-D0	CM1PW
4	D12-D1	CM3PB

CM2IS

TERM	LOCATION	WIRE NAME
1	D15-	

CM2S

TERM	LOCATION	WIRE NAME
1	D13-C3	CM4PW
2	D13-C1	CM2PB
3	D13-C1	CM2PW
4	D13-C3	CM4PB

CM2-S

TERM	LOCATION	WIRE NAME
1	D13-C3	CM4PW
2	D13-C1	CM2PB
3	D13-C1	CM2PW
4	D13-C3	CM4PB

CM3IS

TERM	LOCATION	WIRE NAME
1	D15-D5	C3MB15
2		
3	D15-C5	C3MW15
4		

CM3-S

TERM	LOCATION	WIRE NAME
1	D12-	
2	D12-D1	CM3PB
3	D12-D1	CM3PW
4		

CM4IS

TERM	LOCATION	WIRE NAME
1	D15-	
2		
3		
4		

CM4-S

TERM	LOCATION	WIRE NAME
1	D13-	
2	D13-C3	CM4PB
3	D13-C3	CM4PW
4		

CMAFS

TERM	LOCATION	WIRE NAME
1	B-E4	ALFB12
2	B-E4	ALFR12
3	B-E4	ALFW12
4		

CM1S

TERM	LOCATION	WIRE NAME
1	D12-D1	CM3PW
2	D12-D0	CM1PB
3	D12-D0	CM1PW
4	D12-D1	CM3PB

CM2S

TERM	LOCATION	WIRE NAME
1	D15-	

CMTS

TERM	LOCATION	WIRE NAME
1	B-A3	ALTB12
2	B-A3	ALTR12
3	B-A4	ALTW12
4		

CNX1S

TERM	LOCATION	WIRE NAME
A	D5-F0	SS1T2W1
B	D5-F0	SS1T1G1
C	D5-F0	SS1T3G1
D	D5-F0	SS1T4W1
E	D5-F1	SS1T5G1
F	D5-F1	SS1T6W1
G	D5-F1	SS1TS1
H	D23-D0	SS3T2W1
J	D23-D0	SS3T1G1
K	D23-D0	SS3T3G1
L	D23-D1	SS3T4W1
M	D23-D1	SS3T5G1
N	D23-D1	SS3T6W1
O	D23-D2	SS3TS1
P	D23-D2	SS4T2W1
Q	D23-D3	SS4T1G1
R	D23-F3	SS4T3G1
S	D23-D3	SS4T4W1
T	D23-D4	SS4T5G1
U	D23-D4	SS4T6W1
V	D23-D4	SS4TS1
W	D23-D0	SS3T8W1
X	D23-D2	SS4T8W1
Y		
Z		
a		
b		
c	D33-C2	MPRESW1
d		
e		
f		
g		
h		
j	D11-F0	SS2T2W1
k	D11-F1	SS2T1G1
l		
m	D11-F1	SS2T3G1
n	D11-F1	SS2T4W1
o		
p	D11-F1	SS2T5G1
q	D11-F2	SS2T6W1
r	D11-F2	SS2TS1
s	D22-D5	ESSB1
t	D29-C5	RTPIND1
u		
v	D33-C4	RUMPBS1
w	D22-D5	ESSS1
x	D22-D5	ESSW1
y		
z		

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CONT. ON SH.94 SH. NO. 93

PANEL-PLUG PIN LOCATION

SECT. S3

PANEL-PLUG PIN LOCATION

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DSCBP			DSCBP			DSCCP			DSCCP			DSCDP			DSCDP			DSCEP			DSCEP		
TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME	TERM	LOCATION	WIRE NAME
20	D16-E4	ALFW12	73	D17-E5	B1AB15	19			72	D24-D3	TRKBW2	18	D29-C4	SPD1ST1	71			17	E1-B1	7RTNW	70	D2-C0	11P24V1
21	D16-A3	ALTB12	74	D17-E2	B1AW15	20			73			19	D29-C4	SPD2ST1	72			18	E1-B2	8RTNW	71	D17-E3	BMF24B1
22	D16-A4	ALTW12	75	D21-C3	AMBB17	21			74			20	D29-C5	RTDIND1	73			19	D15-E8	C3F24B1	72	D18-E2	LG24W1
23	D17-F2	B2AB15	76	D21-C3	AMBW17	22			75	D23-C8	FRSP03	21			74			20	D15-E3	C1F24W1	73	D20-E4	1POTRFB1
24	D17-F2	B2AW15	77	D19-F6	TERTS13	23			76	D23-C8	FRSP04	22			75			21	D29-C8	1CPRL	74	D20-F4	1POTRFW1
25	D22-F2	AFSEF1B	78			24			77	D5-F6	TMBS-S	23	E6-E9	CRUS1	76			22			75	E2-B5	55RTN1
26	D22-F2	AFSEF1W	79			25			78	D23-D0	SS3T1G1	24	D25-D1	MDSB1	77			23			76	E2-B6	56RTN1
27	D22-F3	AFSEF2S	80			26			79	D23-D2	SS3TS1	25	D25-D1	MDSR1	78			24			77	D30-C1	6DSCJP1
28	D33-D0	1GFPU1	81			27	D24-D7	SPA004S	80	D23-D1	SS3TS5G1	26	D25-D2	DIDB1	79	D35-D5	HSAPWR1	25			78	E1-B3	16P15VB
29	D33-D1	CGPS16	82			28	D11-F1	SS2T1G1	81	D23-D1	SS3TG6W1	27	D25-D2	DIDR1	80			26			79	E1-B4	17P15VB
30			83			29	D11-F2	SS2TS1	82			28			81			27	D34-D7	2DSCJP1	80	D8-C0	12P24V1
31			84	D18-E5	LNKR17	30	D11-F1	SS2TS5G1	83			29			82	D29-F5	7RTN1	28	E1-B2	4P5VB	81	D2-C5	11N24V1
32			85	D18-E4	LNKS17	31	D11-F2	SS2TG6W1	84			30	D33-C4	DUMPBS1	83	D30-F7	16RTN1	29	E1-B3	9P5VB	82	D17-E2	BMF24W1
33	D30-F1	11RTN1	86			32			85			31			84	D30-F7	17RTN1	30	E1-B2	4RTNW	83		
34	D21-C8	7BATP1	87	D18-E7	LDBS16	33	D24-D0	ENRPMR1	86			32	D30-F2	12RTN1	85	D30-F8	18RTN1	31	E1-B3	16RTNW	84		
35	D21-E0	31RTN14	88	D21-D6	BARS1	34			87			33	D30-F5	13RTN1	86	D30-F9	PSBTLR1	32	D16-E3	CMF24B1	85		
36	D21-E2	AFSTS12	89	D21-C6	BARB1	35			88			34	E6-E7	ACPB1	87	D31-F1	HSABKCM1	33	D15-E9	C3F24W1	86		
37	D19-F7	FLDB13	90	D19-F2	A3PPW13	36	D22-D5	ESSS1	89	D23-C8	FRSP-S	35	E6-E7	ACPW1	88	D31-F1	WSYSPM1	34			87		
38	D19-F7	FLOW13	91	D19-F2	A3PPS13	37	D24-D4	SPA003B	90	D5-F6	TMBS01	36	E6-E7	RPIS1	89			35			88		
39	D33-A4	2CPS1	92			38	D24-D5	SPA003W	91	D5-F7	TMBS02	37	E6-E8	RLVB1	90	D31-F5	19RTN1	36			89	D33-D0	PSSTAT1
40			93			39	D1-B7	4DSCJP1	92	D23-D0	SS3T2W1	38	E6-G8	RLVW1	91			37			90		
41			94			40	D1-B7	4DSCJP1	93	D23-D0	SS3T8W1	39	D34-C0	DATST1	92	D35-D6	HSAWHLK1	38	D30-C3	1DSCJP1	91	D30-C4	3DSCJP1
42			95			41	D11-F0	SS2T2W1	94	D23-D0	SS3T3G1	40	D34-C1	RESE1	93	D36-B1	ACCINH1	39	E1-B5	19N15VB	92	E1-B4	18P15VB
43			96			42	D11-F1	SS2T3G1	95	D23-D1	SS3T4W1	41	D34-C8	BKTREQ1	94			40	E1-B6	22N15VB	93	D29-D0	5DSCJP1
44			97	D18-E5	LNKB17	43	D11-F1	SS2T4W1	96			42	D34-C2	RSTREQ1	95	D29-C5	REVIND1	41	E1-B6	P5VSEN	94	D8-C5	12N24V1
45	D30-F3	SCR3CM12	98	D18-E5	LNKW17	44	D24-D0	ENRPMB1	97			43			96			42	E1-B4	17RTNW	95	E1-B7	P15VSEN
46	D21-E0	2GNDFF14	99	D18-E7	LDBB16	45	D24-D0	ENRPMW1	98			44	D30-B2	ENGRPH1	97			43	E1-B5	18RTNW	96		
47	D21-F0	AFSTB12	100	D18-E8	LDBW16	46			99			45	D30-B5	ENGRPM1	98	D30-B8	SYSRUN01	44	D2-C5	1N24SN1	97		
48	D21-F1	AFSTW12	101	D21-C6	BARR1	47	D22-D5	ESSB1	100			46	E6-G8	ACPS1	99	D30-B9	PSBTL1	45	D16-E2	CMF24W1	98		
49	D19-F7	FLDS13	102	D21-C6	BARW1	48	D22-D5	ESSW1	101			47	E6-E7	RPB1	100	D31-D1	10BAT1	46			99		
50	D19-F8	GRDBL13	103	D19-F1	A3PPB13	49	D24-D5	SPA003S	102	D23-C7	FRSP01	48	E6-E7	RPIW1	101	D31-D1	11BAT1	47			100		
51			104	D19-F1	A3PPR13	50	D30-B6	REVTUR1	103	D23-C8	FRSP02	49	E6-E8	RLVS1	102			48			101		
52			51	D30-F6	15RTN1	51	D30-F6	15RTN1	104			50	D25-D3	DSCB1	103	D31-B5	SBTL1	49			102		
53	D19-F4	LINKS13	52	D24-D1	TRKBS1	52	D24-D1	TRKBS1				51	D25-D4	DSCR1	104			50			103	D34-E2	26BAT1
54			53	D35-D2	HSASOL1	53	D35-D2	HSASOL1				52	D35-D1	EES1				51					
55			54	D24-D1	TRKBB1	54	D24-D1	TRKBB1				53	D25-D4	DSCW1				52	E1-B7	N15VSEN			
56			55	D24-D1	TRKBW1	55	D24-D1	TRKBW1				54						53	D30-C4	25BAT1			
57			56	D5-F0	SS1T1G1	56	D5-F0	SS1T1G1				55	D36-B3	EWUREQ1				54	E1-B5	19RTNW			
58			57	D5-F1	SS1TS1	57	D5-F1	SS1TS1				56	D34-C1	CRUISE1				55	E1-B6	22RTNW			
59			58	D5-F1	SS1TS5G1	58	D5-F1	SS1TS5G1				57	D34-C4	WSSDC1				56	E2-B2	51RTN1			
60			59	D5-F1	SS1T6W1	59	D5-F1	SS1T6W1				58	D33-C2	MPRESW1				57	E1-B0	6RTNW			
61			60	D24-D3	TRKBS2	60	D24-D3	TRKBS2				59						58	E1-B3	9RTNW			
62			61			61						60						59	D15-E3	C1F24B1			
63			62			62						61						60	D18-E3	LG24B1			
64			63			63						62						61	D20-F4	1POTRFS1			
65			64	D11-F7	TMBS03	64	D11-F7	TMBS03				63						62	E2-B2	52RTN1			
66			65	D11-F7	TMBS04	65	D11-F7	TMBS04				64						63	E2-B3	54RTN1			
67			66	D30-B5	FORTRU1	66	D30-B5	FORTRU1				65						64	D30-C1	6DSCJP1			
68			67	D30-F5	14RTN1	67	D30-F5	14RTN1				66	D25-D4	DSCS1				65	D30-C4	3DSCJP1			
69			68	D5-F0	SS1T2W1	68	D5-F0	SS1T2W1				67						66					
70			69	D5-F0	SS1T3G1	69	D5-F0	SS1T3G1				68						67					
71			70	D5-F0	SS1T4W1	70	D5-F0	SS1T4W1				69						68	E2-B7	58RTN1			
72			71	D24-D3	TRKBB2	71	D24-D3	TRKBB2				70						69	E2-B1	48RTN1			

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CONT. ON SH.95 SH. NO. 94

PANEL-PLUG PIN LOCATION

SECT. S4

PANEL-PLUG PIN LOCATION

THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED, NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.

GDP-2

TERM	LOCATION	WIRE NAME
A	D13-E1	P21CNB
B	D13-E1	P21CNW
C	D13-F1	P22CNB
D	D13-F1	P22CNW
E	D13-C1	CM2PB
F	D13-C1	CM2PW
G		
H		
J		
K		
L		
M		

GDP-3

TERM	LOCATION	WIRE NAME
1	D12-C0	27BAT1
2	D12-C0	28BAT1
3	D12-C0	29BAT1
4		
5	D12-A0	27RTN1
6	D12-B0	28RTN1
7	D12-B0	29RTN1
8		
9		

LT1P

TERM	LOCATION	WIRE NAME
A	K-D4	1LTS1
B	K-E4	71RTN1

LT2P

TERM	LOCATION	WIRE NAME
A	K-D5	2LTS14
B	K-E5	72RTN14

LT3P

TERM	LOCATION	WIRE NAME
A	K-D6	3LTS14
B	K-E6	73RTN14

LT4P

TERM	LOCATION	WIRE NAME
A	K-D7	4LTS15
B	K-E7	74RTN15

PS2S

TERM	LOCATION	WIRE NAME
1	E3-E0	14BAT1
2	E3-E7	22BAT1
3		
4		
5	E3-F7	23BAT1
6	E3-B7	2BAT13

PS2S

TERM	LOCATION	WIRE NAME
7	E3-C4	25BAT1
8	E3-E2	26BAT1
9	E3-E2	15BAT1

VAM1S

TERM	LOCATION	WIRE NAME
1	B6-E1	LK1B13
2	B6-E1	LK1W13
3		
4	B10-E3	P1R13
5		
6		
7		
8	B10-B5	35RTN3
9	B10-B5	34RTN13

VAM2S

TERM	LOCATION	WIRE NAME
1	B8-E1	LK2B13
2	B8-E1	LK2W13
3	B11-E3	P2B13
4	B11-E3	P2R13
5	B11-E3	P2W13
6		
7		
8	B11-B5	35RTN3
9		

VAM3S

TERM	LOCATION	WIRE NAME
1	B1-F1	A3PPB13
2	B1-F1	A3PPR13
3	B1-F2	A3PPW13
4	B1-B7	LINKB13
5	B1-F3	LINKW13
6		
7		
8	B1-C7	37RTN3
9	B1-F2	A3PPS13

VAM4S

TERM	LOCATION	WIRE NAME
1	B-F5	TERTB13
2	B-F6	TERTW13
3	B-F7	FLDB13
4	B-F7	FLDW13
5	B4-F8	GRDBL13
6		
7		
8	B4-C7	37RTN3
9	B4-D3	36RTN13

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CONT. ON SH.96 SH. NO.95

RELAY AND CONTACTOR LOCATION

SECT. T

RELAY AND CONTACTOR LOCATION

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NOMENCLATURE	DESCRIPTION	PART NUMBER	COMPARTMENT LOCATION	SCHEMATIC REFERENCE					
				COIL PINS	POLE 1	POLE 2	POLE 3	POLE 4	POLE 5
GF	ALTERNATOR CONTACTOR	41C663321G3	CA3	D29-B6	B-E6	X	X	X	X
GFR	GENERATOR FIELD RELAY	17LV66J46	CA3	D29-B7	E-C4	E-D4	E-D4	X	X

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TITLE: NHL 240 TAC QC II SCHEMATIC
FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE

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MADE BY: MATHI.V
ISSUED BY: MATHI.V

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DATE: 06APR2022

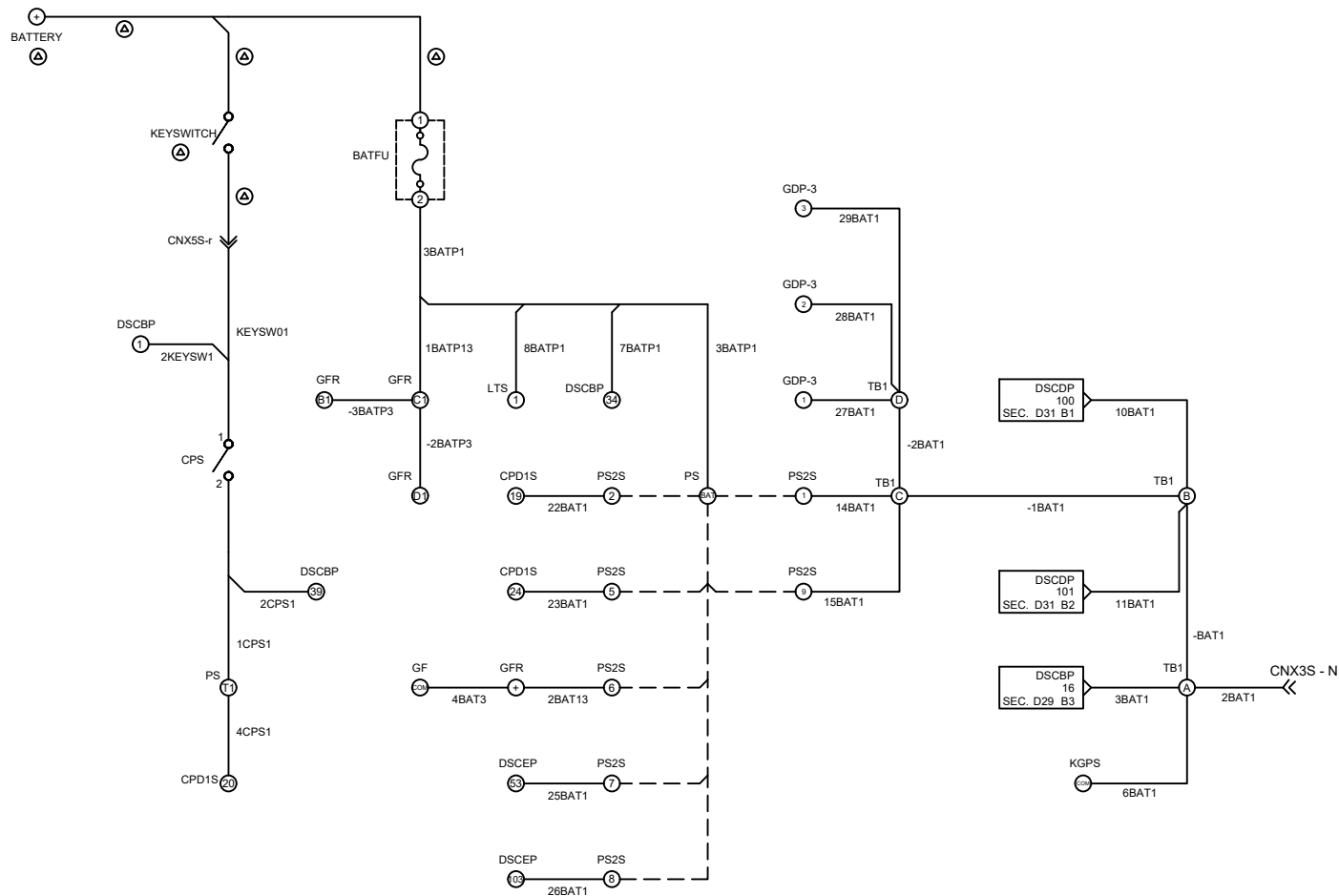
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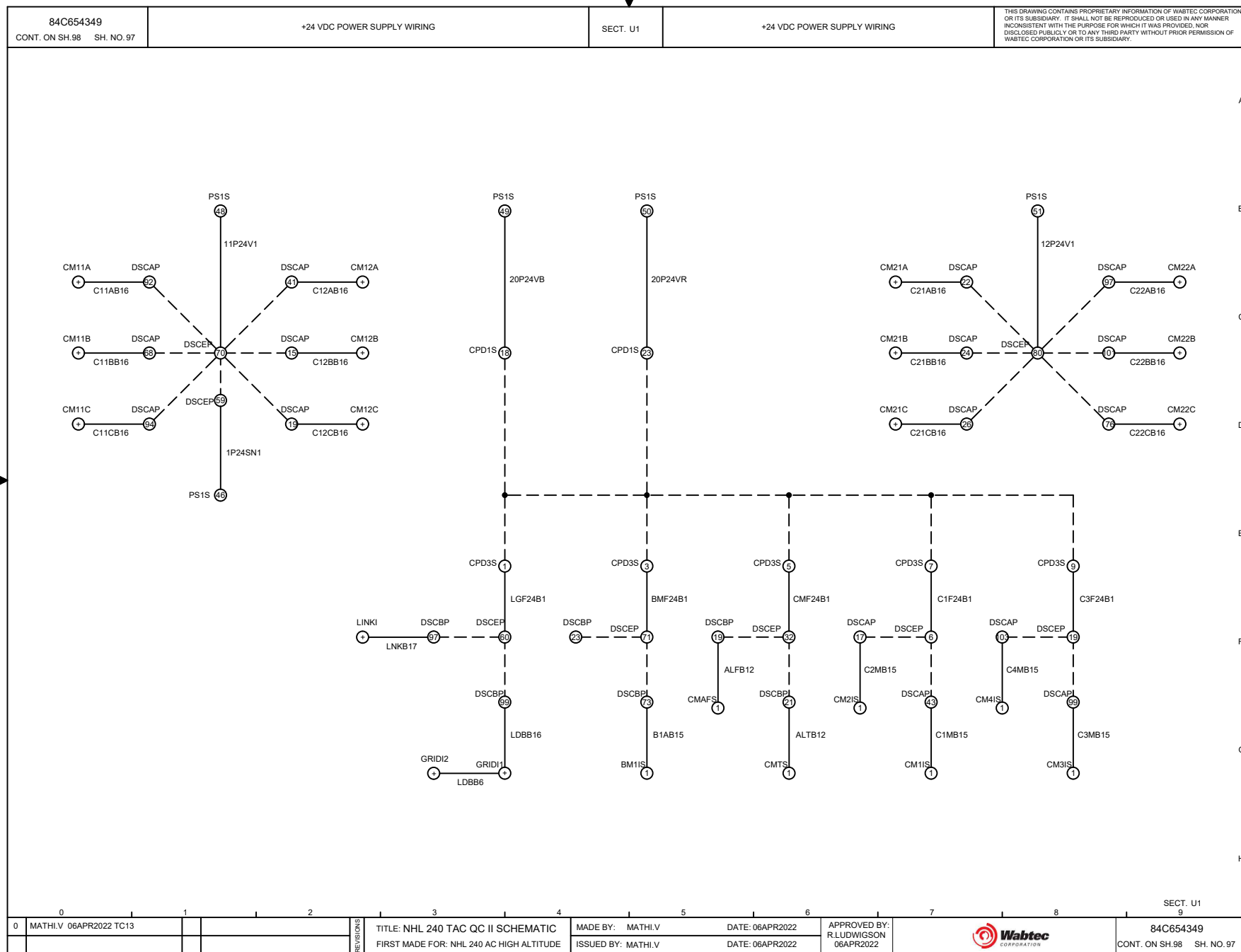
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APPROVED BY:
R.LUDWIGSON
06APR2022

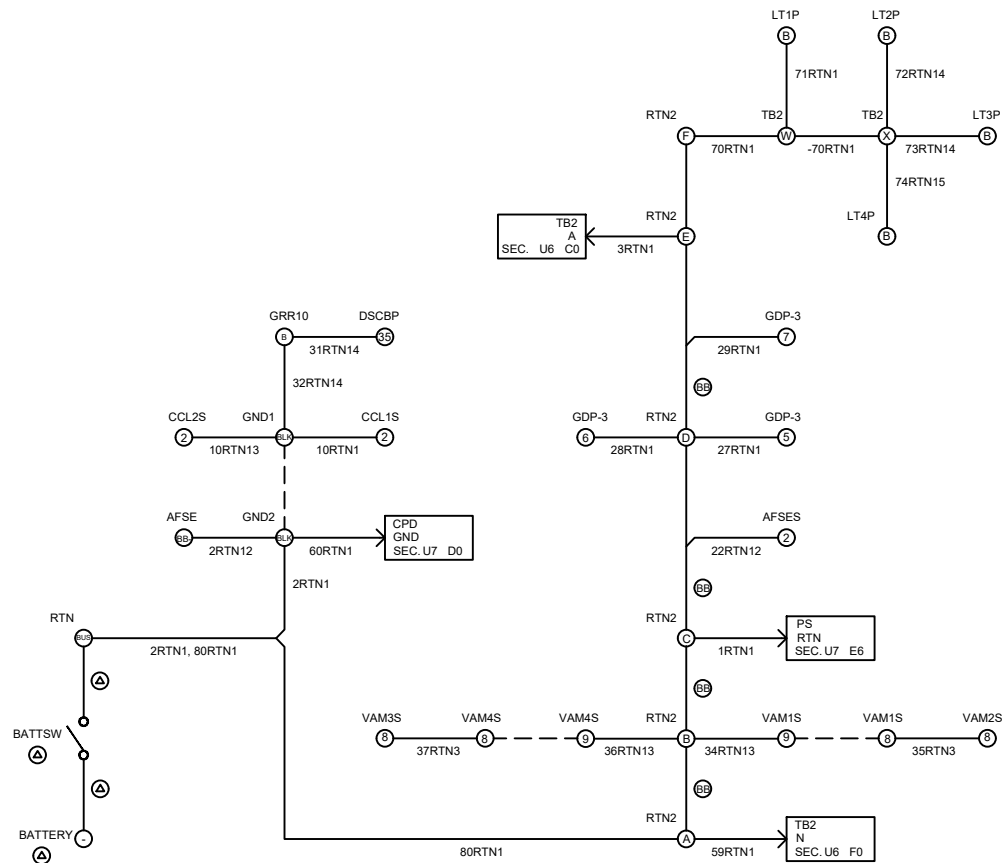
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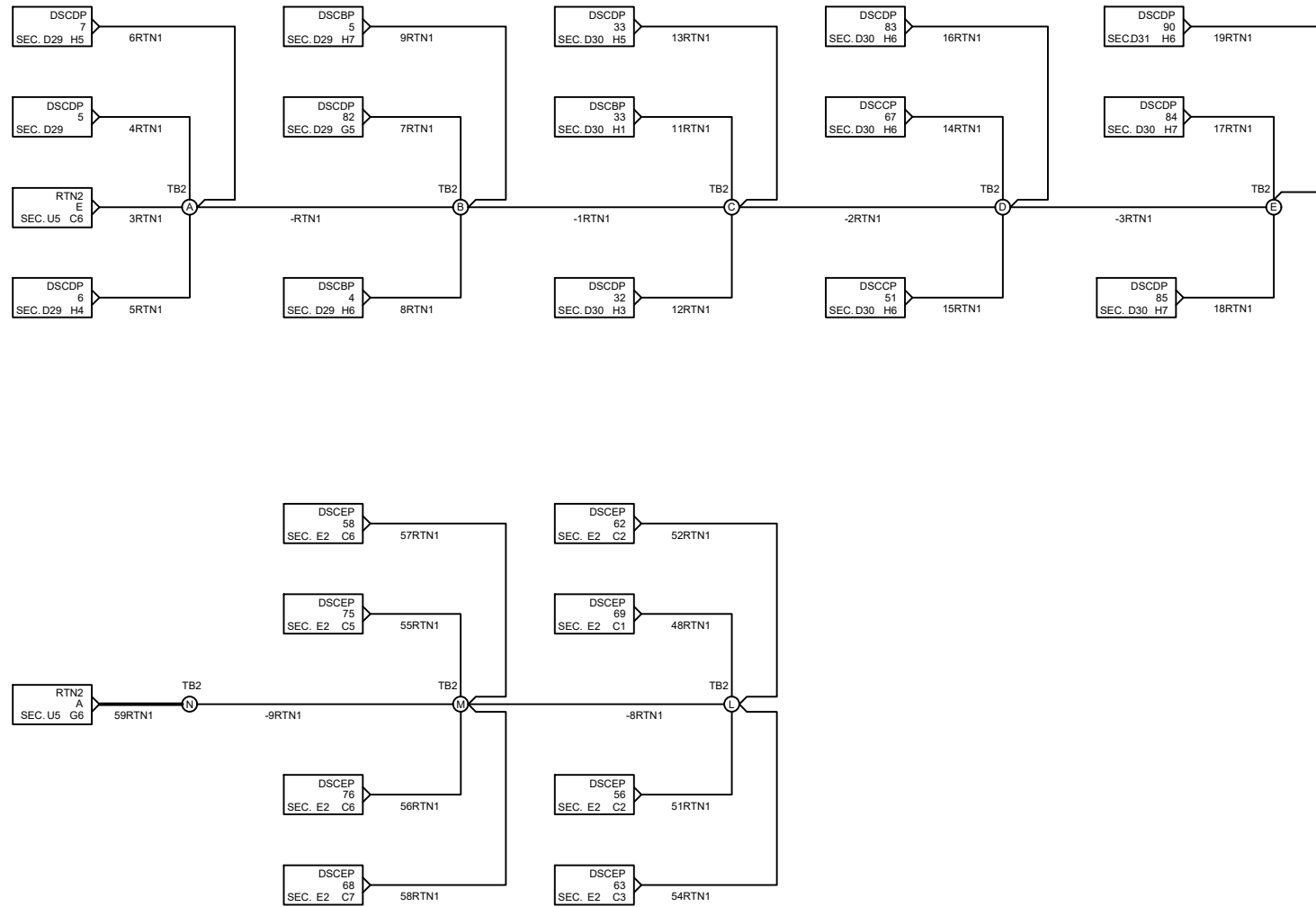

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CONT. ON SH.96 SH. NO.95

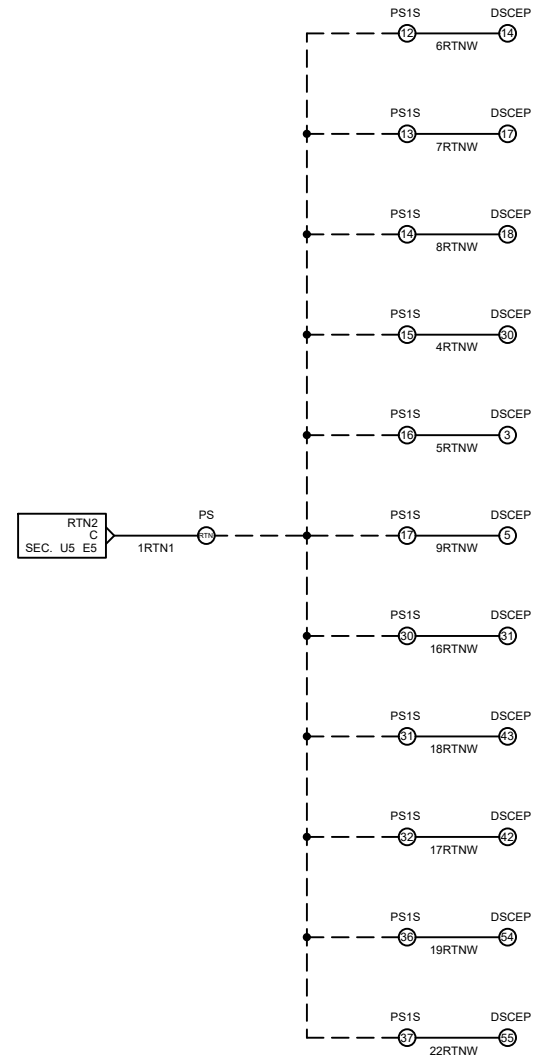
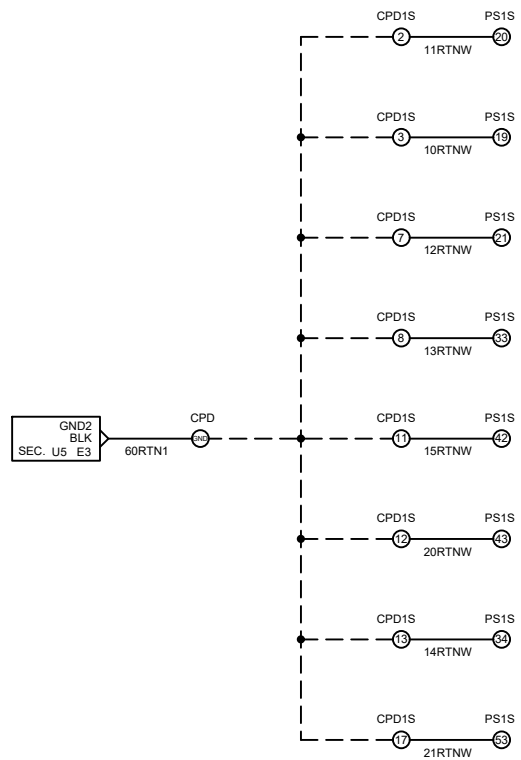
ACES











84C654349		DEVICE TABLE			SECT. V	DEVICE TABLE			THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.	
CONT. ON SH.105 SH. NO.104										

NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC		NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC	
11A+S	INVERTER #11 PHASE A+ MODULE CONNECTOR	41A271471P9	CA4	D12-E7,E8,F7,F8		BAROS	BAROMETRIC PRESSURE SENSOR CONNECTOR	41A296326DA8	CA1	D21-C6,D6	
11A-S	INVERTER #11 PHASE A- MODULE CONNECTOR	41A271471P9	CA4	D12-E6,E7,F6,F7		BATFU	SYSTEM FUSE	41A296321BEP1	CA1	E-D1,D2	
11B+S	INVERTER #11 PHASE B+ MODULE CONNECTOR	41A271471P9	CA4	D12-E4,E5,F4,F5		BATTSW	BATTERY CUTOOUT SWITCH	OEM SUPPLIED	TRK	E-E1	
11B-S	INVERTER #11 PHASE B- MODULE CONNECTOR	41A271471P9	CA4	D12-E3,E4,F3,F4		BM1	GRID BLOWER MOTOR #1	5GY19L8	RGB	B4-C5	
11C+S	INVERTER #11 PHASE C+ MODULE CONNECTOR	41A271471P9	CA4	D12-E1,E2,F1,F2		BM1I	GRID BLOWER MOTOR #1 CURRENT SENSOR	41A296328AKP10	CA5	B4-D2	
11C-S	INVERTER #11 PHASE C- MODULE CONNECTOR	41A271471P9	CA4	D12-E0,E1,F0,F1		BM1IS	GRID BLOWER #1 CURRENT SENSOR CONNECTOR	41A271471P9	CA5	B4-E2	
12A+S	INVERTER #12 PHASE A+ MODULE CONNECTOR	41A271471P9	CA4	D12-F7,F8,G7,G8		BM2	GRID BLOWER MOTOR #2	5GY19L8	RGB	B4-C6	
12A-S	INVERTER #12 PHASE A- MODULE CONNECTOR	41A271471P9	CA4	D12-F6,F7,G6,G7		BM2I	GRID BLOWER MOTOR #2 CURRENT SENSOR	41A296328AKP10	CA5	B4-E2	
12B+S	INVERTER #12 PHASE B+ MODULE CONNECTOR	41A271471P9	CA4	D12-F4,F5,G4,G5		BM2IS	GRID BLOWER #2 CURRENT SENSOR CONNECTOR	41A271471P9	CA5	B4-F2,F3	
12B-S	INVERTER #12 PHASE B- MODULE CONNECTOR	41A271471P9	CA4	D12-F3,F4,G3,G4		BMFF	GRID BLOWER MOTOR FIELD BUS BAR	84B535634ADP1	CA5	B4-D1,F1	
12C+S	INVERTER #12 PHASE C+ MODULE CONNECTOR	41A271471P9	CA4	D12-F1,F2,G1,G2		BMFF3	GRID BLOWER MOTOR #1 CABLE CONNECTION BUS BAR	84B519786P1	CA5	B4-D3	
12C-S	INVERTER #12 PHASE C- MODULE CONNECTOR	41A271471P9	CA4	D12-F0,F1,G0,G1		C11AS	INVERTER #11 PHASE A CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B10-A2,B2	
21A+S	INVERTER #21 PHASE A+ MODULE CONNECTOR	41A271471P9	CA5	D13-E8,E9		C11BS	INVERTER #11 PHASE B CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B10-C2,D1	
21A-S	INVERTER #21 PHASE A- MODULE CONNECTOR	41A271471P9	CA5	D13-E7,E8,E9		C11CS	INVERTER #11 PHASE C CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B10-F1	
21B+S	INVERTER #21 PHASE B+ MODULE CONNECTOR	41A271471P9	CA5	D13-E6,E7		C12AS	INVERTER #12 PHASE A CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B10-B1,B2,C1	
21B-S	INVERTER #21 PHASE B- MODULE CONNECTOR	41A271471P9	CA5	D13-E4,E5		C12BS	INVERTER #12 PHASE B CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B10-E1	
21C+S	INVERTER #21 PHASE C+ MODULE CONNECTOR	41A271471P9	CA5	D13-E3,E4		C12CS	INVERTER #12 PHASE C CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B10-G1,H1	
21C-S	INVERTER #21 PHASE C- MODULE CONNECTOR	41A271471P9	CA5	D13-E2,E3		C21AS	INVERTER #21 PHASE A CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B11-A1,B1	
22A+S	INVERTER #22 PHASE A+ MODULE CONNECTOR	41A271471P9	CA4	D13-F9		C21BS	INVERTER #21 PHASE B CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B11-C1,D1	
22A-S	INVERTER #22 PHASE A- MODULE CONNECTOR	41A271471P9	CA4	D13-F7,F8		C21CS	INVERTER #21 PHASE C CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B11-F1	
22B+S	INVERTER #22 PHASE B+ MODULE CONNECTOR	41A271471P9	CA4	D13-F6,F7		C22AS	INVERTER #22 PHASE A CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B11-B1,C1	
22B-S	INVERTER #22 PHASE B- MODULE CONNECTOR	41A271471P9	CA4	D13-F4,F5		C22BS	INVERTER #22 PHASE B CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B11-E0,E1	
22C+S	INVERTER #22 PHASE C+ MODULE CONNECTOR	41A271471P9	CA4	D13-F3,F4		C22CS	INVERTER #22 PHASE C CURRENT SENSOR CONNECTOR	41A271471P9	CA6	B11-G1,H1	
22C-S	INVERTER #22 PHASE C- MODULE CONNECTOR	41A271471P9	CA4	D13-F2,F3		CAN1P	CAN #1 CONTROL GROUP CONNECTOR	P/O 84A238510P13	CA1	D26-C0	
AFSE	ALTERNATOR FIELD EXCITOR PANEL	17FM689A2	CA2	B-B0,B3,C0,C3,D3,E3		CAN2P	CAN #2 CONTROL GROUP CONNECTOR	P/O 84A238510P14	CA1	D26-C2,C3	
AFSES	ALTERNATOR FIELD EXCITOR PANEL CONNECTOR	41A271471P22	CA2	B-F0,F1,F2,F3		CAN3P	CAN #3 CONTROL GROUP CONNECTOR	P/O 84A238510P15	CA1	D26-C5,C6	
ALT	ALTERNATOR	5GTA41K	TRK	B-D7		CCF1	CHOPPER LINE FILTER CAPACITOR #1	41A296302CDP6	CA5	B3-D7	
AMBTS	AMBIENT TEMPERATURE SENSOR	41C668507G1	CA7	D21-B3,C3,D3		CCF2	CHOPPER LINE FILTER CAPACITOR #2	41A296302CDP6	CA5	B3-D8	
AN1S	CAN #1 PANEL CONNECTOR	P/O 84A238510P13	CA1	D26-D0,D1		CCL1	CAPACITOR CHARGE LIGHT #1	41C666570G1	CA1	D14-B5	
AN2S	CAN #2 PANEL CONNECTOR	P/O 84A238510P14	CA1	D26-D2,D3,D4							
AN3S	CAN #3 PANEL CONNECTOR	P/O 84A238510P15	CA1	D26-D5,D6,D7							
AXLP	AXLE BOX PRESSURE SWITCH	41B512742P3	TRK	D33-B2							
BARO	BAROMETRIC PRESSURE SENSOR	41A317303ACP38	CA1	D21-B6							

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0		MATH.V 06APR2022 TC13						TITLE: NHL 240 TAC QC II SCHEMATIC		MADE BY: MATH.V		DATE: 06APR2022		APPROVED BY: R.LUDWIGSON		06APR2022		84C654349	
								FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE		ISSUED BY: MATH.V		DATE: 06APR2022						CONT. ON SH.105 SH. NO. 104	



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84C654349		DEVICE TABLE			SECT. V1	DEVICE TABLE			THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED, NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.		
CONT. ON SH.106 SH. NO. 105											
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84C654349		DEVICE TABLE			SECT. V2	DEVICE TABLE			THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED. NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.		
CONT. ON SH.107 SH. NO. 106											
NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC	NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC		
CVBD	CHOPPER MODULE #2 CABLE CONNECTION BUS BAR	84B535634ANP1	CA5	B2-D7	F1	ALTERNATOR FIELD INSULATED CONNECTION POINT F1	41A26012BP1	CA3	B-B5	A	
DCN	NEGATIVE DC LINK HORIZONTAL BUS BAR	84C604237P1	CA4	B1-G4,G5,G7,G8,G9, B2-G5, B6-H0,H7, B7-H6, B8-H0,H6, B9-H6	F11A1	INVERTER #11 PHASE A FUSE #1	41A296321BSP2	CA4	B6-B3		
DCP	POSITIVE DC LINK HORIZONTAL BUS BAR	84C604236P1	CA4	B1-A4,A5,A7,A8,A9, B2-B1, B4-C0, B6-A0,A8, B7-A7, B8-A0,A8, B9-A8	F11A2	INVERTER #11 PHASE A FUSE #2	41A296321BSP2	CA4	B6-B3		
DID	DIAGNOSTIC INFORMATION DISPLAY	17FM693B5	TRK	E6-B6	F11B1	INVERTER #11 PHASE B FUSE #1	41A296321BSP2	CA4	B6-B5	B	
DID-S	DIAGNOSTIC INFORMATION DISPLAY CONNECTOR	41A296326AKP2	TRK	D25-D2, E6-A6	F11B2	INVERTER #11 PHASE B FUSE #2	41A296321BSP2	CA4	B6-B6		
DSC	DRIVE SYSTEM CONTROL PANEL	17FL454B1	CA1	D10-E0, D11-F0, D15-E0,E5, D16-E0, D17-E0, D18-E0, D19-F4, D1-B0, D20-F5, D21-E0, D22-F0, D23-F0, D24-F0, D25-F0, D26-D0, D27-D1, D28-E0, D29-D0, D2-C0, D30-C0, D31-D1, D32-D1, D33-D0, D34-D0, D35-E0, D36-D0, D3-E0, D4-E0, D5-F0, D6-C0, D7-C0, D8-C0, D9-E1, D-C0, E1-A0, E2-A1, E6-G6, P11-C1	F11C1	INVERTER #11 PHASE C FUSE #1	41A296321BSP2	CA4	B6-B8		
DSC1: FODC1	INVERTER #1 FIBER OPTIC CARD SLOT 1	84C603188ADG1	CA1	P11-A5	F11C2	INVERTER #11 PHASE C FUSE #2	41A296321BSP2	CA4	B6-B8	C	
DSC2: FODC2	INVERTER #2 FIBER OPTIC CARD SLOT 2	84C603188AEG1	CA1	P11-A5	F12A1	INVERTER #12 PHASE A FUSE #1	41A296321BSP2	CA4	B7-B2		
DSC3: TMC	INVERTER #1 TMC CARD SLOT 3	17FB190D1F	CA1	P11-A5	F12A2	INVERTER #12 PHASE A FUSE #2	41A296321BSP2	CA4	B7-B2		
DSC4: TMC	INVERTER #2 TMC CARD SLOT 4	17FB190D1F	CA1	P11-A5	F12B1	INVERTER #12 PHASE B FUSE #1	41A296321BSP2	CA4	B7-B4	D	
DSC5: ANALG	DSC ANALOG CARD SLOT 5	17FB173C6F	CA1	P11-B5	F12B2	INVERTER #12 PHASE B FUSE #2	41A296321BSP2	CA4	B7-B5		
DSC6: I/O	DSC DIGITAL I/O CARD SLOT 6	17FB104B2F	CA1	P11-B5	F12C1	INVERTER #12 PHASE C FUSE #1	41A296321BSP2	CA4	B7-B7		
DSC7: I/O	DSC DIGITAL I/O CARD SLOT 7	17FB104B2F	CA1	P11-B5	F12C2	INVERTER #12 PHASE C FUSE #2	41A296321BSP2	CA4	B7-B7	E	
DSC8: ETH	DSC ETHERNET CARD SLOT 8	17FB191B1F	CA1	P11-B5	F2	ALTERNATOR FIELD INSULATED CONNECTION POINT F2	41A26012BP1	CA3	B-E7		
DSC9: CPU	DSC CPU CARD SLOT 9	17FB197A3F	CA1	P11-B5	F21A1	INVERTER #21 PHASE A FUSE #1	41A296321BSP2	CA5	B8-B2		
DSCAP	DSC PANEL CONNECTOR A	41A267080P8	CA1	D10-E1,E3,E4, D15-E0,E1,E2,E3,E5,E6,E7,E8, D2-C1,C2,C3,C6,C7,C8, D3-E0,E2,E3,E4,E5,E6,E7,E8, D4-E1,E2,E3,E4, D8-C1,C2,C3,C6,C7,C8, D9-E1,E3,E4,E5,E6,E8,E9	F21A2	INVERTER #21 PHASE A FUSE #2	41A296321BSP2	CA5	B8-B2	F	
DSCBP	DSC PANEL CONNECTOR B	41A267080P8	CA1	D16-E4,E5,E7,E8, D17-E4,E5,E7,E8, D18-E4,E5,E7,E8, D19-F1,F2,F3,F4,F5,F6,F7,F8, D21-E0,E1,E2,E3,E6,E7,E8, D22-F1,F2,F3, D29-D3,D6,D7,F3,F6,F7, D30-C1,F1, D33-D0,D1,D5	F21B1	INVERTER #21 PHASE B FUSE #1	41A296321BSP2	CA5	B8-B4		
DSCCP	DSC PANEL CONNECTOR C	41A267080P8	CA1	D11-B3,C3,F0,F1,F2,F7, D1-B7, D21-E4,E5, D22-F5,F6,F7,F8, D23-E0,E2,F0,F1,F2,F3,F4,F5,F6,F7,F8, D24-E0,F0,F1,F3,F4,F5,F6,F7, D30-C5,C6,F5,F6, D35-E2, D5-B2,C2,F0,F1,F6,F7	F21B2	INVERTER #21 PHASE B FUSE #2	41A296321BSP2	CA5	B8-B5		
DSCDP	DSC PANEL CONNECTOR D	41A267080P8	CA1	D25-F1,F2,F3,F4, D29-D2,D4,D5,F2,F4,F5, D30-C2,C5,C8,C9,F2,F5,F7,F8,F9, D31-D1,D5,F1,F5, D33-D2,D3,D4,D6,D7, D34-D0,D1,D2,D4,D8, D35-E1,E5,E6, D36-D1,D3, E6-G7,G8,G9	F21C1	INVERTER #21 PHASE C FUSE #1	41A296321BSP2	CA5	B8-B7	G	
DSCEP	DSC PANEL CONNECTOR E	41A267080P8	CA1	D15-D4,D9,E3,E4,E8,E9, D16-D2,E2,E3, D17-D2,E2,E3, D18-D2,E2,E3, D20-G6, D29-D0,D1,F1, D2-C0,C5, D30-C1,C3,C4, D33-D0, D34-D4,D7, D8-C0,C5, E1-B0,B1,B2,B3,B4,B5,B6,B7,E0,E1,E2,E3,E4,E5,E6,E7, E2-B1,B2,B3,B5,B6,B7	F21C2	INVERTER #21 PHASE C FUSE #2	41A296321BSP2	CA5	B8-B7		
EN2S	ETHERNET #2 PANEL CONNECTOR	P/O 84A238510P10	CA1	D27-D3,D4,D5	F22A1	INVERTER #22 PHASE A FUSE #1	41A296321BSP2	CA5	B9-B2		
EN4S	ETHERNET #4 PANEL CONNECTOR	P/O 84A238510P12	CA1	D28-B9,E4,E5,E6	F22A2	INVERTER #22 PHASE A FUSE #2	41A296321BSP2	CA5	B9-B2	H	
ESS	ENGINE SPEED SENSOR	41A262893P3	TRK	D22-B5	F22B1	INVERTER #22 PHASE B FUSE #1	41A296321BSP2	CA5	B9-B4		
ESS-P	ENGINE SPEED SENSOR CONNECTOR	41A296326DTP2	TRK	D22-C5	F22B2	INVERTER #22 PHASE B FUSE #2	41A296321BSP2	CA5	B9-B5		
ETC2S	ETHERNET #2 CONTROL GROUP CONNECTOR	P/O 84A238510P10	CA1	D27-C3,C4,C5	F22C1	INVERTER #22 PHASE C FUSE #1	41A296321BSP2	CA5	B9-B7		
ETC4S	ETHERNET #4 CONTROL GROUP CONNECTOR	P/O 84A238510P12	CA1	D28-A9,C5,C6,D4	F22C2	INVERTER #22 PHASE C FUSE #2	41A296321BSP2	CA5	B9-B7		
					FB11F	INVERTER #11 FILTER BUS BAR	41A296332BLP21	CA4	B6-C0,F0		
					FB12F	INVERTER #12 FILTER BUS BAR	41A296332BLP21	CA4	B7-C0,F0		
					FB21F	INVERTER #21 FILTER BUS BAR	41A296332BLP21	CA5	B8-C0,F0		
					FB22F	INVERTER #22 FILTER BUS BAR	41A296332BLP21	CA5	B9-C0,F0		
					FBCF	CHOPPER FILTER BUS BAR	41A296332BLP21	CA5	B3-C7,C8,E7,E8		
					FDR	FILTER DISCHARGE RESISTOR PANEL	17FR402A1	CA7	B1-B4,C4,D4,E4		
					FP	THREE PHASE FILTER PANEL	41C662412G2	CA7	B1-D0,E0		
SECT. V2											
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84C654349		DEVICE TABLE			SECT. V3	DEVICE TABLE			THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED, NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.	
CONT. ON SH.108 SH. NO. 107										
NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC	NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC	
GBMC	GRID BLOWER MOTOR CAPACITOR	41A296302CDP8	CA5	B4-C0,D0	KEYSW	KEY SWITCH	OEM SUPPLIED	TRK	E-A1	
GBMR	GRID BLOWER MOTOR CAPACITOR DISCHARGE RESISTOR	482A444APP44	CA4	B4-D0	KGPS	CONTROL GROUP PRESSURE SWITCH	41B512742P3	CA6	D33-C1	
GDP	GATE DRIVE POWER CONVERTER	41A296352CYP5	CA1	D12-A0, D13-A0	LINKI	DC LINK CURRENT SENSOR	41A296328AKP7	CA7	B1-G3	
GDP-1	GATE DRIVE POWER CONVERTER CONNECTOR #1	41A216641AMP84	CA1	D12-D0,E0,F0,G0	LT1	CONTROL AREA LIGHT #1	84A226679P1	CA1	K-E4	
GDP-2	GATE DRIVE POWER CONVERTER CONNECTOR #2	41A216641AMP84	CA1	D13-C1,E1,F1	LT1P	CONTROL AREA LIGHT #1 CONNECTOR	41A296326DAP5	CA1	K-D4,E4	
GDP-3	GATE DRIVE POWER CONVERTER CONNECTOR #3	41A271474P17	CA1	D12-A0,B0,C0	LT2	POWER AREA LIGHT #2	84A226679P1	CA4	K-E5	
GF	ALTERNATOR CONTACTOR	41C663321G3	CA3	B-E5,E6, D29-B6,B8,C6,C8	LT2P	POWER AREA LIGHT #2 CONNECTOR	41A296326DAP5	CA4	K-D5,E5	
GFCO	GENERATOR FIELD CUTOFF SWITCH	41A267741P2	CA1	D29-A1,A2	LT3	POWER AREA LIGHT #3	84A226679P1	CA5	K-E6	
GFR	GENERATOR FIELD RELAY	17LV66,J46	CA3	D29-B7,C7, E-C4,D4	LT3P	POWER AREA LIGHT #3 CONNECTOR	41A296326DAP5	CA5	K-D6,E6	
GFRS	GENERATOR FIELD RELAY SUPPRESSION MODULE	41B566971G1	CA3	D29-B7	LT4	POWER AREA LIGHT #4	84A226679P1	CA5	K-E7	
GFS	ALTERNATOR FIELD CONTACTOR SUPPRESSION MODULE	41B566971G1	CA3	D29-B6	LT4P	POWER AREA LIGHT #4 CONNECTOR	41A296326DAP5	CA5	K-D7,E7	
GND1	GROUND BLOCK #1	41B565073G1	CA1	E-G2,G3	LTS	AREA LIGHTS SWITCH	41A254891P1	CA1	K-C3	
GND2	GROUND BLOCK #2	41B565073G1	CA1	E-E4	P11A-	INVERTER #11 PHASE A- MODULE	84A241532A1	CA4	B6-E2,F2, D12-E6	
GRD1	GRID CURRENT SENSOR #1	41A296328AKP7	CA6	B4-B2,B3	P11A+	INVERTER #11 PHASE A+ MODULE	84A241531A1	CA4	B6-B2,C2, D12-E7	
GRD12	GRID CURRENT SENSOR #2	41A296328AKP7	CA6	B4-B1	P11B-	INVERTER #11 PHASE B- MODULE	84A241532A1	CA4	B6-E4,F5, D12-E3	
GRR	GROUND RESISTOR PANEL	17FR402A1	CA7	B1-B5,C5,D5,E5	P11B+	INVERTER #11 PHASE B+ MODULE	84A241531A1	CA4	B6-B5,C5, D12-E5	
GRR10	GROUND RESISTOR	41A281404P2	CA4	B1-E6,F6	P11C-	INVERTER #11 PHASE C- MODULE	84A241532A1	CA4	B6-E7,F7, D12-E1	
GRR9	GROUND RESISTOR	482A444APP14	CA4	B1-D6	P11C+	INVERTER #11 PHASE C+ MODULE	84A241531A1	CA4	B6-B7,C7, D12-E2	
I11NB	INVERTER #11 VERTICAL TO DC HORIZONTAL NEGATIVE JUMPER BUS BAR	84B520977ABP1	CA4	B6-G7	P12A-	INVERTER #12 PHASE A- MODULE	84A241532A1	CA4	B7-E1,F1, D12-F6	
I11PB	INVERTER #11 VERTICAL TO DC HORIZONTAL POSITIVE JUMPER BUS BAR	84B520976P1	CA4	B6-A8	P12A+	INVERTER #12 PHASE A+ MODULE	84A241531A1	CA4	B7-B1,C1, D12-F7	
I11VB	INVERTER #11 VERTICAL BUS BAR	84A201786ABP27	CA4	B6-A0,A8,D3,D5,D8,G0,G7	P12B-	INVERTER #12 PHASE B- MODULE	84A241532A1	CA4	B7-E4,F4, D12-F3	
I12NB	INVERTER #12 VERTICAL TO DC HORIZONTAL NEGATIVE JUMPER BUS BAR	84B520977ABP1	CA4	B7-G6	P12B+	INVERTER #12 PHASE B+ MODULE	84A241531A1	CA4	B7-B4,C4, D12-F5	
I12PB	INVERTER #12 VERTICAL TO DC HORIZONTAL POSITIVE JUMPER BUS BAR	84B520976P1	CA4	B7-A7	P12C-	INVERTER #12 PHASE C- MODULE	84A241532A1	CA4	B7-E6,F6, D12-F1	
I12VB	INVERTER #12 VERTICAL BUS BAR	84A201786ABP27	CA4	B7-A0,A7,D2,D4,D7,G0,G6	P12C+	INVERTER #12 PHASE C+ MODULE	84A241531A1	CA4	B7-B6,C6, D12-F2	
I21NB	INVERTER #21 VERTICAL TO DC HORIZONTAL NEGATIVE JUMPER BUS BAR	84B520977ABP1	CA5	B8-G6	P21A-	INVERTER #21 PHASE A- MODULE	84A241532A1	CA5	B8-E1,F1, D13-D8	
I21PB	INVERTER #21 VERTICAL TO DC HORIZONTAL POSITIVE JUMPER BUS BAR	84B520976P1	CA5	B8-A7	P21A+	INVERTER #21 PHASE A+ MODULE	84A241531A1	CA5	B8-B1,C1, D13-D9	
I21VB	INVERTER #21 VERTICAL BUS BAR	84A201786ABP27	CA5	B8-A0,A8,D2,D4,D5,D7,G0,G6	P21B-	INVERTER #21 PHASE B- MODULE	84A241532A1	CA5	B8-E4,F4, D13-D5	
I22NB	INVERTER #22 VERTICAL TO DC HORIZONTAL NEGATIVE JUMPER BUS BAR	84B520977ABP1	CA5	B9-G6	P21B+	INVERTER #21 PHASE B+ MODULE	84A241531A1	CA5	B8-B4,C4, D13-D6	
I22PB	INVERTER #22 VERTICAL TO DC HORIZONTAL POSITIVE JUMPER BUS BAR	84B520976P1	CA5	B9-A7	P21C-	INVERTER #21 PHASE C- MODULE	84A241532A1	CA5	B8-E6,F6, D13-D2	
I22VB	INVERTER #22 VERTICAL BUS BAR	84A201786ABP27	CA5	B9-A0,A8,D2,D4,D5,D7,G0,G6	P21C+	INVERTER #21 PHASE C+ MODULE	84A241531A1	CA5	B8-B6,C6, D13-D4	
					P22A-	INVERTER #22 PHASE A- MODULE	84A241532A1	CA4	B9-E1,F1, D13-F8	
					P22A+	INVERTER #22 PHASE A+ MODULE	84A241531A1	CA5	B9-B1,C1, D13-F9	

84C654349		DEVICE TABLE		SECT. V4		DEVICE TABLE		THIS DRAWING CONTAINS PROPRIETARY INFORMATION OF WABTEC CORPORATION OR ITS SUBSIDIARY. IT SHALL NOT BE REPRODUCED OR USED IN ANY MANNER INCONSISTENT WITH THE PURPOSE FOR WHICH IT WAS PROVIDED, NOR DISCLOSED PUBLICLY OR TO ANY THIRD PARTY WITHOUT PRIOR PERMISSION OF WABTEC CORPORATION OR ITS SUBSIDIARY.							
CONT. ON SH.109 SH. NO. 108															
NOMENCL					DESCRIPTION					PART NUMBER	LOC	SCHEMATIC REF LOC			
P22B-					INVERTER #22 PHASE B- MODULE					84A241532A1	CA4	B9-E4,F4, D13-F5			
P22B+					INVERTER #22 PHASE B+ MODULE					84A241531A1	CA5	B9-B4,C4, D13-F6			
P22C-					INVERTER #22 PHASE C- MODULE					84A241532A1	CA4	B9-E6,F6, D13-F2			
P22C+					INVERTER #22 PHASE C+ MODULE					84A241531A1	CA5	B9-B6,C6, D13-F4			
PS					+5VDC, ± 15VDC, ± 24VDC POWER SUPPLY					17FH43A3	CA1	E1-G0, E2-F0, E3-F0, E-B8,C8			
PS1S					+5VDC, ± 15VDC, ± 24VDC POWER SUPPLY CONNECTOR #1					41A296326ADP9	CA1	E1-F0,F1,F2,F3,F4,F5,F6,F7,G0,G1,G2,G3,G4,G5,G6,G7,G8,G9, E2-F0,F1,F2,F3,F4,F5,F6,F7			
PS2S					UNREGULATED 24VDC POWER SUPPLY CONNECTOR #2					41A271471P18	CA1	E3-E0,E1,E2			
R1					BATTERY BOOST RESISTOR					41A278283G2	CA3	B-C4,D4			
RDA					PHASE A RECTIFIER DIODE PANEL					17FM792A2	CA7	B1-B2,C2,F2			
RDB					PHASE B RECTIFIER DIODE PANEL					17FM792A2	CA7	B1-B3,D2,E2,E3,F3			
RDC					PHASE C RECTIFIER DIODE PANEL					17FM792A2	CA7	B1-B3,E2,F2,F3			
RG1A					RETARD GRID RESISTOR 1A (17EM146)					17EA33J732	RGB	B4-E7			
RG1B					RETARD GRID RESISTOR 1B (17EM146)					17EA33J651	RGB	B4-D7			
RG1C					RETARD GRID RESISTOR 1C (17EM146)					17EA33J565	RGB	B4-C7			
RG1D					RETARD GRID RESISTOR 1D (17EM146)					17EA33J965A	RGB	B4-B7			
RG2A					RETARD GRID RESISTOR 2A (17EM146)					17EA33J651	RGB	B4-B7			
RG2B					RETARD GRID RESISTOR 2B (17EM146)					17EA33J651	RGB	B4-C7			
RG2C					RETARD GRID RESISTOR 2C (17EM146)					17EA33J565	RGB	B4-D7			
RG2D					RETARD GRID RESISTOR 2D (17EM146)					17EA33J320	RGB	B4-E7			
RG3A					RETARD GRID RESISTOR 3A (17EM146)					17EA33J965A	RGB	B4-B8			
RG3B					RETARD GRID RESISTOR 3B (17EM146)					17EA33J965A	RGB	B4-C8			
RG3C					RETARD GRID RESISTOR 3C (17EM146)					17EA33J965A	RGB	B4-D8			
RG4A					RETARD GRID RESISTOR 4A (17EM146)					17EA33J965A	RGB	B4-D8			
RG4B					RETARD GRID RESISTOR 4B (17EM146)					17EA33J965A	RGB	B4-C8			
RG4C					RETARD GRID RESISTOR 4C (17EM146)					17EA33J965A	RGB	B4-B8			
RGB					RETARDING GRID BOX					17EM146	TRK	B4-B4			
RTN					RETURN GROUND STRIP					41A327436P1	CA1	E-E0,E2			
RTN2					RETURN 2 TERMINAL BOARD					84B530598ABG1	CA1	E-,G4,G5,G6,G7,G8,G9			
SS1					LEFT REAR WHEEL SPEED SENSOR					41A328039EBP4	TRK	D5-B8			
SS2					RIGHT REAR WHEEL SPEED SENSOR					41A328039EBP4	TRK	D11-B8			
SS3					LEFT FRONT WHEEL SPEED SENSOR					41A328039EBP4	TRK	D23-A1			
SS4					RIGHT FRONT WHEEL SPEED SENSOR					41A328039EBP4	TRK	D23-A3			
NOMENCL					DESCRIPTION					PART NUMBER	LOC	SCHEMATIC REF LOC			
T19					ALTERNATOR TERTIARY INSULATED CONNECTION POINT T19					41A260128P1	CA3	B-A5			
T20					ALTERNATOR TERTIARY INSULATED CONNECTION POINT T20					41A260128P1	CA3	B-B0			
TA11					INVERTER #11 PHASE A INSULATED CONNECTION POINT					41A260128P1	CA6	B10-A4			
TA12					INVERTER #12 PHASE A INSULATED CONNECTION POINT					41A260128P1	CA6	B10-B4			
TA21					INVERTER #21 PHASE A INSULATED CONNECTION POINT					41A260128P1	CA6	B11-A4			
TA22					INVERTER #22 PHASE A INSULATED CONNECTION POINT					41A260128P1	CA6	B11-B4			
TB1					TERMINAL BOARD #1					41A250095G10	CA1	D20-E3,E4,F3,F4, D29-A1,B1, K-C4,C5, U-E6,E8,F8			
TB11					INVERTER #11 PHASE B INSULATED CONNECTION POINT					41A260128P1	CA6	B10-C4			
TB12					INVERTER #12 PHASE B INSULATED CONNECTION POINT					41A260128P1	CA6	B10-D4			
TB2					TERMINAL BOARD #2					41A250095G10	CA1	K-F4,F5, U6-C1,C3,C4,C6,C8,F1,F3,F4			
TB21					INVERTER #21 PHASE B INSULATED CONNECTION POINT					41A260128P1	CA6	B11-C4			
TB22					INVERTER #22 PHASE B INSULATED CONNECTION POINT					41A260128P1	CA6	B11-D4			
TC11					INVERTER #11 PHASE C INSULATED CONNECTION POINT					41A260128P1	CA6	B10-F4			
TC12					INVERTER #12 PHASE C INSULATED CONNECTION POINT					41A260128P1	CA6	B10-G4			
TC21					INVERTER #21 PHASE C INSULATED CONNECTION POINT					41A260128P1	CA6	B11-F4			
TC22					INVERTER #22 PHASE C INSULATED CONNECTION POINT					41A260128P1	CA6	B11-G4			
TH1					ALTERNATOR FIELD THYRITE (VARISTOR)					41A259026P2	CA3	B-C6,C7			
TM1					AC TRACTION MOTOR (LEFT REAR WHEEL)					5GE25A	TRK	B10-D8			
TM2					AC TRACTION MOTOR (RIGHT REAR WHEEL)					5GE25A	TRK	B11-D8			
VAM1					INVERTER #1 LINK AND PHASE VOLTAGE MODULE #1					17FM681A1	CA3	B10-A4,A5,B4,B5, B6-D0,D1			
VAM1S					INVERTER #1 LINK AND PHASE VOLTAGE MODULE #1 CONNECTOR					41A271471P18	CA3	B10-A5,B5, B6-D1			
VAM2					INVERTER #2 LINK AND PHASE VOLTAGE MODULE #2					17FM681A1	CA3	B11-A4,A5,B4,B5, B8-D0			
VAM2S					INVERTER #2 LINK AND PHASE VOLTAGE MODULE #2 CONNECTOR					41A271471P18	CA3	B11-A5,B5, B8-D0			
VAM3					ALTERNATOR PHASE AND LINK VOLTAGE MODULE #3					17FM681A1	CA3	B1-B7,C7,F1,F2,G1,G2			
VAM3S					ALTERNATOR PHASE AND LINK VOLTAGE MODULE #3 CONNECTOR					41A271471P18	CA3	B1-B7,C6,C7,F1,G1			
VAM4					ALTERNATOR FIELD & TERTIARY AND GRID BLOWER VOLTAGE MODULE #4					17FM681A1	CA3	B4-C2, B-A1,B1,D5			
VAM4S					ALTERNATOR FIELD & TERTIARY AND GRID BLOWER VOLTAGE MODULE #4 CONNECTOR					41A271471P18	CA3	B4-C3,D3, B-A1,B1,B2,D6			
V-CN1					CAN #1 CABLE					84A238510P13	CA1	D26-C0			
V-CN2					CAN #2 CABLE					84A238510P14	CA1	D26-C1			
V-CN3					CAN #3 CABLE					84A238510P15	CA1	D26-C4			

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DEVICE TABLE

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NOMENCL	DESCRIPTION	PART NUMBER	LOC	SCHEMATIC REF LOC
V-ESS	VENDOR ENGINE SPEED SENSOR CABLE	84C620849G43	TRK	D22-C4
V-ET2	ETHERNET #2 CABLE	84A238510P10	CA1	D27-C3
V-ET4	ETHERNET #4 CABLE	84A238510P12	CA1	D28-D3

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0	MATHI.V 06APR2022 TC13			TITLE: NHL 240 TAC QC II SCHEMATIC	MADE BY: MATHI.V	DATE: 06APR2022	APPROVED BY: R.LUDWIGSON		84C654349
				FIRST MADE FOR: NHL 240 AC HIGH ALTITUDE	ISSUED BY: MATHI.V	DATE: 06APR2022	06APR2022		

ACES

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